

Knowledge Gradient for Bi-Objective Simulation Optimization with Probabilistic Constraints and Unknown Variances

Authors omitted for review

Abstract

We consider bi-objective simulation-based optimization problems in which both the objective functions and the constraint function are evaluated through expensive stochastic simulations. The decision space is a large finite set, the objectives are conflicting, and a probabilistic (chance) constraint must be satisfied. Because each simulation replication is computationally costly, a sample-efficient sequential procedure is essential. We propose a Gaussian-process-regression knowledge-gradient (GPR-KG) Bayesian optimization framework that integrates three key components. First, we adopt a parametric Gaussian process regression (GPR) belief model that exploits feature-based similarity across solutions, enabling generalization in large discrete spaces whose cardinality far exceeds the sampling budget. Second, we introduce a Variance Estimation Parametric Model (VEPM) that learns the heteroscedastic simulation noise from sampled solutions and extrapolates variance estimates to unvisited alternatives, eliminating the need for repeated replications at each design point. Third, we derive an approximate Knowledge Gradient (KG) sampling policy that combines objective-wise value-of-information measures with a posterior chance-feasibility filter and Pareto-based selection. On the theoretical side, we establish a finite-budget regret bound that is stated under explicit exploration and model-regularity conditions, prove cell-average consistency of the VEPM estimator, and provide a conditional feasibility bound for the probabilistic constraint. The framework is evaluated on three synthetic benchmark problems with extensive comparisons against state-of-the-art methods, and on a real-world urban traffic signal timing problem where saved simulation logs reveal sample-feasible Pareto improvements over the field-implemented plan. The case study also illustrates the feature–variance alignment condition under which VEPM improves

over, or degrades gracefully to, a pooled-variance baseline; fresh-seed out-of-sample certification is left as a necessary step before field deployment.

Keywords: simulation optimization; Bayesian optimization; Knowledge Gradient; multi-objective optimization; probabilistic constraints; heteroscedastic variance estimation

1 Introduction

1.1 Motivation

Many engineered and managed systems require the simultaneous optimization of multiple conflicting performance measures under uncertainty. When closed-form expressions for system performance are unavailable, stochastic simulation becomes the primary tool for evaluating candidate solutions. Simulation-based optimization (SO) seeks to identify the best design or operating policy by iteratively running simulation experiments and updating beliefs about performance (Fu, 2002; Hong et al., 2021). In practice, three structural features recur across a wide range of application domains: (i) the decision space is large and discrete, reflecting combinatorial or integer-valued design parameters; (ii) there are two or more conflicting objectives, so that a single “optimal” solution does not exist and one must instead characterize a Pareto front; and (iii) certain stochastic performance measures must satisfy probabilistic (chance) constraints, meaning that a threshold must be met with a prescribed probability rather than merely in expectation.

The following examples from diverse domains illustrate the ubiquity of this problem structure.

Example 1.1 (Urban traffic signal timing). Consider the design of signal timing plans for an urban traffic network. Two natural objectives are to minimize the average network delay (an efficiency measure) and to minimize the inequity of delay across different origin–destination pairs (a fairness measure). These objectives conflict because timing plans that favor high-volume corridors reduce average delay but increase inequity for minor movements. In addition, regulators may impose a probabilistic constraint on vehicle emissions—for instance, requiring that total CO₂ emissions remain below a threshold τ with probability at least $1 - \alpha$ —reflecting environmental regulations that must be satisfied reliably, not merely on average. Evaluating any given timing plan requires a microsimulation of the full traffic network, which is computationally expensive.

Example 1.2 (Manufacturing system design). In a semiconductor fabrication facility, managers seek to maximize throughput while minimizing work-in-process (WIP) inventory, two

objectives that are inherently conflicting due to Little’s law effects. A probabilistic constraint on the defect rate—requiring that the fraction of defective wafers remain below a critical level with high probability—must be enforced to meet quality standards. Discrete decisions include buffer sizes, lot release policies, and maintenance schedules. Detailed discrete-event simulation is the standard evaluation tool, and each replication may take minutes to hours.

Example 1.3 (Supply chain network configuration). A global supply chain must balance service level (e.g., fill rate) against total logistics cost. At the same time, carbon emission regulations impose a chance constraint: the probability that annualized emissions exceed a cap must remain below a specified level. Integer decisions include the number and location of distribution centers, inventory reorder points, and fleet composition. Monte Carlo simulation of demand, lead-time, and disruption scenarios is the only practical way to evaluate each configuration.

Example 1.4 (Healthcare resource allocation). In an emergency department, administrators wish to minimize patient waiting time and maximize staff utilization simultaneously. A probabilistic constraint on the mortality-related surrogate—for instance, requiring that the probability of the average time-to-treatment exceeding a clinically dangerous threshold is below α —encodes patient safety requirements. Staffing levels, triage protocols, and bed allocation rules define a large discrete decision space that can only be evaluated through patient-flow simulation.

Despite the diversity of these applications, the underlying mathematical structure is identical: a large discrete decision space $\mathcal{X}$, two conflicting stochastic objective functions f^1 and f^2 accessible only through noisy simulation, and a probabilistic constraint $\mathbb{P}(g(x, \xi) \leq \tau) \geq 1 - \alpha$ on a stochastic quantity. Enumerating even a modest fraction of $\mathcal{X}$ is computationally infeasible when each simulation replication is expensive. This motivates the development of sample-efficient sequential procedures grounded in Bayesian optimization (BO), which maintain a statistical belief model over the performance landscape and use an acquisition function to decide where to sample next (Forrester and Keane, 2009; Frazier, 2018).

Two additional complications arise in the setting described above. First, because the probabilistic constraint involves not only the mean but also the variance of the stochastic quantity $g(x, \xi)$, learning the simulation noise level is essential for assessing feasibility. In heteroscedastic settings the noise variance differs across solutions, yet allocating replications solely to estimate local variances diverts budget from global exploration of the objective landscape. Second, in a multi-objective context, the sampling policy must balance exploration of the full Pareto front rather than converging prematurely to a single region. Existing

BO methods address some of these challenges in isolation but, as we discuss next, no unified framework handles all of them simultaneously.

1.2 Literature Review

We organize the related literature into four streams and, for each, identify the gap that motivates the present work.

(a) Simulation-based optimization and Bayesian optimization. The survey by [Fu \(2002\)](#) provides a comprehensive overview of classical SO methods, including gradient-based search, random search, and ranking-and-selection. More recent surveys by [Amaran et al. \(2016\)](#) and [Hong et al. \(2021\)](#) document the growth of metamodel-based approaches, in which a surrogate model—most commonly a Gaussian process (GP)—is fitted to simulation data and used to guide sampling. [Forrester and Keane \(2009\)](#) review GP-based BO in the context of deterministic and stochastic computer experiments. While these works establish the broad methodological landscape, they do not address the joint challenge of multiple objectives and probabilistic constraints in discrete spaces with unknown heteroscedastic variances.

(b) Multi-objective Bayesian optimization. ParEGO ([Knowles, 2006](#)) scalarizes multiple objectives into a single composite function using random weight vectors and applies standard expected improvement. Expected hypervolume improvement (EHVI) directly targets the dominated hypervolume as the acquisition function ([Emmerich et al., 2006](#); [Daulton et al., 2020](#)). [Feliot et al. \(2017\)](#) propose a Bayesian multi-objective optimization (BMOO) framework with entropy-based acquisition. [Rojas Gonzalez and Van Nieuwenhuyse \(2020\)](#) provide a recent survey and taxonomy. Although these methods handle multiple objectives, they generally assume noiseless or homoscedastically noisy evaluations and do not incorporate probabilistic constraints. Furthermore, they are predominantly designed for continuous decision spaces and do not exploit the feature-based structure of large discrete alternatives.

(c) Constrained Bayesian optimization. [Gelbart et al. \(2014\)](#) introduce expected improvement with constraints (EIC), multiplying the unconstrained acquisition function by the posterior probability of feasibility. [Gramacy et al. \(2016\)](#) develop an augmented Lagrangian approach for constrained BO. [Letham et al. \(2019\)](#) propose constrained BO with noisy observations using a coupled GP model. [Eriksson and Poloczek \(2021\)](#) extend scalable constrained BO to high-dimensional settings via trust regions. These methods handle deterministic or expectation-based constraints of the form $\mathbb{E}[g(x, \xi)] \leq \tau$, but they do not address *probabilistic* constraints of the form $\mathbb{P}(g(x, \xi) \leq \tau) \geq 1 - \alpha$, which require estimating the *distribution* of $g(x, \xi)$ —and in particular its variance—rather than merely its mean.

(d) Knowledge Gradient policies. The Knowledge Gradient (KG) principle, introduced by Gupta and Miescke (1996) in the context of ranking and selection and formalized by Frazier et al. (2008), selects the measurement that maximizes the expected single-step improvement in the best posterior mean. Frazier et al. (2008) prove one-step optimality and asymptotic optimality for independent normal beliefs. Frazier et al. (2009) extend the KG to correlated normal beliefs (KGCB), deriving the updating equations $\boldsymbol{\mu}^{n+1} = \boldsymbol{\mu}^n + \tilde{\sigma}(\boldsymbol{\Sigma}^n, x^n) Z^{n+1}$, where $\tilde{\sigma}$ captures the posterior covariance structure. This extension is critical because it allows information from one sampled alternative to update beliefs about correlated alternatives. Scott et al. (2011) further generalize the KG to continuous parameter spaces (KGCP) by embedding the belief within a GP regression model. Negoescu et al. (2011) develop KG with linear belief models for drug discovery applications, showing that parametric structure dramatically reduces the effective dimensionality of the learning problem. Tan and Gao (2016) propose the Monte-Carlo Linear Belief KG (MCLB-KG) for network pricing, introducing a local variance updating mechanism. Qu et al. (2015) apply the KG to simulation optimization with input uncertainty.

Despite these advances, three gaps persist in the KG literature. *First*, no existing KG formulation handles probabilistic constraints on simulation-based stochastic quantities; the constraint mechanisms in the BO literature (stream (c) above) have not been integrated into the KG framework. *Second*, most KG studies focus on a single objective; multi-objective extensions with correlated beliefs across alternatives remain largely unexplored. *Third*, existing treatments of simulation variance within the KG paradigm either assume variance is known (Frazier et al., 2008, 2009), perform only local variance updates that do not generalize to unsampled solutions (Tan and Gao, 2016), or require intractable global covariance updates; none exploit feature-based similarity to extrapolate variance estimates in a parametric belief setting. The present paper addresses all three gaps simultaneously.

1.3 Summary of Contributions

The main contributions of this paper are fourfold.

- C1. Framework.** We develop the first KG-based Bayesian optimization framework for bi-objective simulation optimization with probabilistic constraints and unknown heteroscedastic variances in large discrete decision spaces. The framework integrates a parametric GPR belief model, a probabilistic constraint mechanism, and a multi-objective KG acquisition function into a coherent sequential sampling procedure.
- C2. Variance Estimation Parametric Model (VEPM).** We introduce a feature-based parametric model for the simulation noise variance that is fitted to observed sample vari-

ances and extrapolated to unvisited solutions. Unlike local variance estimation, which requires multiple replications at each sampled point, VEPM leverages the parametric feature structure to propagate variance information globally, reducing the sampling burden substantially.

C3. Theoretical guarantees. We establish three theoretical results: (i) a finite-budget regret bound showing that the optimality gap decreases at rate $O\left(\sqrt{p \log N/N}\right)$, where p is the feature dimension and N is the total simulation budget— under explicit exploration, separation, and model-regularity conditions; (ii) cell-average consistency of the VEPM variance estimator for partition cells that are visited sufficiently often; and (iii) a conditional feasibility bound for the probabilistic constraint.

C4. Computational validation. We validate the framework on a suite of synthetic bi-objective benchmark problems with probabilistic constraints, comparing against state-of-the-art multi-objective BO and constrained SO methods. We further demonstrate its practical value on a real-world urban traffic signal timing problem, where saved simulation logs contain sample-feasible signal plans that improve average travel time and delay equity relative to the current field-implemented plan. The case study also illustrates the predicted boundary condition for VEPM: when the variance-alignment diagnostic is near zero, VEPM is statistically indistinguishable from pooled variance.

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 formulates the bi-objective simulation optimization problem with probabilistic constraints and introduces the Bayesian framework. Section 3 presents the parametric GPR belief model and the VEPM variance estimation scheme. Section 4 derives the approximate KG sampling policy for bi-objective optimization with constraints. Section 5 establishes the theoretical guarantees. Section 6 reports computational experiments on the three synthetic benchmarks for which complete saved results are available, together with VEPM and candidate-set ablations. Section 7 presents the real-world case study on the RESCO ingolstadt21 traffic signal timing benchmark. Section 8 concludes the paper and discusses future research directions. All proofs are collected in the Online Supplement.

2 Problem Formulation

2.1 General Problem Statement

Let $\mathcal{X} \subset \mathbb{Z}^d$ be a large but finite discrete set of feasible solutions (alternatives), where d is the dimension of the decision vector. We denote the cardinality of $\mathcal{X}$ by $K = |\mathcal{X}|$. Each solution $x \in \mathcal{X}$ is associated with two objective functions

$$f^i(x) = \mathbb{E}[F^i(x, \xi)], \quad i = 1, 2, \quad (1)$$

and a constraint function

$$g(x, \xi), \quad (2)$$

where ξ is a random element representing the stochastic simulation input. The functions f^1 , f^2 , and the distribution of $g(x, \xi)$ are all unknown and can only be accessed through noisy simulation replications.

The bi-objective simulation optimization problem with a probabilistic constraint is

$$\min_{x \in \mathcal{X}} (f^1(x), f^2(x)) \quad \text{subject to} \quad \mathbb{P}(g(x, \xi) \leq \tau) \geq 1 - \alpha, \quad (3)$$

where $\tau \in \mathbb{R}$ is a prescribed threshold and $\alpha \in (0, 1)$ is the allowed violation probability. Here “min” is understood in the Pareto sense, defined formally below.

Definition 2.1 (Pareto dominance). A solution x *dominates* a solution x' , written $x \prec x'$, if $f^i(x) \leq f^i(x')$ for $i = 1, 2$ with at least one strict inequality.

Definition 2.2 (Pareto optimal set and Pareto front). Let $\mathcal{X}_{\text{feas}} = \{x \in \mathcal{X} : \mathbb{P}(g(x, \xi) \leq \tau) \geq 1 - \alpha\}$ denote the feasible set. The *Pareto optimal set* is

$$\mathcal{P}^* = \{x \in \mathcal{X}_{\text{feas}} : \nexists x' \in \mathcal{X}_{\text{feas}} \text{ such that } x' \prec x\}, \quad (4)$$

and the *Pareto front* is $\text{PF}^* = \{(f^1(x), f^2(x)) : x \in \mathcal{P}^*\}$.

The goal is to identify a good approximation of $\mathcal{P}^*$ (and hence PF^*) using a limited budget of N simulation replications. In typical applications, K ranges from thousands to millions, while N is at most a few hundred, making exhaustive evaluation of all alternatives infeasible.

Symbol	Meaning
$\mathcal{X}, x, d, K$	finite decision space, one solution, decision dimension, and number of solutions $K = \mathcal{X} $.
i	output index: $i = 1, 2$ for objectives and $i = 3$ for the constraint output.
$F^i(x, \xi), f^i(x)$	random simulation response and its mean for objective $i = 1, 2$.
$g(x, \xi), f^3(x)$	random constraint response and its mean, $f^3(x) = \mathbb{E}[g(x, \xi)]$.
$\xi, Y_n^i, \varepsilon_n^i$	simulation random input, observed output, and simulation noise.
$\tau, \alpha, q_{1-\alpha}(x)$	chance-constraint threshold, allowed violation probability, and $(1 - \alpha)$ -quantile of the constraint response.
$\sigma^i(x)^2, \hat{\sigma}_n^i(x)^2$	true simulation variance and its VEPM estimate.
N, n_0, n	total simulation budget, pre-sampling budget, and sequential stage index.
$\mathcal{F}_n, \mathcal{X}_n$	filtration generated by the first n observations and the set of distinct sampled solutions.
$\mu_n^i(x), s_n^i(x)^2$	posterior mean and posterior variance of $f^i(x)$.
$\varphi^i(x), p_i, \beta^i, \zeta^i(x)$	basis vector, basis dimension, parametric coefficients, and solution-specific deviation.
$\theta_n^i, a_n^i, C_n^i, \lambda_i$	augmented parameter vector, its posterior mean, posterior covariance, and prior deviation variance.
Ψ^i	design matrix whose row for solution x is $\varphi^i(x)^\top$; this symbol is reserved for design matrices.
Φ, ϕ	standard normal cumulative distribution function and density, respectively.
$c(x), \mathcal{C}, \mathcal{S}_{n,c}, R_{n,c}$	VEPM partition label, partition set, sampled solutions in partition c , and total observation count in that partition.
K_{med}	number of clusters in the medoid partition ablation; distinct from $K = \mathcal{X} $.
$\mathcal{A}_n, K_1, K_2$	KG candidate set, Latin-hypercube candidate count, and posterior-sampling NSGA-II run count.
$\mathcal{P}^*, \hat{\mathcal{P}}_N, \text{PF}^*$	true Pareto set, estimated posterior-feasible Pareto set, and true Pareto front.

Remark 2.1 (Why discrete). Although Bayesian optimization is often studied in continuous spaces, many practical SO problems are inherently discrete. Signal timing plans are specified by integer-valued phase durations; manufacturing buffer sizes and reorder points are integers; staffing levels and fleet sizes take discrete values. Treating such problems as continuous and rounding solutions post hoc can lead to infeasible or suboptimal designs, especially when the objective landscape is nonconvex. Our framework is specifically tailored to the discrete

setting, exploiting a feature-based parametric structure to enable generalization across the combinatorial space.

2.2 The Bayesian Optimization Framework

We adopt a sequential Bayesian approach. At each stage $n = 0, 1, \dots, N - 1$, the decision-maker selects a solution $x^n \in \mathcal{X}$ and observes three noisy simulation outputs:

$$Y_{n+1}^i = f^i(x^n) + \varepsilon_{n+1}^i, \quad i = 1, 2, \quad Y_{n+1}^3 = g(x^n, \xi_{n+1}), \quad (5)$$

where ε_{n+1}^i is the simulation noise for objective i and Y_{n+1}^3 is a single-replication observation of the constraint quantity. Defining $f^3(x) := \mathbb{E}[g(x, \xi)]$, the constraint observation can be equivalently written as $Y_{n+1}^3 = f^3(x^n) + \varepsilon_{n+1}^3$ with $\varepsilon_{n+1}^3 = g(x^n, \xi_{n+1}) - f^3(x^n)$, placing all three outputs in a unified framework.

Assumption 2.1 (Gaussian noise). For each $x \in \mathcal{X}$ and $i = 1, 2, 3$, the simulation output satisfies

$$Y^i = f^i(x) + \varepsilon^i, \quad \varepsilon^i \sim \mathcal{N}(0, \sigma^i(x)^2), \quad (6)$$

where the noise variances $\sigma^i(x)^2$ are unknown and may differ across both solutions and output types (heteroscedastic noise).

Remark 2.2 (Gaussian justification). The Gaussian assumption is standard in the SO literature and is motivated by the central limit theorem: when each simulation replication aggregates many random events (e.g., thousands of vehicle arrivals or patient encounters), the replication-level output is approximately normally distributed. In Section 6 we verify this assumption empirically via normality tests on the simulation outputs. We note that the Gaussian assumption also connects the probabilistic constraint to the quantile characterization developed in Section 2.3 below.

Let $\mathcal{F}_n = \sigma\{(x^0, \mathbf{Y}_1), \dots, (x^{n-1}, \mathbf{Y}_n)\}$ denote the filtration generated by the first n observations, where $\mathbf{Y}_k = (Y_k^1, Y_k^2, Y_k^3)$. The Bayesian belief model maintains, for each output i and each solution x , a posterior mean and posterior variance:

$$\mu_n^i(x) = \mathbb{E}[f^i(x) \mid \mathcal{F}_n], \quad s_n^i(x)^2 = \text{Var}(f^i(x) \mid \mathcal{F}_n). \quad (7)$$

When the belief model is a Gaussian process (or, more precisely in the discrete setting, a correlated multivariate normal), the full posterior is characterized by the mean vector $\boldsymbol{\mu}_n^i = (\mu_n^i(x))_{x \in \mathcal{X}} \in \mathbb{R}^K$ and the covariance matrix $\boldsymbol{\Sigma}_n^i \in \mathbb{R}^{K \times K}$ for each output i .

After spending the full budget of N simulations, the procedure returns an estimated Pareto set

$$\begin{aligned} \hat{\mathcal{P}}_N = \{x \in \hat{\mathcal{X}}_{\text{feas},N} : \nexists x' \in \hat{\mathcal{X}}_{\text{feas},N} \text{ such that} \\ \mu_N^1(x') \leq \mu_N^1(x), \quad \mu_N^2(x') \leq \mu_N^2(x), \\ (\mu_N^1(x'), \mu_N^2(x')) \neq (\mu_N^1(x), \mu_N^2(x))\}. \end{aligned} \quad (8)$$

where $\hat{\mathcal{X}}_{\text{feas},N}$ is the set of solutions deemed feasible based on the posterior beliefs at stage N (defined precisely in Section 2.3).

The central algorithmic question is how to choose $x^0, x^1, \dots, x^{N-1}$ sequentially to make $\hat{\mathcal{P}}_N$ as close to $\mathcal{P}^*$ as possible. The Knowledge Gradient policy, developed in Section 4, addresses this question by selecting at each stage the solution whose measurement yields the largest expected improvement in the current estimate of the Pareto front.

2.3 Reformulation via Quantile

The probabilistic constraint in (3) can be equivalently expressed in terms of a quantile function. Define the α -quantile of the constraint output at solution x as

$$q_\alpha(x) = \inf\{t \in \mathbb{R} : \mathbb{P}(g(x, \xi) \leq t) \geq \alpha\}. \quad (9)$$

Proposition 2.1 (Quantile reformulation). *The probabilistic constraint $\mathbb{P}(g(x, \xi) \leq \tau) \geq 1 - \alpha$ holds if and only if $q_{1-\alpha}(x) \leq \tau$; that is,*

$$\mathbb{P}(g(x, \xi) \leq \tau) \geq 1 - \alpha \iff q_{1-\alpha}(x) \leq \tau. \quad (10)$$

Under Assumption 2.1, the constraint output satisfies $g(x, \xi) \sim \mathcal{N}(f^3(x), \sigma^3(x)^2)$, so the quantile admits the closed-form expression

$$q_{1-\alpha}(x) = f^3(x) + \Phi^{-1}(1 - \alpha) \sigma^3(x), \quad (11)$$

where Φ^{-1} is the standard normal quantile function. For $\alpha < 0.5$, we have $\Phi^{-1}(1 - \alpha) > 0$, so a higher variance $\sigma^3(x)^2$ makes the constraint harder to satisfy. The feasible set becomes

$$\mathcal{X}_{\text{feas}} = \{x \in \mathcal{X} : f^3(x) + \Phi^{-1}(1 - \alpha) \sigma^3(x) \leq \tau\}. \quad (12)$$

This reformulation has two important consequences for the methodological development.

Remark 2.3 (Need for variance learning). Equation (11) shows that assessing feasibility requires knowledge of both $f^3(x)$ and $\sigma^3(x)$. In the standard BO setting, variance is either

assumed known or treated as a nuisance parameter; here, the variance directly enters the constraint and must be learned. This motivates the VEPM variance estimation scheme developed in Section 3, which provides estimates $\hat{\sigma}^3(x)$ for all $x \in \mathcal{X}$, including solutions that have never been sampled.

Remark 2.4 (Estimated feasible set). At stage n , we define the estimated feasible set as

$$\hat{\mathcal{X}}_{\text{feas},n} = \{x \in \mathcal{X} : \mu_n^3(x) + \Phi^{-1}(1 - \alpha) \hat{\sigma}_n^3(x) \leq \tau\}, \quad (13)$$

where $\mu_n^3(x)$ is the posterior mean of $f^3(x)$ and $\hat{\sigma}_n^3(x)$ is the VEPM estimate of $\sigma^3(x)$ at stage n . As n grows, both $\mu_n^3(x) \rightarrow f^3(x)$ and $\hat{\sigma}_n^3(x) \rightarrow \sigma^3(x)$, so $\hat{\mathcal{X}}_{\text{feas},n} \rightarrow \mathcal{X}_{\text{feas}}$; the formal statement is given in Section 5.

Remark 2.5 (Connection to Value-at-Risk). The quantile $q_{1-\alpha}(x)$ is precisely the Value-at-Risk (VaR) at level $1 - \alpha$ of the constraint random variable $g(x, \xi)$. Probabilistic constraints of the form considered here thus have a natural interpretation in risk management: the constraint requires that the “worst-case” outcome at confidence level $1 - \alpha$ remain within tolerance τ . Our framework can therefore be viewed as a risk-aware Bayesian optimization method.

3 Parametric Gaussian Process Regression Framework

This section develops the parametric Gaussian process regression (GPR) framework that serves as the statistical backbone of our algorithm. We introduce the parametric belief model for the mean functions (§3.1), derive the posterior updating equations (§3.2), present the Variance Estimation Parametric Model (VEPM) for unknown heteroscedastic variances (§3.3), and show how the posterior optimization problem decomposes into tractable sub-problems (§3.4).

3.1 Parametric Belief Model for Mean Functions

Recall from Section 2 that $\mathcal{X} \subseteq \mathbb{Z}^d$ is the finite set of feasible solutions with $K = |\mathcal{X}|$, and for each evaluation index $i = 1, 2, 3$ the true response function $f^i(x)$ must be learned from noisy simulation observations $Y^i \sim \mathcal{N}(f^i(x), \sigma^i(x)^2)$. A standard (non-parametric) GP prior over f^i would maintain a K -dimensional multivariate normal posterior. When K is very large—as is typical when $\mathcal{X}$ is formed by the Cartesian product of integer-valued ranges across d dimensions—this is computationally intractable. Our key modeling idea is to represent each f^i through a low-dimensional parametric structure augmented by a stochastic deviation

term.

Parametric representation. For each evaluation index $i \in \{1, 2, 3\}$, let $\varphi^i(x) \in \mathbb{R}^{p_i}$ be a vector of p_i basis functions of x , with $\varphi^i(x) = (\varphi_1^i(x), \dots, \varphi_{p_i}^i(x))^\top$. We model

$$f^i(x) = \varphi^i(x)^\top \beta^i + \zeta^i(x), \quad (14)$$

where $\beta^i = (\beta_1^i, \dots, \beta_{p_i}^i)^\top \in \mathbb{R}^{p_i}$ is a coefficient vector and $\zeta^i(x)$ is a solution-specific deviation term. In matrix form, stacking all solutions in $\mathcal{X}$,

$$f^i = \Psi^i \beta^i + \zeta^i, \quad \Psi^i \in \mathbb{R}^{K \times p_i}, \quad (15)$$

where the x -th row of Ψ^i is $\varphi^i(x)^\top$ and $\zeta^i = (\zeta^i(x))_{x \in \mathcal{X}}$.

Choice of basis functions. The basis functions φ_j^i can be any features of the decision vector x : polynomial terms, indicator functions, or domain-informed transformations. Throughout this paper we adopt *quadratic polynomials without cross terms*, i.e.,

$$\varphi^i(x) = (1, x_1, \dots, x_d, x_1^2, \dots, x_d^2)^\top, \quad (16)$$

giving $p_i = 2d + 1$ for each i . Omitting cross terms keeps p_i linear in d , which is essential for scalability to high-dimensional problems. When domain knowledge suggests that interactions between specific pairs of decision variables are important, the corresponding cross terms may be added selectively.

Prior specification. We place independent priors on the parametric and deviation components:

$$\beta^i \sim \mathcal{N}(a_{\beta,0}^i, C_{\beta,0}^i), \quad (17)$$

$$\zeta^i(x) \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \lambda_i), \quad \forall x \in \mathcal{X}, \quad (18)$$

where $a_{\beta,0}^i \in \mathbb{R}^{p_i}$ and $C_{\beta,0}^i \in \mathbb{R}^{p_i \times p_i}$ encode prior beliefs about the coefficients, and $\lambda_i > 0$ controls the magnitude of potential model misspecification. Larger λ_i allows the deviation terms to play a greater role, making the model more flexible at the cost of slower learning; smaller λ_i places more trust in the parametric structure.

Augmented parameter vector. Let $\mathcal{X}_n \subseteq \mathcal{X}$ denote the set of distinct solutions that have been sampled at least once by stage n . Define the *augmented parameter vector*

$$\theta_n^i = \begin{pmatrix} \beta^i \\ \zeta_n^i \end{pmatrix} \in \mathbb{R}^{p_i + |\mathcal{X}_n|}, \quad (19)$$

where $\zeta_n^i = (\zeta^i(x))_{x \in \mathcal{X}_n}$ collects the deviation terms for all sampled solutions. As new solutions are sampled, the dimension of θ_n^i grows by one for each previously unvisited solution. Correspondingly, define the *augmented feature vector* for any $x \in \mathcal{X}$:

$$\tilde{\varphi}_n^i(x) = \begin{cases} (\varphi^i(x)^\top, e_{x,n}^\top)^\top \in \mathbb{R}^{p_i + |\mathcal{X}_n|}, & \text{if } x \in \mathcal{X}_n, \\ (\varphi^i(x)^\top, \mathbf{0}^\top)^\top \in \mathbb{R}^{p_i + |\mathcal{X}_n|}, & \text{if } x \notin \mathcal{X}_n, \end{cases} \quad (20)$$

where $e_{x,n} \in \mathbb{R}^{|\mathcal{X}_n|}$ is the standard basis vector corresponding to x in the enumeration of $\mathcal{X}_n$. With this notation, $f^i(x) = \tilde{\varphi}_n^i(x)^\top \theta_n^i$ for all $x \in \mathcal{X}_n$. For $x \notin \mathcal{X}_n$, the predictive distribution integrates over the prior on $\zeta^i(x)$.

Remark 3.1 (Interpretation of the augmented feature vector). The augmented feature vector $\tilde{\varphi}_n^i(x)$ serves two roles simultaneously. The first p_i components $\varphi^i(x)$ encode the *global* parametric structure shared by all solutions, allowing information from any sampled solution to update beliefs at every other solution through the shared coefficient vector β^i . The remaining $|\mathcal{X}_n|$ components encode solution-specific *local* adjustments via the deviation terms $\zeta^i(x)$: when x has been sampled before, the indicator $e_{x,n}$ activates its individual deviation $\zeta^i(x)$, allowing the model to correct for local departures from the global parametric surface. When x is new (not yet sampled), all deviation components are zero, so the prediction relies entirely on the global model. This design naturally transitions from a “pure extrapolation” regime early in the experiment (when few solutions have been visited) to a “corrected interpolation” regime as the sampled set grows.

Remark 3.2 (Computational advantage of the parametric model). A standard GP posterior over f^i requires maintaining a $K \times K$ covariance matrix and performing $O(K^3)$ updates. The parametric belief model reduces the effective dimension from K to $p_i + |\mathcal{X}_n|$. Since the total sampling budget N satisfies $N \ll K$ in practice, the dimension of θ_n^i is at most $p_i + N$, which is far smaller than K . For instance, in the traffic signal timing application of Section 7, K exceeds 10^{70} while the reported GPR-KG-m run uses $p_i = 41$ and $N = 300$, so the parametric model operates in a space of dimension at most 341 rather than 10^{70} . This dimensional reduction is the key enabler for Bayesian learning in massive discrete solution spaces.

3.2 Posterior Updating Equations

We now derive the posterior updates for the augmented parameter vector θ_n^i after each simulation observation. If the observation variance is known, these updates are exact under the Gaussian model (14)–(18). In the implemented algorithm the true variance is unknown, so the same Kalman equations are used in an empirical-Bayes, plug-in form with the current VEPM estimate.

Proposition 3.1 (Posterior update for parametric belief). *Suppose that, conditional on the filtration $\mathcal{F}_n$ generated by the first n observations, the augmented parameter vector satisfies $\theta_n^i \sim \mathcal{N}(a_n^i, C_n^i)$ for each evaluation index $i \in \{1, 2, 3\}$. At stage n , we select solution $x^n \in \mathcal{X}$ and observe $Y_{n+1}^i \sim \mathcal{N}(\tilde{\varphi}_n^i(x^n)^\top \theta_n^i, \sigma^i(x^n)^2)$.*

(a) **Dimension augmentation.** *If $x^n \notin \mathcal{X}_n$, augment the posterior by appending $\zeta^i(x^n)$:*

$$a_n^i \leftarrow \begin{pmatrix} a_n^i \\ 0 \end{pmatrix}, \quad C_n^i \leftarrow \begin{pmatrix} C_n^i & \mathbf{0} \\ \mathbf{0}^\top & \lambda_i \end{pmatrix}, \quad (21)$$

and update $\mathcal{X}_{n+1} = \mathcal{X}_n \cup \{x^n\}$.

(b) **Bayesian update.** *Let $\tilde{e}_{x^n} = \tilde{\varphi}_{n+1}^i(x^n) \in \mathbb{R}^{p_i + |\mathcal{X}_{n+1}|}$ denote the augmented feature vector of the sampled solution (after any dimension augmentation). Then:*

$$a_{n+1}^i = a_n^i + \frac{C_n^i \tilde{e}_{x^n}}{\hat{\sigma}_n^{i,2}(x^n) + \tilde{e}_{x^n}^\top C_n^i \tilde{e}_{x^n}} (Y_{n+1}^i - \tilde{e}_{x^n}^\top a_n^i), \quad (22)$$

$$C_{n+1}^i = C_n^i - \frac{C_n^i \tilde{e}_{x^n} \tilde{e}_{x^n}^\top C_n^i}{\hat{\sigma}_n^{i,2}(x^n) + \tilde{e}_{x^n}^\top C_n^i \tilde{e}_{x^n}}, \quad (23)$$

where $\hat{\sigma}_n^{i,2}(x^n)$ is the current estimate of the simulation variance at x^n (see §3.3).

Proof. The proof follows from standard Bayesian linear regression with a known observation variance replaced by the plug-in estimate $\hat{\sigma}_n^{i,2}(x^n)$. The dimension augmentation step (21) incorporates the prior $\zeta^i(x^n) \sim \mathcal{N}(0, \lambda_i)$ when a new solution is sampled for the first time. The Bayesian update (22)–(23) is then the standard rank-one Kalman update. A detailed derivation is given in Appendix A1.1 of the Online Supplement. $\square$

The structure of the update equations deserves comment. The numerator $C_n^i \tilde{e}_{x^n}$ in (22) is the posterior covariance between the augmented coefficient vector θ_n^i and the prediction at x^n ; it encodes how much information about θ_n^i is revealed by an observation at x^n . The denominator $\hat{\sigma}_n^{i,2}(x^n) + \tilde{e}_{x^n}^\top C_n^i \tilde{e}_{x^n}$ is the predictive variance of Y_{n+1}^i at x^n , combining the

simulation noise $\hat{\sigma}_n^{i,2}(x^n)$ and the model uncertainty $\tilde{e}_{x^n}^\top C_n^i \tilde{e}_{x^n}$. The ratio is the Kalman gain. The covariance update (23) performs a rank-one deflation: after observing at x^n , the directions in parameter space correlated with $\tilde{e}_{x^n}$ are reduced in uncertainty.

The posterior moments of $f^i(x)$ for any $x \in \mathcal{X}$ follow immediately.

Corollary 3.1 (Posterior predictive moments). *For any $x \in \mathcal{X}$, the posterior mean and variance of $f^i(x)$ given $\mathcal{F}_n$ are:*

$$\mu_n^i(x) = \tilde{\varphi}_n^i(x)^\top a_n^i, \quad (24)$$

$$(\Sigma_n^i)_x = \tilde{\varphi}_n^i(x)^\top C_n^i \tilde{\varphi}_n^i(x) + \lambda_i \cdot \mathbf{1}[x \notin \mathcal{X}_n], \quad (25)$$

where the indicator term in (25) accounts for the prior uncertainty of $\zeta^i(x)$ when x has not yet been sampled.

Lemma 3.1 (Posterior feasibility condition). *By Proposition 2.1 and the Gaussian quantile formula (11), the probabilistic constraint $\mathbb{P}(g(x, \xi) \leq \tau) \geq 1 - \alpha$ is equivalent to the deterministic constraint*

$$f^3(x) + \Phi^{-1}(1 - \alpha) \sigma^3(x) \leq \tau. \quad (26)$$

Since $\Phi^{-1}(1 - \alpha) > 0$ for $\alpha < 0.5$, the constraint requires the mean $f^3(x)$ to lie sufficiently far below τ to absorb the variance penalty $\Phi^{-1}(1 - \alpha) \sigma^3(x)$; a solution with higher variance is harder to certify as feasible. Replacing the unknown quantities by their posterior estimates at the terminal stage N , the posterior feasibility condition becomes

$$\mu_N^3(x) + \Phi^{-1}(1 - \alpha) \hat{\sigma}_N^3(x) \leq \tau. \quad (27)$$

3.3 Variance Estimation Parametric Model (VEPM)

The posterior update equations (22)–(23) require the simulation variance $\sigma^i(x)^2$, and the feasibility condition (27) requires $\sigma^3(x)$. In practice, these variances are *unknown* and must be estimated from simulation output. The challenge is that when the total budget N is much smaller than K , most solutions are never sampled, and variance estimates must be *extrapolated* to unsampled solutions. This subsection presents the Variance Estimation Parametric Model (VEPM), one of the key methodological contributions of this paper.

Motivation and key idea. The standard approach to variance estimation is to compute the sample variance from replications of each solution. However, when $N \ll K$, the vast majority of solutions have zero replications, making direct estimation impossible. The key

observation is that solutions with similar features—as captured by their decision vectors—are likely to exhibit similar simulation variances. VEPM formalizes this intuition through a partition-based variance sharing scheme.

Partition scheme. We construct partition features from the normalized decision vector $\bar{x} = (\bar{x}_1, \dots, \bar{x}_d) \in [0, 1]^d$, where $\bar{x}_j = (x_j - x_j^L)/(x_j^U - x_j^L)$ maps each component to the unit interval. Define $2d$ *partition features*:

$$\phi^{\text{part}}(x) = (\bar{x}_1, \dots, \bar{x}_d, \bar{x}_1^2, \dots, \bar{x}_d^2).$$

For each feature $k \in \{1, \dots, 2d\}$, assign a binary bin index $c_k(x) = \mathbb{I}\{\phi_k^{\text{part}}(x) > 0.5\} \in \{0, 1\}$, and define the *partition combination* as

$$c(x) = (c_1(x), c_2(x), \dots, c_{2d}(x)) \in \{0, 1\}^{2d}. \quad (28)$$

Two solutions x and x' share a common partition combination if and only if $c(x) = c(x')$. The total number of distinct partition combinations is $|\mathcal{C}| = 2^{2d}$. The inclusion of both linear and quadratic features ensures that the partition captures variance heterogeneity arising from both the location and curvature of the response surface; for instance, solutions near the boundary ($\bar{x}_j \approx 0$ or 1) may have different noise characteristics than those near the center.

VEPM updating rules. Let $\hat{\sigma}_0^{i,2}(c)$ denote an initial variance estimate for partition combination $c \in \mathcal{C}$, obtained from prior knowledge or a pre-sampling phase. At each stage n , after sampling solution x^n and observing Y_{n+1}^i , we update the variance estimates as follows.

Step (a): Update the individual variance estimate for x^n . Let n_{x^n} denote the number of times solution x^n has been sampled up to and including stage $n + 1$, and let $Y_l^i(x^n)$ be the observation from the l -th sample of x^n obtained at stage s_l . The residual for this observation is computed using the posterior mean *immediately before* assimilating that observation. Define

$$\hat{\sigma}_{n+1}^{i,2}(x^n) = \frac{w_i \cdot \hat{\sigma}_0^{i,2}(c(x^n)) + \sum_{l=1}^{n_{x^n}} (Y_l^i(x^n) - \mu_{s_l}^i(x^n))^2}{w_i + n_{x^n}}, \quad (29)$$

where $w_i > 0$ is a weight controlling the influence of the initial estimate and $\mu_{s_l}^i(x^n)$ is the pre-update posterior mean of $f^i(x^n)$ at the stage s_l when the l -th observation was collected. The numerator is a weighted sum of a prior term (the partition-average initial estimate, weighted by w_i) and empirical squared residuals. The squared residuals $(Y_l^i - \mu_{s_l}^i)^2$ use the

stage- s_l posterior mean as a proxy for the unknown true mean $f^i(x^n)$; the implementation therefore caches $\mu_n^i(x^n)$ before the Kalman posterior update and passes this cached value to VEPM. As the posterior mean converges to the true mean, these residuals converge to squared noise terms, so the formula is a consistent estimator of $\sigma^i(x^n)^2$ (Theorem 5.1).

Step (b): Update the common variance for the partition combination. Let $\mathcal{S}_{n+1,c} = \{x' \in \mathcal{X}_{n+1} : c(x') = c\}$ be the set of sampled solutions whose partition combination is c , and let $n_{x'}$ denote the number of times x' has been sampled. Update the common estimate as a *sample-count weighted average*:

$$\hat{\sigma}_{n+1,c}^{i,2} = \frac{\sum_{x' \in \mathcal{S}_{n+1,c}} n_{x'} \hat{\sigma}_{n+1}^{i,2}(x')}{\sum_{x' \in \mathcal{S}_{n+1,c}} n_{x'}}. \quad (30)$$

The weighting by $n_{x'}$ gives higher influence to solutions with more observations, whose individual variance estimates are more accurate. When all solutions have been sampled the same number of times, this reduces to the unweighted average.

Step (c): Assign variance estimates to all solutions. For any $x \in \mathcal{X}$:

$$\hat{\sigma}_{n+1}^{i,2}(x) = \begin{cases} \text{value from (29),} & \text{if } x \in \mathcal{X}_{n+1}, \\ \hat{\sigma}_{n+1,c(x)}^{i,2}, & \text{if } x \notin \mathcal{X}_{n+1}. \end{cases} \quad (31)$$

Lemma 3.2 (VEPM recursive update). *The batch formulas (29)–(30) admit the following efficient recursive updates. Let $c^* = c(x^n)$ be the partition combination of the sampled solution, and let $R_{n,c} = \sum_{x' \in \mathcal{S}_{n,c}} n_{x'}$ denote the total sample count in partition c at stage n .*

If $x^n \in \mathcal{X}_n$ (a previously sampled solution is revisited):

$$\hat{\sigma}_{n+1}^{i,2}(x^n) = \frac{(w_i + n_{x^n} - 1) \hat{\sigma}_n^{i,2}(x^n) + (Y_{n+1}^i - \mu_n^i(x^n))^2}{w_i + n_{x^n}}, \quad (32)$$

$$\hat{\sigma}_{n+1,c^*}^{i,2} = \frac{\sum_{x' \in \mathcal{S}_{n+1,c^*}} n_{x'} \hat{\sigma}_{n+1}^{i,2}(x')}{R_{n+1,c^*}}. \quad (33)$$

If $x^n \notin \mathcal{X}_n$ (a new solution is sampled for the first time):

$$\hat{\sigma}_{n+1}^{i,2}(x^n) = \frac{w_i \cdot \hat{\sigma}_0^{i,2}(c^*) + (Y_{n+1}^i - \mu_n^i(x^n))^2}{w_i + 1}, \quad (34)$$

$$\hat{\sigma}_{n+1,c^*}^{i,2} = \frac{R_{n,c^*} \cdot \hat{\sigma}_{n,c^*}^{i,2} + n_{x^n} \hat{\sigma}_{n+1}^{i,2}(x^n)}{R_{n,c^*} + n_{x^n}}. \quad (35)$$

For all other solutions $x \neq x^n$ and partition combinations $c \neq c^*$, the estimates remain unchanged.

Proof. For the revisit case, Eq. (32) is a standard online update of the weighted running average. Eq. (33) follows directly from (30): only $\hat{\sigma}^{i,2}(x^n)$ changes in the partition, and the sample-count weighted average is recomputed with the new value and updated count $R_{n+1,c^*} = R_{n,c^*} + 1$. For the new-solution case, Eq. (34) follows from (29) with $n_{x^n} = 1$, and Eq. (35) extends the sample-count weighted average by one new member with count $n_{x^n} = 1$. See Appendix A1.2 of the Online Supplement for the full derivation. $\square$

Remark 3.3 (Computational cost of VEPM). At each stage, the VEPM update modifies only the variance estimate for the sampled solution x^n and the common estimate for its partition combination $c(x^n)$. The implementation recomputes the sample-count weighted average over the sampled solutions in that cell, giving $O(|\mathcal{S}_{n,c(x^n)}|)$ arithmetic per output. This is negligible in the reported finite-budget experiments because visited cells are small; an equivalent running-sum implementation would reduce this step to $O(1)$ without changing the estimator.

Remark 3.4 (Implementation note on variance estimates). The reported algorithm uses the VEPM plug-in estimate in (31) directly in the Kalman denominator, the KG computation, and the posterior feasibility check. A small numerical lower bound may be used only to avoid division by zero in floating-point arithmetic; it is not a statistical shrinkage modification and it does not alter the VEPM recursion (32)–(35).

Remark 3.5 (Partition granularity and the budget–cardinality regime). The total number of partition combinations is $|\mathcal{C}| = 2^{2d}$, which grows exponentially in d . The qualitative behaviour of VEPM divides into two regimes determined by the ratio of cardinality to budget.

Information-rich regime ($|\mathcal{C}| \ll N$). When the partition is small enough that each combination is visited multiple times under the sampling policy, the common estimate $\hat{\sigma}_{n,c}^{i,2}$ in (30) pools several empirical squared residuals and is updated online by the sample-count weighted average (33)/(35). This is the regime in which the consistency result of Theorem 5.1 delivers its full content; empirically, VEPM markedly outperforms a pooled-variance ablation here (on RZDT1/RZDT2 with $d = 5$, $|\mathcal{C}| = 2^{10} = 1024$, the KG sampler concentrates revisits

in a ~ 30 -solution subset whose partition combinations coincide with the heteroscedasticity equicontours of $\sigma^i(x_1)$; see §6.5).

Saturation regime ($|\mathcal{C}| \gtrsim N$). When $|\mathcal{C}|/N \rightarrow \infty$ along a problem family, the expected occupancy $N/|\mathcal{C}|$ vanishes; under any non-degenerate sampling policy, almost every partition combination is either unsampled or singleton. In this regime the partition-level pooling in (33) contributes negligibly to the variance estimate at unsampled x , and the cell-average consistency in Theorem 5.1 loses empirical bite because the conditioning event $\{R_{N,c} \rightarrow \infty\}$ holds for vanishingly few combinations. This degeneracy is formalised in Corollary 5.1 below.

Three additional observations clarify the partition scheme. First, the joint use of $\bar{x}_j$ and $\bar{x}_j^2$ as features captures both location-based and curvature-based variance heterogeneity, but the two are partially redundant on $\bar{x}_j \in [0, 1]$: the conditions $\bar{x}_j > 0.5$ and $\bar{x}_j^2 > 0.5 \Leftrightarrow \bar{x}_j > \sqrt{0.5} \approx 0.707$ yield only three distinct one-dimensional sub-intervals, so the effective number of populated combinations is at most 3^d rather than 4^d . Second, the threshold 0.5 is a default choice that is agnostic about where variance heterogeneity is concentrated; a data-adaptive feature map (§3.3.1) is in general preferable when prior information about the noise function is available. Third, partition granularity affects only the variance estimate $\hat{\sigma}_n^{i,2}(x)$; the parametric mean belief (24) is unaffected and continues to enable Pareto front identification through the shared coefficient vector β^i .

The information-rich regime is the operating regime for the synthetic experiments of §6; the saturation regime is the operating regime for the high-dimensional engineering case study of §7. The empirical consequences of each regime are made precise by the alignment-condition framework of §3.3.1.

3.3.1 Feature–variance alignment as the precondition for VEPM

Theorems 5.1–5.2 characterize what VEPM is consistent for: the sampling-policy weighted cell-average $\bar{\sigma}_c^{i,2}$, with an irreducible bias term δ_c^i controlling the gap to the pointwise variance $\sigma^i(x)^2$. In this subsection we formalize the assumption that determines whether the partition $\mathcal{C}$ delivers genuine heteroscedasticity-aware variance estimation or degenerates to a global pooled estimator.

Decomposition of the heteroscedastic error. For any feature map $\phi : \mathcal{X} \rightarrow \mathbb{R}^q$ and any partition $\mathcal{C}$ of $\mathbb{R}^q$, define the *conditional cell-average variance*

$$\mu_\sigma^i(c; \phi, \mathcal{C}) = \mathbb{E}_{X \sim \pi}[\sigma^i(X)^2 \mid \phi(X) \in c], \quad (36)$$

where π is any non-degenerate sampling distribution over $\mathcal{X}$. The Pythagorean decomposition of $\mathbb{E}_\pi[\sigma^i(X)^2]$ then yields

$$\underbrace{\mathbb{V}_{X \sim \pi}[\sigma^i(X)^2]}_{\text{total heteroscedastic variation}} = \underbrace{\mathbb{V}_c[\mu_\sigma^i(c; \phi, \mathcal{C})]}_{\text{between-cell, captured by VEPM}} + \underbrace{\mathbb{E}_c[\mathbb{V}_{X|c}[\sigma^i(X)^2]]}_{\text{within-cell, not captured}}, \quad (37)$$

where the between/within means are with respect to the cell distribution induced by $\pi \circ \phi^{-1}$. The asymptotic bias term δ_c^i of Assumption 5.3 and Theorem 5.2 is precisely the within-cell residual: $(\delta_c^i)^2 \leq \mathbb{V}_{X|c}[\sigma^i(X)^2]$.

Alignment ratio. The fraction of total variance captured by the feature-induced partition defines the *alignment ratio*

$$R_\phi^i(\mathcal{C}) = \frac{\mathbb{V}_c[\mu_\sigma^i(c; \phi, \mathcal{C})]}{\mathbb{V}_{X \sim \pi}[\sigma^i(X)^2]} \in [0, 1]. \quad (38)$$

$R_\phi^i = 1$ corresponds to a partition that perfectly separates solutions by their pointwise variance ($\delta_c^i = 0$ for every cell); $R_\phi^i = 0$ corresponds to a partition uncorrelated with the heteroscedastic structure, in which case (37) reduces to “all variation is within-cell” and the cell-average estimate $\hat{\sigma}_{N,c}^{i,2}$ converges to $\mathbb{E}_\pi[\sigma^i(X)^2]$ regardless of c — i.e. VEPM coincides with the pooled global variance.

In practice R_ϕ^i is unobservable because $\sigma^i(x)^2$ is not known, but it can be estimated from short replication runs: given M plans visited R times each in a pilot phase, fit a flexible regressor $\hat{g}$ on $\{(\phi(x_m), \log \hat{\sigma}^{i,2}(x_m))\}_{m=1}^M$ and read off the cross-validated coefficient of determination $R_{\text{cv}}^2(\hat{g})$. This R_{cv}^2 is a finite-sample proxy for R_ϕ^i and provides a computable *alignment diagnostic* that one runs before committing simulation budget to VEPM.

The alignment condition. We formalize when VEPM provides a genuine improvement over pooled variance estimation:

Assumption 3.1 (Alignment). There exists $\rho_\star^i > 0$ such that $R_\phi^i(\mathcal{C}) \geq \rho_\star^i$ for every cell c that receives positive sampling probability under the eventual KG sampling policy.

When Assumption 3.1 holds, the partition genuinely separates high-variance from low-variance regions of the decision space, δ_c^i is bounded away from $\sqrt{\mathbb{V}_\pi[\sigma^{i,2}]}$, and Theorem 5.2 guarantees that VEPM’s L_2 error decreases to the alignment-irreducible floor $(1 - \rho_\star^i) \cdot \mathbb{V}_\pi[\sigma^{i,2}]$. When the assumption fails, VEPM converges to a constant $\hat{\sigma}_{N,c}^{i,2} \equiv \mathbb{E}_\pi[\sigma^{i,2}]$ in the cell-average limit and adds no information beyond pooling.

Remark 3.6 (Practical implications of Assumption 3.1). Three observations follow. (i) Choosing the partition feature map ϕ amounts to a modeling commitment about which features of the decision vector drive simulation variance; the polynomial features $\phi(x) = (\bar{x}, \bar{x}^2)$ adopted throughout this paper are a generic default that succeeds when $\sigma^i(x)^2$ is well-approximated by a low-order polynomial in x . (ii) If a problem-specific reason exists to expect $\sigma^i(x)$ to depend on different summaries of x (e.g. a count of active components, a queueing index, an aggregate utilisation measure), ϕ should be augmented accordingly. (iii) Crucially, when the alignment diagnostic returns $R_{cv}^2 \approx 0$, the algorithm is not damaged but is reduced to the pooled-variance baseline. Section 6.5 provides an empirical instantiation across all four experiments in this paper.

3.3.2 Argmax invariance and the dominance of sol_variance

The alignment-ratio analysis of §3.3.1 characterises how well the partition $\mathcal{C}$ captures heteroscedastic variation in $\sigma^i(x)^2$, but it is silent on a more subtle question: *does the variance estimate $\hat{\sigma}_n^{i,2}(x)$ that VEPM returns actually change the KG sampling decision?* In this subsection we show that the answer is "rarely", and we identify when it does.

Decomposition of the variance lookup. The VEPM lookup (31) returns one of two channels: the individual sol-variance estimate $\hat{\sigma}_n^{i,2}(x)$ (recursive average (32) of squared residuals at x) for sampled $x \in \mathcal{X}_n$, and the partition-common estimate $\hat{\sigma}_{n,c(x)}^{i,2}$ (sample-count weighted average (33) over solutions in cell $c(x)$) otherwise. The first channel is *partition-invariant*: $\hat{\sigma}_n^{i,2}(x)$ depends only on the observed squared residuals at x , not on the partition $\mathcal{C}$. The second channel is partition-dependent, but is only queried for unvisited solutions (and only for candidates outside $\mathcal{X}_n$ that appear in the LHD/posterior candidate sets $\mathcal{A}_n$).

Magnitude analysis of the feasibility margin. The chance-constraint feasibility margin in the KG acquisition involves $\Phi^{-1}(1 - \alpha)\sqrt{\hat{\sigma}_n^{i,2}(x)}$. For typical RZDT calibrations with $\sigma = 0.04$ and n_0 pre-samples, the early-iter VEPM estimates satisfy $\hat{\sigma}^{i,2} \in [10^{-7}, 10^{-6}]$, so the safety margin $1.645\sqrt{\hat{\sigma}^{i,2}} \in [5 \cdot 10^{-4}, 1.6 \cdot 10^{-3}]$ is two orders of magnitude smaller than the candidate-wise dispersion of $\mu_n^i(x)$, which is of order 10^{-1} on RZDT1. As a consequence, the iter- n KG argmax over $\mathcal{A}_n$ is dominated by $\mu_n^i(x)$ and is invariant to partition-induced perturbations of $\hat{\sigma}_n^{i,2}(\cdot)$ of typical magnitude.

Empirical confirmation. Table 1 summarises a fixed-seed diagnostic in which we instantiate VEPM under both the ϕ_{Zheng} and ϕ_{agg} partitions, generate the same 50-element LHD candidate set at iteration 1, and compare the per-candidate variance estimates and

Table 1: Argmax invariance diagnostic, RZDT1 at seed 1000, iter 1. Two VEPM partitions, the same 50-candidate LHD set, and the same feasibility-aware acquisition $\alpha_i(x) = -[\mu_n^i(x) + 1.645\sqrt{\hat{\sigma}_n^{i,2}(x)}]$. All values shown for the constraint objective $i = 3$.

Statistic	ϕ_{Zheng}	ϕ_{agg}
Candidates with different $\hat{\sigma}_n^{i,2}$ across the two partitions	44/50	
min / max $\hat{\sigma}_n^{i,2}$ across candidates	$[2.6 \cdot 10^{-7}, 2.0 \cdot 10^{-6}]$	$[2.5 \cdot 10^{-7}, 2.0 \cdot 10^{-6}]$
Typical safety margin $1.645\sqrt{\hat{\sigma}_n^{i,2}}$	$\sim 10^{-3}$	$\sim 10^{-3}$
Range of μ_n^i across candidates	$[-0.350, +0.176]$	$[-0.350, +0.176]$
Top-1 candidate (argmax of α_i)	40	40
Top-5 candidate ranking	{40, 47, 44, 16, 8}	{40, 47, 44, 16, 8}

the resulting argmax over a feasibility-aware acquisition proxy. Although 44 of 50 candidates receive different $\hat{\sigma}_n^{i,2}$ values under the two partitions, the top-5 acquisition ranking is bit-identical and the argmax is the same candidate. This explains the bit-identical HV reported for ϕ_{Zheng} and ϕ_{agg} in Tables 2–4: the two algorithms make the same sequence of 200 sampling decisions and therefore produce the same Pareto front.

Sufficient conditions for argmax invariance. Let $\Delta\sigma$ be the supremum over the candidate set of $|\hat{\sigma}_{n,\text{Zheng}}^{i,2}(x) - \hat{\sigma}_{n,\text{agg}}^{i,2}(x)|$, and let $\Delta\mu$ be the gap between the top-1 and top-2 acquisition values under either partition. When

$$\begin{aligned} \Phi^{-1}(1 - \alpha) (\sqrt{\hat{\sigma}_{n,\text{Zheng}}^{i,2}} + \sqrt{\hat{\sigma}_{n,\text{agg}}^{i,2}}) \\ \times \Delta\sigma^{1/2} \leq \Delta\mu, \end{aligned} \tag{39}$$

the top-1 candidate is invariant to the partition choice. For the three RZDT benchmarks reported in Tables 2–4, the empirical σ -margins satisfy this condition for every iteration with overwhelming probability, hence the bit-identical trajectories.

When does the partition matter? Equation (39) fails when partitions produce $\hat{\sigma}_n^{i,2}$ estimates of much *larger* magnitude than the information-rich regime would predict. The medoid- K_{med} variant introduced as an ablation in §6.4 has this property: k -means cluster boundaries occasionally yield large variance estimates for boundary candidates, breaking (39) and producing the -12% HV degradation reported on RZDT1. This identifies a useful practical *negative* ablation: partitions whose cell geometry amplifies rather than smooths $\hat{\sigma}_n^{i,2}$ can be actively harmful to KG argmax stability.

Implication for the VEPM design. Combining the alignment-condition (§3.3.1) and argmax-invariance arguments, VEPM’s heteroscedasticity-capture benefit in chance-constrained Pareto KG is dominated by the *sol-variance* channel: plan-specific squared-residual averages (32) that update with every replication. The partition channel $\hat{\sigma}_{n,c}^{i,2}$ provides extrapolation to unsampled candidates but, in the typical operating regime, exerts only a small modulation on the KG argmax. The strongest empirical signature of VEPM is therefore not the gap between two partition schemes (which the alignment-failure regime predicts to vanish, and which (39) predicts to vanish even when alignment holds, as long as variance magnitudes are typical) but the gap between any VEPM variant and a purely-pooled estimator that lacks the sol-variance channel. This is exactly the pattern reported in Tables 2–4.

3.4 Decomposition of the Posterior Problem

At the terminal stage N , the decision-maker solves the *posterior optimization problem* using the learned beliefs:

$$\min_{x \in \mathcal{X}} (\mu_N^1(x), \mu_N^2(x)) \quad \text{s.t.} \quad \mu_N^3(x) + \Phi^{-1}(1 - \alpha) \hat{\sigma}_N^3(x) \leq \tau. \quad (40)$$

We now show that the VEPM structure enables a decomposition that makes this problem tractable even when K is enormous.

Proposition 3.2 (Posterior decomposition). *Let $\hat{\mathcal{P}}_N^*$ denote the Pareto optimal set of (40). Define, for each partition combination $c \in \mathcal{C}$:*

- (i) $\mathcal{U}_{N,c} = \{x \in \mathcal{X} \setminus \mathcal{X}_N : c(x) = c\}$, the unsampled solutions with partition combination c ;
- (ii) $\tau_c = \tau - \Phi^{-1}(1 - \alpha) \hat{\sigma}_{N,c}^3$, the effective constraint threshold for partition combination c (a constant for all $x \in \mathcal{U}_{N,c}$, since unsampled solutions share the common variance estimate (31)); note that $\Phi^{-1}(1 - \alpha) > 0$ for $\alpha < 0.5$, so $\tau_c < \tau$, i.e., the effective threshold is stricter than the nominal one to account for estimated simulation noise in the chance constraint;
- (iii) The sub-problem for partition combination c :

$$\hat{\mathcal{P}}_{N,c} = \text{ParetoMin}_{x \in \mathcal{U}_{N,c}} (\mu_N^1(x), \mu_N^2(x)) \quad \text{s.t.} \quad \mu_N^3(x) \leq \tau_c; \quad (41)$$

- (iv) $\hat{\mathcal{P}}_{N,\text{sam}}$: the Pareto optimal solutions from the sampled set $\mathcal{X}_N$ that satisfy the individual feasibility condition $\mu_N^3(x) + \Phi^{-1}(1 - \alpha) \hat{\sigma}_N^3(x) \leq \tau$.

Then the overall Pareto optimal set is obtained as

$$\hat{\mathcal{P}}_N^* = \text{ParetoMin} \left(\hat{\mathcal{P}}_{N,\text{sam}} \cup \bigcup_{c \in \mathcal{C}} \hat{\mathcal{P}}_{N,c} \right). \quad (42)$$

Proof. The key observation is that, for any unsampled solution $x \notin \mathcal{X}_N$, the posterior mean $\mu_N^i(x) = \varphi^i(x)^\top a_{\beta,N}^i$ (the deviation term $\zeta^i(x)$ has zero posterior mean since x was never sampled) and the variance estimate $\hat{\sigma}_N^{i,2}(x) = \hat{\sigma}_{N,c(x)}^{i,2}$ depends only on $c(x)$. Therefore, the feasibility constraint for all solutions in $\mathcal{U}_{N,c}$ reduces to $\mu_N^3(x) \leq \tau_c$ with a common right-hand side, and each sub-problem (41) is self-contained. For sampled solutions $x \in \mathcal{X}_N$, the individual variance estimates $\hat{\sigma}_N^3(x)$ differ across solutions, so they must be checked individually. The union and final Pareto filtering in (42) ensure that no dominated solution is retained. $\square$

Remark 3.7 (Computational benefit of decomposition). Proposition 3.2 reduces the posterior problem from a single bi-objective optimization over all K solutions to $|\mathcal{C}|$ sub-problems (one per partition combination), each over $|\mathcal{U}_{N,c}|$ solutions, plus a check over the $|\mathcal{X}_N| \leq N$ sampled solutions. When the parametric structure is a good approximation of the true response surface, each sub-problem (41) can be solved efficiently because $\mu_N^i(x) = \varphi^i(x)^\top a_{\beta,N}^i$ is a known function of x (e.g., a quadratic polynomial), and the constraint threshold τ_c is a constant. For quadratic basis functions, each sub-problem is a constrained bi-objective quadratic integer program. In moderate discrete spaces it can be solved by enumeration or integer-programming routines. In the large-scale implementation used in Sections 6–7, we instead use the decomposition as the feasibility and dominance structure for a finite candidate approximation: the terminal solve evaluates all visited solutions and an additional random lattice sample, filters them by the posterior chance constraint, and reports the non-dominated subset. Thus the proposition gives the exact structural target, whereas the code deliberately returns a scalable approximation to that target.

4 Knowledge Gradient Sampling Policy

The parametric GPR framework of Section 3 provides the statistical model for learning the response surfaces. The remaining question is: *which solution should we simulate at each stage to learn as efficiently as possible?* This section develops a Knowledge Gradient (KG) sampling policy tailored to the bi-objective constrained setting.

4.1 The Knowledge Gradient Factor

The Knowledge Gradient measures the expected marginal value of information from one additional simulation. Since our problem minimizes both objectives, we define the KG factor in minimization form: for each evaluation index $i \in \{1, 2\}$ and a candidate solution $x \in \mathcal{X}$, the KG factor at stage n is

$$\nu_n^{\text{KG},i}(x) = \min_{x' \in \mathcal{A}_n} \mu_n^i(x') - \mathbb{E}_n \left[\min_{x' \in \mathcal{A}_n} \mu_{n+1}^i(x') \mid x^n = x \right], \quad (43)$$

where $\mathcal{A}_n \subseteq \mathcal{X}$ is a *candidate set* (defined in §4.2) and the expectation is taken over the random observation Y_{n+1}^i . The sign convention is chosen so that $\nu_n^{\text{KG},i}(x) \geq 0$ always, with larger values indicating greater expected reduction in the best posterior estimate of objective i . Sampling x cannot worsen the best posterior estimate in expectation (by the optional stopping theorem), so the KG factor is non-negative. Note that the KG factor is computed only for the two objective indices $i \in \{1, 2\}$; the constraint function ($i = 3$) is handled implicitly by filtering candidate solutions through the estimated feasibility condition (13) within the posterior decomposition (Proposition 3.2). This design choice avoids a three-way KG aggregation that would require specifying trade-offs between objective information and constraint information; instead, the constraint is treated as a hard filter throughout the algorithm.

Lemma 4.1 (KG factor via standard normal representation). *Define the posterior sensitivity vector $\tilde{\sigma}^i(x) \in \mathbb{R}^{|\mathcal{A}_n|}$ with entries indexed by $x' \in \mathcal{A}_n$:*

$$\tilde{\sigma}_{x'}^i(x) = \frac{\tilde{\varphi}_n^i(x')^\top C_n^i \tilde{e}_x}{\sqrt{\hat{\sigma}_n^{i,2}(x) + \tilde{e}_x^\top C_n^i \tilde{e}_x}}, \quad (44)$$

where $\tilde{e}_x = \tilde{\varphi}_n^i(x)$ is the augmented feature vector of the sampled solution. Intuitively, $\tilde{\sigma}_{x'}^i(x)$ measures how strongly the posterior mean at x' is correlated with the next observation at x : if x' and x share similar features, $\tilde{\varphi}_n^i(x')^\top C_n^i \tilde{e}_x$ is large, so sampling x provides substantial information about x' .

Then, conditional on $\mathcal{F}_n$ and $x^n = x$, the updated posterior means satisfy

$$\mu_{n+1}^i(x') = \mu_n^i(x') + \tilde{\sigma}_{x'}^i(x) \cdot Z, \quad Z \sim \mathcal{N}(0, 1), \quad (45)$$

for all $x' \in \mathcal{A}_n$, where Z is a standard normal random variable representing the normalized innovation. This representation shows that all posterior means shift linearly in the same random variable Z , with amplitudes given by $\tilde{\sigma}_{x'}^i(x)$. Consequently, in the minimization

convention (43),

$$\nu_n^{\text{KG},i}(x) = h(-\mu_n^i(\mathcal{A}_n), -\tilde{\sigma}^i(x)), \quad (46)$$

where $\mu_n^i(\mathcal{A}_n) = (\mu_n^i(x'))_{x' \in \mathcal{A}_n}$ and $h(\mathbf{a}, \mathbf{b}) = \mathbb{E}[\max_j(a_j + b_j Z)] - \max_j a_j$ is the standard KG computation function of [Frazier et al. \(2009\)](#). (The negation converts minimization to maximization so that the h -function formula applies directly.)

Proof. From the posterior update (22):

$$\mu_{n+1}^i(x') = \tilde{\varphi}_n^i(x')^\top a_{n+1}^i = \mu_n^i(x') + \frac{\tilde{\varphi}_n^i(x')^\top C_n^i \tilde{e}_x}{\hat{\sigma}_n^{i,2}(x) + \tilde{e}_x^\top C_n^i \tilde{e}_x} \cdot (Y_{n+1}^i - \tilde{e}_x^\top a_n^i).$$

Conditional on $\mathcal{F}_n$, the innovation $Y_{n+1}^i - \tilde{e}_x^\top a_n^i \sim \mathcal{N}(0, \hat{\sigma}_n^{i,2}(x) + \tilde{e}_x^\top C_n^i \tilde{e}_x)$, so dividing by its standard deviation yields a standard normal Z , giving (45). Taking the expectation of the minimum after substitution gives (46). The h -function is computed by the sort-based algorithm of [Frazier et al. \(2009\)](#) in $O(|\mathcal{A}_n| \log |\mathcal{A}_n|)$ time. Details are provided in Appendix A2.2 of the Online Supplement. $\square$

4.2 Candidate Set Generation

Computing the KG factor over the entire solution space $\mathcal{X}$ is infeasible when K is large. Instead, we evaluate the KG factor over a carefully constructed candidate set $\mathcal{A}_n \subset \mathcal{X}$ at each stage. The reported algorithm uses two problem-independent components:

$$\mathcal{A}_n = \mathcal{A}_n^{\text{LHD}} \cup \mathcal{A}_n^{\text{post}}, \quad (47)$$

where:

Latin Hypercube component $\mathcal{A}_n^{\text{LHD}}$. Sample K_1 solutions via Latin Hypercube Design (LHD), rounded to the integer grid $\mathcal{X}$. LHD provides better space-filling coverage than simple random sampling while ensuring that the candidate set spans all regions of the decision space.

Posterior-guided component $\mathcal{A}_n^{\text{post}}$. Generate solutions by sampling from the posterior distribution and solving a bi-objective optimization on the sampled posterior. This component exploits the current belief structure to focus candidates near potentially Pareto-optimal regions. The construction is repeated K_2 times:

1. For each objective $i \in \{1, 2, 3\}$, sample posterior parametric coefficients $\tilde{\beta}^i \sim \mathcal{N}(\mathbf{a}_{\beta,n}^i, C_{\beta\beta,n}^i)$, where $\mathbf{a}_{\beta,n}^i$ and $C_{\beta\beta,n}^i$ denote the parametric components of the posterior mean and covariance.

2. Solve the sampled bi-objective problem

$$\min_{u \in [0,1]^d} (\varphi(x(u))^\top \tilde{\beta}^1, \varphi(x(u))^\top \tilde{\beta}^2) \quad (48)$$

using NSGA-II with population size 100 and 50 generations, where $x(u)$ maps a continuous search point to the nearest feasible integer-grid solution before evaluating the sampled quadratic belief. When the current iteration exceeds a threshold n_{thr} , a probabilistic feasibility constraint $\varphi(x(u))^\top \tilde{\beta}^3 + \Phi^{-1}(1 - \alpha) \hat{\sigma}_n^3(x(u)) \leq \tau$ is imposed using the current VEPM variance lookup for the candidate solution.

3. Add each integer-grid Pareto-optimal solution $x(u)$ to $\mathcal{A}_n^{\text{post}}$.

By sampling the posterior coefficients and re-optimizing, the algorithm generates diverse candidates that reflect uncertainty in the current belief model. This is a finite-candidate approximation to KG over the full discrete space: all KG factors, feasibility checks, and posterior evaluations after candidate generation are performed on integer-grid solutions in $\mathcal{X}$. In our experiments, we use $K_1 = 50$ and $K_2 = 2$.

4.3 Bi-Objective KG Sampling Decision

In the bi-objective setting, each candidate solution $x \in \mathcal{A}_n$ has two KG factors, $\nu_n^{\text{KG},1}(x)$ and $\nu_n^{\text{KG},2}(x)$, corresponding to the two objective functions. We adopt the *Pareto-KG* approach to select the next solution to simulate.

Pareto-KG rule.

1. For each $x \in \mathcal{A}_n$, compute the KG factor pair $(\nu_n^{\text{KG},1}(x), \nu_n^{\text{KG},2}(x))$.
2. Identify the non-dominated set in KG-factor space:

$$\mathcal{N}_n = \{x \in \mathcal{A}_n : \nexists x' \in \mathcal{A}_n \text{ with } \nu_n^{\text{KG},i}(x') \geq \nu_n^{\text{KG},i}(x) \forall i \in \{1, 2\}, \text{ strict for some } i\}. \quad (49)$$

3. Select x^n from $\mathcal{N}_n$ using crowding distance in KG-factor space.

4.4 The Complete Algorithm

We now assemble the parametric belief model (§3.1), posterior updates (§3.2), VEPM (§3.3), posterior decomposition (§3.4), and Pareto-KG sampling (§4.3) into the complete GPR-KG algorithm.

Remark 4.1 (Computational complexity per iteration). The per-iteration computational cost of Algorithm 1 is composed of the following terms:

- *Candidate set generation:* $O(K_1 d + K_2 P G C_{\text{eval}})$, where P and G are the NSGA-II population size and generation count and C_{eval} is the cost of evaluating the sampled posterior objectives and feasibility expression at one candidate.
- *KG factor computation:* For each candidate $x \in \mathcal{A}_n$, the vector $\tilde{\sigma}^i(x)$ in (44) requires a matrix-vector product in $O(\dim(\theta_n^i) \cdot |\mathcal{A}_n|)$ time, and the h -function evaluation takes $O(|\mathcal{A}_n| \log |\mathcal{A}_n|)$ time (Frazier et al., 2009). Over all candidates, this yields $O(|\mathcal{A}_n|^2 \log |\mathcal{A}_n|)$ per objective.
- *Posterior update:* The rank-one update (22)–(23) requires $O(\dim(\theta_n^i)^2)$ per evaluation index.
- *VEPM update:* $O(|\mathcal{S}_{n,c(x^n)}|)$ per evaluation index in the reported implementation, or $O(1)$ with an equivalent running-sum bookkeeping implementation (Remark 3.3).

The total per-iteration cost is therefore $O(|\mathcal{A}_n|^2 \log |\mathcal{A}_n| + \dim(\theta_n)^2)$, where $\dim(\theta_n) = p + |\mathcal{X}_n| \leq p + N$. Crucially, this cost is *independent of the total number of solutions* K , which can be astronomically large.

5 Theoretical Analysis

This section provides a rigorous theoretical foundation for the GPR-KG framework. The analysis concerns the statistical model and an exploration-augmented version of the sampling policy; the implementation used in Sections 6–7 is the corresponding finite-candidate approximation described in §4.2. This separation is deliberate: the theory identifies the conditions under which the posterior and VEPM components are consistent, while the experiments evaluate the computationally scalable policy used in practice. We begin by stating the assumptions that underpin the analysis (§5.1), then establish consistency and convergence rates for the VEPM variance estimator (§5.2), derive finite-budget regret bounds for the posterior Pareto front (§5.3), give a probabilistic feasibility guarantee (§5.4), prove asymptotic convergence (§5.5), discuss the one-step optimality of the Pareto-KG sampling rule (§5.6), and finally analyze robustness to model misspecification (§5.7).

5.1 Assumptions

Assumption 5.1 (Gaussian Observations with Bounded Variances). This assumption strengthens Assumption 2.1 by requiring uniformly bounded variances. For each evaluation index $i \in$

$\{1, 2, 3\}$ and every solution $x \in \mathcal{X}$, the simulation output satisfies $Y^i(x) \sim \mathcal{N}(f^i(x), \sigma^i(x)^2)$, where $f^i(x)$ is the unknown true mean and $\sigma^i(x)^2$ is the unknown true variance, with

$$0 < \sigma_{\min}^2 \leq \sigma^i(x)^2 \leq \sigma_{\max}^2 < \infty \quad \forall x \in \mathcal{X}, i \in \{1, 2, 3\}.$$

Assumption 5.2 (Well-Specified Parametric Model). For each evaluation index $i \in \{1, 2, 3\}$, there exist true coefficients $\beta^{i,*} \in \mathbb{R}^{p_i}$ and true deviation values $\zeta^{i,*}(x)$ such that

$$f^i(x) = \varphi^i(x)^\top \beta^{i,*} + \zeta^{i,*}(x) \quad \forall x \in \mathcal{X}.$$

Moreover, the basis functions are uniformly bounded: $|\varphi_j^i(x)| \leq B_\varphi$ for all $x \in \mathcal{X}$, $i \in \{1, 2, 3\}$, and $j \in \{1, \dots, p_i\}$.

Assumption 5.3 (VEPM Model). For each evaluation index i and each partition combination c , the within-partition variance heterogeneity is bounded:

$$\max_{x, x': c(x)=c(x')=c} |\sigma^i(x)^2 - \sigma^i(x')^2| \leq \delta_c^i,$$

where $\delta_c^i \geq 0$ is the variance variation within partition c . When $\delta_c^i = 0$ for all c (i.e., constant variance within each partition), the assumption reduces to exact homoscedasticity.

Assumption 5.4 (Exploration). For the asymptotic statements, the sampling policy includes an exploration component such that, for every fixed solution $x \in \mathcal{X}$ and every stage n , there exists a problem-dependent constant $\rho_x > 0$ (independent of n) such that

$$\Pr(x^n = x \mid \mathcal{F}_n) \geq \rho_x > 0.$$

Because $\mathcal{X}$ is finite for a fixed problem, this implies $\rho = \min_{x \in \mathcal{X}} \rho_x > 0$, although ρ may be extremely small and is not claimed to be uniform across problem dimensions.

Remark 5.1. Assumptions 5.1–5.4 are ordered roughly from weakest to strongest. Assumption 5.1 is standard in simulation-based optimization when simulation outputs are averages or are well approximated by Gaussian errors; independence and finite moments alone do not imply normality, so non-Gaussian finite-sample behavior should be viewed as model misspecification. Assumption 5.2 requires that the parametric basis be rich enough to capture the true response surface; the nonparametric deviation $\zeta^{i,*}(x)$ provides flexibility, and partial misspecification is analyzed in §5.7. Assumption 5.3 asks for approximate homoscedasticity *within* each partition combination. When this condition is violated, the VEPM estimate converges to a sample-count weighted average of the true variances in the partition; the

resulting bias and its implications are discussed in §5.7 and quantified in Theorem 5.2. Assumption 5.4 is a sufficient asymptotic exploration condition rather than a finite-budget performance claim. It is enforced by an explicit ϵ -exploration mixture that occasionally samples a lattice point independently of the KG score. The reported implementation uses Latin-hypercube and posterior-guided NSGA-II candidate generation followed by Pareto-KG selection; this scalable finite-candidate policy is the engineering implementation used in the experiments. The asymptotic statements below should therefore be read as guarantees for the exploration-augmented version of the algorithm, while the numerical experiments assess the finite-budget implementation.

5.2 VEPM Consistency and Convergence Rate

We first establish the statistical validity of the Variance Estimation via Partition Modeling (VEPM) procedure.

Theorem 5.1 (VEPM cell-average consistency). *Fix a partition $\mathcal{C}$, an evaluation index $i \in \{1, 2, 3\}$ and a partition combination $c \in \mathcal{C}$. Suppose Assumptions 5.1–5.3 hold and that, along the sequence of algorithmic stages, the total within-partition observation count $R_{N,c} = \sum_{x' \in \mathcal{S}_{N,c}} n_{x'}$ satisfies $R_{N,c} \rightarrow \infty$ almost surely. Define the sampling-policy weighted cell-average true variance*

$$\bar{\sigma}_c^{i,2} = \lim_{N \rightarrow \infty} \frac{\sum_{x' \in \mathcal{S}_{N,c}} n_{x'} \sigma^i(x')^2}{\sum_{x' \in \mathcal{S}_{N,c}} n_{x'}}.$$

Then

$$\hat{\sigma}_{N,c}^{i,2} \xrightarrow{a.s.} \bar{\sigma}_c^{i,2} \quad \text{as } N \rightarrow \infty.$$

Moreover, for every x with $c(x) = c$, $|\bar{\sigma}_c^{i,2} - \sigma^i(x)^2| \leq \delta_c^i$, where $\delta_c^i \geq 0$ is the within-partition variance variation from Assumption 5.3. When $\delta_c^i = 0$ (within-cell homoscedasticity), the limit equals the true point variance $\sigma_c^{i,2}$ and the VEPM estimate is therefore strongly consistent for $\sigma^i(x)^2$ at every x with $c(x) = c$.

Scope of Theorem 5.1. Two qualifications govern the finite-budget use of this result. (i) *Pointwise vs. cell-average.* The limit $\bar{\sigma}_c^{i,2}$ is a sampling-policy weighted mean of the true variances at solutions visited inside cell c ; it equals the pointwise variance $\sigma^i(x)^2$ only when within-cell heterogeneity vanishes ($\delta_c^i = 0$). Pointwise consistency requires a refining partition sequence $\{\mathcal{C}_N\}_{N \geq 1}$ with cell diameter $\text{diam}(c_N(x)) \rightarrow 0$ and within-cell sample count $N_{c_N(x)} \rightarrow \infty$, so that both δ_c^i and the finite-sample term in Theorem 5.2 vanish jointly. (ii) *Cell visitation.* The hypothesis $R_{N,c} \rightarrow \infty$ holds along a typical sample path only for cells that

retain positive sampling probability under the eventual KG policy. In the saturation regime (Corollary 5.1), this condition fails for almost all cells, so the theorem says nothing about them and the algorithm reverts to the initialization/global fallback for those cells.

Proof sketch. The full proof is given in the Online Supplement. We outline the three main steps.

Step (i). By Bayesian consistency of the parametric belief model (Assumption 5.2), the posterior mean estimates $\mu_n^i(x)$ converge almost surely to $f^i(x)$ for every solution x that is sampled infinitely often.

Step (ii). Because $\mu_{s_\ell}^i(x) \rightarrow f^i(x)$ a.s., the squared residuals are asymptotically equivalent to the squared noise terms:

$$(Y_\ell^i(x) - \mu_{s_\ell}^i(x))^2 = (Y_\ell^i(x) - f^i(x))^2 + o_\ell(1) \quad \text{a.s.,}$$

where $\mathbb{E}[(Y_\ell^i(x) - f^i(x))^2] = \sigma^i(x)^2$.

Step (iii). The VEPM common estimate (30) is a sample-count weighted average of the individual estimates. By the strong law applied to each solution's individual estimate (Step ii), and the fact that the sample-count weighted average of convergent sequences converges to the corresponding weighted average of the limits, we obtain $\hat{\sigma}_{N,c}^{i,2} \rightarrow \bar{\sigma}_c^{i,2}$ a.s. The bound $|\bar{\sigma}_c^{i,2} - \sigma^i(x)^2| \leq \delta_c^i$ follows directly from Assumption 5.3 and the convexity of the weighted average. $\square$

Theorem 5.2 (VEPM Convergence Rate under Heterogeneity). *Under Assumptions 5.1–5.3, for any solution x with $c(x) = c$,*

$$\mathbb{E}\left[(\hat{\sigma}_{N,c}^{i,2} - \sigma^i(x)^2)^2\right] \leq (\delta_c^i)^2 + \frac{C_1 \sigma_{\max}^4}{N_c r} + \frac{C_2 p_i \sigma_{\max}^2}{N} + \frac{C_3 w_i^2 \sigma_{\max}^4}{(w_i + r)^2},$$

where the first term captures the irreducible bias from within-partition heterogeneity and the remaining terms are the same as in Theorem 5.3. The bias vanishes when $\delta_c^i = 0$.

Corollary 5.1 (VEPM in the partition-saturation regime). *Consider a sequence of problem instances indexed by d_m with partition cardinalities $K_m = |\mathcal{C}_m|$ and total simulation budgets N_m . Suppose $N_m/K_m \rightarrow 0$ as $m \rightarrow \infty$, and the sampling policy is not concentrated on a vanishing fraction of cells, in the sense that $\max_{c \in \mathcal{C}_m} N_c/N_m \rightarrow 0$ almost surely. Then for any solution x_m that has not been individually sampled ($x_m \notin \mathcal{X}_{N_m}$),*

$$\hat{\sigma}_{N_m, c(x_m)}^{i,2} = \begin{cases} \hat{\sigma}_0^{i,2}(c(x_m)) & \text{if cell } c(x_m) \text{ has been visited at least once,} \\ \hat{\sigma}_{\text{global}, N_m}^{i,2} & \text{if cell } c(x_m) \text{ is unvisited at stage } N_m, \end{cases}$$

with probability tending to one, where $\hat{\sigma}_0^{i,2}(c)$ is the pre-sampling initial estimate for cell c and $\hat{\sigma}_{\text{global},N}^{i,2}$ is the global pooled estimator computed from all observations up to stage N . In particular, the partition-level common estimate $\hat{\sigma}_{N_m,c(x_m)}^{i,2}$ remains essentially frozen at its initialization across the main loop, so the conclusion of Theorem 5.1 provides no operational guidance for unsampled solutions.

Proof sketch. Under $\max_c N_c/N \rightarrow 0$, the indicator that cell c has been revisited beyond its first sample is asymptotically zero with probability one for every fixed cell. The recursive update (33)/(35) modifies $\hat{\sigma}_{n,c}^{i,2}$ only when an individual estimate $\hat{\sigma}_{n+1}^{i,2}(x^n)$ is added to the running average over $\mathcal{S}_{n,c}$, so a single-sample cell carries $\hat{\sigma}_{n,c}^{i,2} = \hat{\sigma}_{n,c}^{i,2}(x^n)$, which by Eq. (34) reduces to a w_i -weighted average of the initial estimate $\hat{\sigma}_0^{i,2}(c)$ and a single residual term; for unvisited x the lookup in (31) is by definition the initialization $\hat{\sigma}_0^{i,2}(c)$. When a cell is entirely unvisited the variance lookup falls through to the global pooled estimator $\hat{\sigma}_{\text{global},N}^{i,2}$ defined in Algorithm 1. Full details are in the Online Supplement. $\square$

Practical implications of Corollary 5.1. At our high-dimensional engineering case study (§7, $d = 44$, $|\mathcal{C}| = 2^{88}$), the saturation regime applies and the VEPM common estimate at a fixed reference plan remains constant across all 300 KG iterations. This motivates the alignment-condition framework of §3.3.1, which characterizes when the partition is expected to deliver heteroscedasticity-aware variance estimation *even before* the algorithm is run.

Theorem 5.3 (VEPM Convergence Rate). *Under Assumptions 5.1–5.3, fix an evaluation index i and a partition combination c . Suppose N_c distinct solutions from partition c have been sampled, each receiving at least r replications, and the total simulation budget is N . Then*

$$\mathbb{E}\left[\left(\hat{\sigma}_{N,c}^{i,2} - \bar{\sigma}_c^{i,2}\right)^2\right] \leq \frac{C_1 \sigma_{\max}^4}{N_c r} + \frac{C_2 p_i \sigma_{\max}^2}{N} + \frac{C_3 w_i^2 \sigma_{\max}^4}{(w_i + r)^2},$$

where $\bar{\sigma}_c^{i,2}$ is the limiting weighted-average variance from Theorem 5.1, $C_1, C_2, C_3 > 0$ are universal constants, and w_i is the initial variance weight in the VEPM prior.

Proof sketch. Full proof in the Online Supplement. The three terms arise from distinct error sources. *Term 1* captures sampling variability: given the true mean, the sample variance of $N_c \times r$ observations has mean squared error $O(\sigma_{\max}^4/(N_c r))$. *Term 2* captures the propagation of mean estimation error into the residuals; by Lemma 5.1 the parametric posterior mean satisfies $\mathbb{E}[(\mu_N^i(x) - f^i(x))^2] = O(p_i \sigma_{\max}^2/N)$. *Term 3* captures the shrinkage bias introduced by the VEPM prior weight w_i : as the number of replications r grows, the data dominate the prior and this term vanishes. $\square$

The three terms in Theorem 5.3 have transparent interpretations:

1. *Sampling variability* (Term 1): decreases as more replications are allocated to each solution.
2. *Mean estimation error propagation* (Term 2): decreases as the total budget N grows and the posterior mean becomes more accurate.
3. *Shrinkage bias from initial estimate* (Term 3): decreases as the number of replications r grows to dominate the prior weight w_i .

Corollary 5.2. *If each sampled solution receives $r = \Theta(\sqrt{N})$ replications, then*

$$\mathbb{E}\left[(\hat{\sigma}_{N,c}^{i,2} - \sigma_c^{i,2})^2\right] = O(N^{-1/2}).$$

Proof. Substituting $r = \Theta(\sqrt{N})$ into Theorem 5.3: Term 1 becomes $O(\sigma_{\max}^4/(N_c\sqrt{N}))$; since $N = N_c \cdot r = N_c \cdot \Theta(\sqrt{N})$ implies $N_c = \Theta(\sqrt{N})$, this yields $O(N^{-1})$. Term 2 is already $O(N^{-1})$. Term 3 becomes $O(w_i^2\sigma_{\max}^4/N) = O(N^{-1})$. All three terms are $O(N^{-1})$, but since N_c may in general be smaller than $\Theta(\sqrt{N})$ (if the budget is not uniformly spread across partitions), the conservative bound is $O(N^{-1/2})$. $\square$

5.3 Posterior Mean Convergence and Regret Bound

Lemma 5.1 (Posterior Mean Convergence Rate). *Under Assumptions 5.1–5.2, consider the parametric belief model for evaluation index i with p_i basis functions. After N observations, the posterior mean of the augmented coefficient vector $\mathbf{a}^i$ satisfies*

$$\mathbb{E}[\|\mathbf{a}_N^i - \mathbf{a}^{i,*}\|^2] \leq \frac{p_i \sigma_{\max}^2}{N} \text{tr}((\mathbf{C}_0^i)^{-1})^{-1},$$

where $\mathbf{a}^{i,*}$ is the true augmented coefficient vector and $\mathbf{C}_0^i$ is the prior precision matrix.

Consequently, for any $x \in \mathcal{X}$:

$$\mathbb{E}[(\mu_N^i(x) - f^i(x))^2] \leq \frac{p_i \sigma_{\max}^2 \|\tilde{\boldsymbol{\varphi}}^i(x)\|^2}{N},$$

where $\tilde{\boldsymbol{\varphi}}^i(x)$ is the augmented basis vector at x .

Proof sketch. Full proof in the Online Supplement. The posterior precision matrix after N observations is $\mathbf{C}_N^i = \mathbf{C}_0^i + \sum_{\ell=1}^N \tilde{\boldsymbol{\varphi}}^i(x_\ell) \tilde{\boldsymbol{\varphi}}^i(x_\ell)^\top / \sigma^i(x_\ell)^2$. The mean squared error of the posterior mean is $\text{tr}((\mathbf{C}_N^i)^{-1})$, which is bounded above by $p_i \sigma_{\max}^2 / N$ after applying the bounded basis function assumption and the lower bound on the prior precision. $\square$

Remark 5.2. The convergence rate in Lemma 5.1 scales with p_i (the number of basis functions), *not* with $K = |\mathcal{X}|$ (the number of solutions). This is the key theoretical advantage of the parametric approach: information obtained at any sampled solution improves the estimate at *every* solution through the shared coefficient vector. In contrast, a nonparametric GP model would require $O(K)$ observations to achieve comparable accuracy across the full solution space.

Theorem 5.4 (Conditional Bi-Objective Simple Regret Bound). *Define the simple regret as the Hausdorff distance between the estimated and true Pareto fronts:*

$$r_N = d_H(\widehat{PF}_N, PF^*).$$

For the exploration-augmented policy satisfying Assumptions 5.1–5.4, and assuming that the VEPM estimates satisfy $\hat{\sigma}_n^{i,2}(x) \geq \sigma_{\min}^2/2$ for all n, x , and i , the expected simple regret is bounded by

$$\mathbb{E}[r_N] \leq C \sqrt{\frac{p \log K}{N}} \cdot \frac{\sigma_{\max}}{\Delta_{\min}},$$

where $K = |\mathcal{X}|$, $p = \max_i p_i$, $\Delta_{\min}$ is the minimum gap between adjacent Pareto optimal points in objective space, and $C > 0$ is a constant that depends on the problem geometry and the prior.

Remark 5.3 (Finite-candidate implementation). The bound in Theorem 5.4 is a conditional statistical result: it requires the sampling coverage in Assumption 5.4. The reported implementation evaluates KG only on the finite set $\mathcal{A}_n$ and does not include an explicit ϵ -exploration step, because the paper targets tight finite simulation budgets. Its behavior is therefore assessed empirically rather than claimed to satisfy the full asymptotic coverage condition. Adding an ϵ -exploration mixture would restore the stated asymptotic coverage without changing the posterior, VEPM, or KG update formulas, but it would change finite-budget sampling paths and is not used in the reported experiments.

Remark 5.4. When the decision-space parameters (d, L) are fixed, $K = L^d$ is a constant and $\log K = d \log L$ does not grow with the budget N . The bound then simplifies to $\mathbb{E}[r_N] = O(\sqrt{p/N})$, which is the parametric rate driven by the p -dimensional coefficient estimation. In the traffic application of Section 7, for instance, $K \approx 10^{70}$ so $\log K \approx 161$, which is absorbed into the constant C . By making the $\log K$ dependence explicit, the bound is valid for any finite K without requiring an auxiliary growth condition.

Proof sketch. Full proof in the Online Supplement. The argument proceeds as follows.

Step (i). By Lemma 5.1, after N observations the posterior mean error for each objective satisfies $\mathbb{E}[(\mu_N^i(x) - f^i(x))^2] = O(p/N)$. To control the *worst-case* error $\max_{x \in \mathcal{X}} |\mu_N^i(x) - f^i(x)|$, we apply a sub-Gaussian maximal inequality over the finite set $\mathcal{X}$: the maximum of K zero-mean sub-Gaussian random variables with variance proxy $O(p/N)$ is of order $O(\sqrt{p \log K/N})$.

Step (ii). A solution x is misclassified (either incorrectly included in, or excluded from, the estimated Pareto set) only when the estimation error in at least one objective exceeds $\Delta_{\min}/2$.

Step (iii). Applying a union bound over the $|\mathcal{P}^*|$ true Pareto optimal solutions and the K total solutions, the probability of any misclassification is bounded by $O(K \exp(-N \Delta_{\min}^2/(Cp)))$.

Step (iv). Assumption 5.4 guarantees that each solution is sampled with frequency at least ρ , providing the coverage needed for the uniform bound.

Combining these steps yields the stated rate. $\square$

Corollary 5.3 (Sample Complexity). *To achieve $\mathbb{E}[r_N] \leq \varepsilon$, it suffices to have*

$$N = O\left(\frac{p \sigma_{\max}^2 \log K}{\varepsilon^2 \Delta_{\min}^2}\right).$$

When (d, L) are fixed, $\log K$ is constant and the sample complexity is $O(p/(\varepsilon^2 \Delta_{\min}^2))$.

Remark 5.5. A nonparametric approach with *independent* beliefs across alternatives (i.e., no spatial correlation structure) would require

$$N = O\left(\frac{K \sigma_{\max}^2}{\varepsilon^2 \Delta_{\min}^2}\right)$$

observations to achieve the same guarantee. The parametric formulation thus saves a factor of K/p , which can be enormous. For instance, in the traffic network case study of Section 6, $K \approx 10^{70}$ whereas $p = 41$, yielding a reduction of roughly 68 orders of magnitude in the effective problem dimension.

5.4 Feasibility Guarantee

Theorem 5.5 (Probabilistic Feasibility Guarantee). *Under Assumptions 5.1–5.3, let the estimated feasible set be*

$$\hat{F}_N = \{x \in \mathcal{X} : \mu_N^3(x) + \Phi^{-1}(1 - \alpha) \hat{\sigma}_N^3(x) \leq \tau\},$$

and define the true feasible set as $F^ = \{x \in \mathcal{X} : \Pr(g(x, \xi) \leq \tau) \geq 1 - \alpha\}$.*

For any $\delta > 0$, with probability at least $1 - \delta$, every solution in the estimated feasible set satisfies an approximate feasibility condition:

$$\forall x \in \hat{F}_N : \quad q_{1-\alpha}(x) \leq \tau + \varepsilon_N(\delta),$$

where

$$\varepsilon_N(\delta) = C_f \sqrt{\frac{p \log(N/\delta)}{N}} (1 + \Phi^{-1}(1 - \alpha) \cdot \sigma_{\max})$$

and $C_f > 0$ is a constant depending on the basis function bound B_φ and the prior.

Proof sketch. Full proof in the Online Supplement.

Step (i). Decompose the feasibility estimation error:

$$|\hat{q}_{1-\alpha}(x) - q_{1-\alpha}(x)| \leq |\mu_N^3(x) - f^3(x)| + \Phi^{-1}(1 - \alpha) \cdot |\hat{\sigma}_N^3(x) - \sigma^3(x)|.$$

Step (ii). Bound the mean estimation error using Lemma 5.1 and a high-probability concentration inequality (sub-Gaussian tail) to obtain $|\mu_N^3(x) - f^3(x)| \leq O(\sqrt{p \log(N/\delta)/N})$ with probability at least $1 - \delta/(2K)$.

Step (iii). Bound the variance estimation error using Theorem 5.3 and Markov's inequality to obtain a high-probability bound.

Step (iv). Apply the union bound over all K solutions and combine the two error terms. $\square$

Corollary 5.4. As $N \rightarrow \infty$, $\varepsilon_N(\delta) \rightarrow 0$ for every fixed $\delta > 0$. Consequently, the estimated feasible set converges to a subset of the true feasible set with probability 1:

$$\Pr(\hat{F}_N \subseteq F^* \text{ for all sufficiently large } N) = 1.$$

Remark 5.6. Theorem 5.5 provides a *conservative* guarantee: solutions identified as feasible are approximately truly feasible, with the approximation error $\varepsilon_N(\delta)$ vanishing at rate $\tilde{O}(N^{-1/2})$. The result does *not* guarantee that all truly feasible solutions are found—such a guarantee would require exhaustive sampling of the entire solution space and is unattainable within a finite budget.

5.5 Asymptotic Convergence

The following theorem strengthens the original convergence result (Theorem 1 in the initial submission) by establishing convergence of all components simultaneously.

Theorem 5.6 (Asymptotic Convergence under Explicit Exploration). *Under Assumptions 5.1–5.4, if the exploration-augmented GPR-KG algorithm is run with $N \rightarrow \infty$, then the following hold almost surely:*

- (i) *The posterior mean $\mu_N^i(x)$ converges to $f^i(x)$ for all $x \in \mathcal{X}$ and $i \in \{1, 2, 3\}$.*
- (ii) *The VEPM estimates $\hat{\sigma}_{N,c}^{i,2}$ converge to $\bar{\sigma}_c^{i,2}$ for all partition combinations c and all i , where $\bar{\sigma}_c^{i,2}$ is the limiting weighted-average variance from Theorem 5.1.*
- (iii) *The estimated Pareto optimal set $\hat{\mathcal{P}}_N$ converges to the true Pareto optimal set $\mathcal{P}^*$.*
- (iv) *The convergence in (iii) holds in the Hausdorff metric: $d_H(\widehat{PF}_N, PF^*) \rightarrow 0$ a.s.*

Proof. Part (i). Assumption 5.4 ensures that every solution $x \in \mathcal{X}$ is sampled infinitely often almost surely (by the second Borel–Cantelli lemma, since the sampling events have probability bounded below by ρ and are conditionally independent given the history). The parametric belief model satisfies Bayesian consistency under Assumption 5.2: with infinitely many observations, the posterior on $\mathbf{a}^i$ concentrates on the true value $\mathbf{a}^{i,*}$, and hence $\mu_N^i(x) = \tilde{\varphi}^i(x)^\top \mathbf{a}_N^i \rightarrow f^i(x)$ a.s.

Part (ii). This is precisely Theorem 5.1.

Part (iii). For each pair of solutions $x, x' \in \mathcal{X}$ and each objective i , the event $\{f^i(x) < f^i(x')\}$ is eventually correctly identified by the posterior means (from (i), the estimation errors become smaller than the gap $|f^i(x) - f^i(x')|$ for all non-tied pairs). Since the Pareto optimal set is a deterministic function of the pairwise dominance relations, $\hat{\mathcal{P}}_N \rightarrow \mathcal{P}^*$ a.s.

Part (iv). Part (iii) implies that $\widehat{PF}_N$ converges point-wise in objective space. Since $\mathcal{X}$ is finite, the Pareto front is a finite point set, and point-wise convergence of a finite collection implies convergence in the Hausdorff metric. $\square$

5.6 One-Step Optimality of Pareto-KG

Proposition 5.1 (Leading-Order Myopic Optimality of Pareto-KG). *For each candidate $x \in A_n$, define the one-step expected improvement in the posterior hypervolume: $\Delta\text{HV}_n(x) = \mathbb{E}_n[\text{HV}(\hat{\mathcal{P}}_{n+1}) - \text{HV}(\hat{\mathcal{P}}_n) \mid x^n = x]$, where $\text{HV}(\cdot)$ is the hypervolume indicator with respect to a fixed reference point. Then*

$$\Delta\text{HV}_n(x) = \sum_{i=1}^2 \nu_n^{\text{KG},i}(x) \cdot \lambda_{i,n}(x) + R_n(x),$$

where $\lambda_{i,n}(x) \geq 0$ is the marginal hypervolume contribution of objective i at the current stage and $R_n(x)$ is a cross-term of order $O((\tilde{\sigma}^i(x))^2)$ that vanishes as $\min_x \|\tilde{\sigma}(x)\| \rightarrow 0$.

The Pareto-KG rule, which selects the non-dominated solution in $(\nu_n^{\text{KG},1}(x), \nu_n^{\text{KG},2}(x))$ space, maximizes the leading-order term of $\Delta\text{HV}_n(x)$.

Proof sketch. Full proof in the Online Supplement. The key step is to express the posterior Pareto front update using the linear update (45): $\mu_{n+1}^i(x') = \mu_n^i(x') + \tilde{\sigma}_{x'}^i(x)Z$. The hypervolume change is a function of Z through the shift in posterior means. A first-order Taylor expansion in the $\tilde{\sigma}_{x'}^i(x)$ vector (which is small when the information gain from one observation is modest relative to the posterior spread) separates the contribution by objective, yielding the additive form above. The remainder $R_n(x)$ captures interaction effects between the two objectives; its magnitude is bounded by $O(\|\tilde{\sigma}(x)\|^2/\Delta_{\min}^2)$, which is negligible in the early and mid stages of optimization. $\square$

Remark 5.7. Unlike single-objective KG, which is *exactly* one-step optimal (Frazier et al., 2008), the bi-objective Pareto-KG rule is an approximation: the knowledge gradient factors for the two objectives are computed independently, neglecting their correlation. The Pareto-KG aggregation is a principled heuristic that avoids the computational burden of directly maximizing the expected hypervolume improvement—a quantity whose exact computation is $\#P$ -hard in general (Bringmann and Friedrich, 2009). In practice, the approximation is highly effective, as our numerical experiments in Section 6 confirm.

5.7 Discussion: Robustness to Model Misspecification

We now discuss the behavior of GPR-KG when the modeling assumptions are violated.

(a) When Assumption 5.2 fails (parametric model misspecification). If the true response surface $f^i(x)$ does not lie exactly in the span of the basis functions $\varphi^i(x)$, the deviation term $\zeta^{i,*}(x)$ absorbs the residual. In the parametric GP framework, $\zeta^i(x)$ is modeled as a Gaussian process, and the posterior contracts to the best projection of f^i onto the augmented basis. If the projection residual is small—specifically, if $\max_x |f^i(x) - \tilde{\varphi}^i(x)^\top \mathbf{a}^{i,*}| \leq \epsilon_{\text{mis}}$ —then the regret bound of Theorem 5.4 degrades gracefully:

$$\mathbb{E}[r_N] \leq C \sqrt{\frac{p \log N}{N}} \cdot \frac{\sigma_{\max}}{\Delta_{\min}} + O(\epsilon_{\text{mis}}).$$

The additional term reflects an irreducible approximation error.

(b) When Assumption 5.3 fails (VEPM model misspecification). When the true variance is heterogeneous within a partition combination c , the VEPM estimate converges

(by the strong law) to the weighted average variance:

$$\hat{\sigma}_{\infty,c}^{i,2} = \bar{\sigma}_c^{i,2} := \frac{\sum_{x: c(x)=c} \sigma^i(x)^2}{|\{x : c(x) = c\}|}.$$

This introduces a bias $\text{bias}_c(x) = \bar{\sigma}_c^{i,2} - \sigma^i(x)^2$ for individual solutions. Finer partitions reduce this bias (since the variance heterogeneity within each cell decreases) but increase the estimation variance (since fewer solutions per cell are available). This is a classical bias–variance tradeoff, and the partition granularity should be chosen to balance these two effects.

(c) When the Gaussian assumption fails. The Gaussian assumption (Assumption 5.1) enters the analysis in two ways. For *mean estimation*, the central limit theorem ensures that the sample mean is approximately Gaussian regardless of the output distribution, so the posterior mean convergence (Lemma 5.1 and Theorem 5.6(i)) remains valid under mild moment conditions.

For *quantile estimation*, departure from normality introduces bias. Specifically, if the true output distribution at x has skewness $\gamma(x)$, the bias of the Gaussian quantile approximation is of order $O(\gamma(x) \sigma^3(x) / \sqrt{n_x})$, where n_x is the number of replications at x . For applications with significantly non-Gaussian outputs, a Cornish–Fisher correction

$$\hat{q}_{1-\alpha}(x) \approx \hat{\mu}^3(x) + \left(\Phi^{-1}(1-\alpha) + \frac{1}{6}((\Phi^{-1}(1-\alpha))^2 - 1)\hat{\gamma}(x) \right) \hat{\sigma}^3(x)$$

can be employed, where $\hat{\gamma}(x)$ is an estimate of the skewness. Under this correction, the feasibility guarantee of Theorem 5.5 extends with an additional error term of order $O(n_x^{-1/2})$.

6 Numerical Experiments

This section validates the GPR-KG framework on three synthetic heteroscedastic test problems with complete saved numerical results: RZDT1, RZDT2, and RZDT5-RR. We assess solution quality, constraint satisfaction, the contribution of VEPM, and the effect of candidate-set size using reproducible result files. All reported synthetic tables are traceable to saved per-replication JSON logs or summary files; a reproducibility manifest and an automated table-audit script are included with the code archive to regenerate the reported table values and figures from those saved logs.

6.1 Experimental Setup

Test problems. We construct three bi-objective test problems by adapting the classical ZDT family (Zitzler et al., 2000) with heteroscedastic stochastic noise and a probabilistic constraint. Each problem has a d -dimensional integer decision vector with problem-specific bounds ($x \in \{0, \dots, 100\}^d$ for RZDT1/2 and mixed bounds for RZDT5-RR), two objectives $f^1(x)$ and $f^2(x)$, and one constraint output $f^3(x)$. Simulation outputs are

$$Y^i(x) = f^i(x) + \varepsilon^i(x), \quad \varepsilon^i(x) \sim \mathcal{N}(0, \sigma^i(x)^2), \quad i = 1, 2, 3, \quad (50)$$

where $\sigma^i(x)$ is *position-dependent*, creating heteroscedastic noise that varies significantly across the decision space. The three primary benchmark problems (RZDT1, RZDT2, RZDT5-RR) are deliberately chosen to probe three qualitatively distinct sources of difficulty: (i) *asymmetric* heteroscedasticity with feasibility split between a quiet and a noisy tail (RZDT1); (ii) *symmetric bell-shaped* heteroscedasticity whose peak lies in the infeasible interior but whose pooled estimate still contaminates the low-noise feasible tails (RZDT2); and (iii) *monotone quadratic* heteroscedasticity with a strict chance constraint, where the feasible Pareto front sits in the low-variance region but the candidate set routinely probes the high-variance tail (RZDT5-RR). All three are built so that Pareto-optimal solutions satisfy $x_j = 0$ for $j \geq 2$, isolating x_1 as the controlling coordinate of both the front and the noise profile.

- **RZDT1** (convex Pareto front, asymmetric square-root noise, split-tail feasibility): Based on ZDT1 (Zitzler et al., 2000), with Pareto front on the convex curve $f^2 = 1 - \sqrt{f^1}$ and constraint $f^3 = -(t_1 - 0.5)^2 + 0.04$ that carves out the middle of the front. The noise has a *square-root* profile $\sigma^i(x) = \sigma(0.5 + 2\sqrt{t_1})$, low at the left tail (0.5σ at $t_1 = 0$) and high at the right (2.5σ at $t_1 = 1$), giving a $25\times$ global variance ratio. After heteroscedastic safety margins, $x_1 \in \{0, \dots, 14\} \cup \{95, \dots, 100\}$ remain feasible (21 TPOS: 15 on the low-noise left tail, 6 on the high-noise right tail). The *asymmetry of the safety margin* is the key difficulty: a pooled variance estimator either admits infeasible right-tail solutions (by under-estimating their noise) or discards feasible ones (by over-estimating it), directly producing CVR errors.
- **RZDT2** (concave Pareto front, bell-shaped noise, symmetric split-tail feasibility): Based on ZDT2 (Zitzler et al., 2000), with Pareto front on the concave curve $f^2 = 1 - (f^1)^2$ and the same quadratic constraint as RZDT1. The noise is *bell-shaped*, $\sigma^i(x) = \sigma(0.5 + 2.5 \sin^2(\pi t_1))$, 0.5σ at $t_1 \in \{0, 1\}$ rising to 3σ at $t_1 = 0.5$ (global variance ratio $36\times$). The peak of the noise sits in the *infeasible* interior, so the 34 TPOS $x_1 \in \{0, \dots, 16\} \cup \{84, \dots, 100\}$ themselves span a comparatively narrow noise range

(variance ratio $\approx 4.6\times$ on the TPOS). A pooled estimator nevertheless fails here because it averages the quiet tails with the noisy centre, systematically *over-estimating* noise at the feasibility boundary ($t_1 \approx 0.17$ and 0.84) and eroding constraint-aware ranking.

- **RZDT5-RR** (hyperbolic Pareto front, monotone quadratic noise, strict chance constraint): Uses $f^1 = t_1$, $f^2 = g/(x_1 + 1)$ with $g = 1 + \sum_{j \geq 2} t_j$ and $g = 1$ on the Pareto front ($x_j = 0$, $j \geq 2$), giving a hyperbolic curve $f^2 = 1/(x_1 + 1)$. Decision space is enlarged and heterogeneous: $x_1 \in \{0, \dots, 100\}$, $x_j \in \{0, \dots, 50\}$ for $j \geq 2$. The noise is *monotone quadratic* in $t_1 = x_1/100$: $\sigma^i(x) = \sigma(0.3 + 2t_1^2)$, rising from 0.3σ at $t_1 = 0$ to 2.3σ at $t_1 = 1$ (global variance ratio $\approx 59\times$). RZDT5-RR keeps the *strict* constraint threshold $\tau = 0$ from the original ZDT formulation on $f^3 = t_1 - 0.5$, which under the heteroscedastic safety margin $\Phi^{-1}(0.95)\sigma(0.3 + 2t_1^2)$ restricts feasibility to $x_1 \in \{0, \dots, 45\}$ and yields 46 TPOS. This combination creates a sharp asymmetry between the feasible and infeasible halves of the front (feasible side entirely in $\sigma \leq 0.71\sigma$; infeasible side up to 2.3σ), making the chance-constrained safety margin $\Phi^{-1}(1 - \alpha)\hat{\sigma}^3(x)$ highly spatially non-uniform—exactly the setting VEPM is designed to handle.

Complete formulations and noise model equations are in Appendix A3. Unless stated otherwise, the default setting is $d = 5$ with baseline noise scale $\sigma = 0.04$ for RZDT1, RZDT2, and RZDT5-RR, all with strict threshold $\tau = 0$.

VEPM advantage analysis. To isolate VEPM’s contribution, we compare GPR-KG and its ablation variant GPR-KG-nV across all three heteroscedastic problems, where the three distinct noise patterns (asymmetric square-root, bell-shaped, and monotone quadratic) each challenge pooled variance estimators in a different way.

The two methods differ only in how they estimate the input-dependent observation variance used by both the GPR update and the posterior chance-feasibility filter. GPR-KG uses VEPM: residuals from sampled solutions are pooled within feature partitions and then extrapolated to unsampled candidates in the same partition, so $\hat{\sigma}_n^{i,2}(x)$ is spatially local even when x has not been replicated. GPR-KG-nV disables this partition model. If a candidate x has at least two observations, nV uses the Bessel-corrected local residual variance at that exact solution; otherwise it uses the pooled mean of all currently available replicated residual variances, with a fixed prior fallback before any replication exists. Thus nV can exploit direct replications but cannot generalize a learned variance pattern to unvisited points. This is the intended component-level contrast: VEPM tests whether spatial variance extrapolation improves posterior calibration beyond a local/pooled residual estimator under the same sampling framework.

Comparison methods. Seven methods are compared:

- (1) **GPR-KG** (proposed): the full framework with parametric GPR belief model, VEPM, and Pareto-KG sampling described in Sections 3–4.4.
- (2) **GPR-KG-nV** (ablation): all algorithmic components of GPR-KG are retained—structured finite-grid pre-sampling, candidate generation via K_1 LHD samples and K_2 NSGA-II posterior-sampling runs, integer-grid candidate evaluation, Pareto-KG acquisition, GPR update, checkpointing, and final posterior recommendation—but VEPM is disabled. The probabilistic constraint check $\mu^3(x) + \Phi^{-1}(1 - \alpha) \hat{\sigma}^3(x) \leq \tau$ uses the local/pooled residual variance described above in place of VEPM. This design isolates the contribution of spatial variance extrapolation from all other algorithmic choices.
- (3) **cEHVI** (Daulton et al., 2020, 2021): one Gaussian process surrogate with Matérn-5/2 kernel is fit to each of the three outputs via maximum marginal likelihood estimation (Balandat et al., 2020). At each iteration the next evaluation is selected by maximising the product of the Monte Carlo estimated Expected Hypervolume Improvement (Daulton et al., 2020) and the probability of feasibility

$$\text{PoF}(x) = \Phi\left(\frac{\tau - \mu_3(x)}{\sigma_3(x)}\right),$$

where $\mu_3(x)$ and $\sigma_3(x)$ are the constraint GP’s posterior mean and standard deviation (Gelbart et al., 2014). The acquisition score at candidate x is therefore $\text{cEHVI}(x) = \text{EHVI}(x) \times \text{PoF}(x)$. *Note:* MLE estimates a single global noise variance per output, implicitly assuming homoscedastic noise; this is a known limitation on the heteroscedastic benchmarks used here.

- (4) **cParEGO** (Knowles, 2006): at each iteration, a weight vector $w = (w_1, w_2)$ is drawn uniformly from the unit simplex, and all observed objective pairs are scalarized by the augmented Chebyshev function

$$s(x) = \max_{i=1,2} (w_i f_i(x)) + \rho \sum_{i=1}^2 w_i f_i(x), \quad \rho = 0.05.$$

A GP surrogate is fit to the scalarized values and a second GP is fit to the constraint output f_3 . The next point is selected by maximising the constrained Expected Improvement

$$\text{cEI}(x) = \text{EI}(x) \times \text{PoF}(x),$$

where $\text{EI}(x) = (s^* - \mu_s(x))\Phi(z) + \sigma_s(x)\phi(z)$, $z = (s^* - \mu_s(x))/\sigma_s(x)$ is the standard closed-form expected improvement for minimisation (Jones et al., 1998), s^* is the best scalarized value among feasible observations, and $\text{PoF}(x) = \Phi((\tau - \mu_3(x))/\sigma_3(x))$ (Gelbart et al., 2014). *Note:* Like cEHVI, the GP noise model is homoscedastic (single MLE-estimated variance per output), which is a systematic disadvantage on the heteroscedastic benchmarks.

- (5) **NSGA-II-K** (Deb et al., 2002): three squared-exponential kernel GP surrogates are fit at each iteration, one for each output f_1 , f_2 , and the constraint f_3 . An inner NSGA-II loop (Deb et al., 2002) (population size 100, 100 generations on the cheap surrogate) runs on the surrogates; the constraint-domination principle is applied with probabilistic constraint violation

$$\text{CV}(x) = \max(0, \mu_3(x) + \Phi^{-1}(1 - \alpha)\sigma_3(x) - \tau).$$

The best unvisited candidate from the EA population is then simulated. The total simulation budget is $N = 150$. As with cEHVI and cParEGO, noise variance is estimated as a single MLE constant (homoscedastic assumption).

- (6) **NSGA-II-D** (Deb et al., 2002): direct NSGA-II with a $10\times$ simulation budget ($10N = 1500$ evaluations) and no surrogate model. Each evaluation calls the simulator directly; constraint violation is $\text{CV}(x) = \max(0, Y^3(x) - \tau)$ using the noisy observed output. The constraint-domination selection principle (Deb et al., 2002) is applied throughout, with SBX crossover ($\eta = 20$) and polynomial mutation as variation operators (Deb and Agrawal, 1995). NSGA-II-D serves as a computationally intensive gold standard.
- (7) **RS**: random search draws $N = 150$ solutions uniformly from the problem’s integer decision space $\mathcal{X}$, simulates each once, and returns the non-dominated set of feasible observed pairs $(Y^1(x), Y^2(x))$ satisfying the noisy constraint $Y^3(x) \leq \tau$. RS provides a lower-bound baseline requiring no modelling assumptions.

Performance metrics. We report the following metrics, all computed with respect to the *true* objective values and the *true* probabilistic constraint:

- **Hypervolume Indicator (HV):** the Lebesgue measure of the dominated region between the output Pareto front and a fixed reference point $r = (r_1, r_2)$. Higher is better.
- **Inverted Generational Distance (IGD):** the average distance from each point on the true Pareto front to the nearest output point, using Euclidean distance in objective

space. Lower is better.

- **Constraint Violation Rate (CVR):** the fraction of solutions in the output Pareto front that violate the true probabilistic constraint $\mathbb{P}(f^3(x) + \varepsilon^3(x) \leq \tau) \geq 1 - \alpha$.
- **Computation Time:** method wall-clock time for one complete optimization replication (in seconds), excluding any external simulator cost.

Statistical protocol. The main seven-method comparisons use 10 independent replications. We report the mean $\pm$ standard error of each metric. Where a statistical statement is made, it is based on a two-sided Wilcoxon signed-rank test on paired replications; otherwise the text describes the observed numerical ordering of the saved runs.

GPR-KG parameters. Unless otherwise noted, the sampling budget is $N = 150$ with $n_0 = 30$ pre-samples consisting of a structured finite-grid initial design (problem-independent center and coordinate-axis endpoint seeds plus random integer-grid points), the VEPM prior weight is $w_i = 1$ (equivalent to one pseudo-observation per partition bin), the candidate set is constructed from $K_1 = 50$ LHD random samples, $K_2 = 2$ NSGA-II posterior-sampling runs (population size 100, 50 generations). The basis functions are quadratic polynomials without cross-terms (16), giving $p_i = 2d + 1$. The prior parameters λ_i (deviation variance) and $C_{\beta,0}^i$ (parametric coefficient covariance) are set in a *data-driven* manner from the n_0 pre-samples: an ordinary least-squares (OLS) fit to the pre-sampled data yields $\hat{\beta}^i$, and we set $C_{\beta,0}^i = \text{Var}(\hat{\beta}^i) \cdot I_{p_i}$ and $\lambda_i = \text{Var}(\text{residuals})$ for each objective i , where $\text{Var}(\hat{\beta}^i)$ and $\text{Var}(\text{residuals})$ denote the sample variance of the OLS coefficient vector and the residual vector, respectively. This data-driven initialisation adapts the prior scale to the problem’s signal-to-noise ratio and avoids the need for hand-tuned prior hyperparameters.

6.2 Performance on Heteroscedastic Benchmark Problems

Tables 2–4 report, for each benchmark problem (RZDT1, RZDT2, RZDT5.RR), a full set of performance indicators on all seven methods: hypervolume (HV and its ratio HV/HV*), inverted generational distance (IGD), constraint violation rate (CVR), together with three structural diagnostics —the number of reported non-dominated solutions (#ND), how many of those are in fact infeasible under the true chance constraint (#Infeas), and the number of *exact* hits of a theoretically Pareto-optimal solution (#TPOS-hit). The TPOS-hit diagnostic requires exact equality of the integer decision vector, namely $x_j = 0$ for all $j \geq 2$ and an analytically feasible value of x_1 ; it is therefore a stringent geometric consistency check rather than a primary quality metric. A fourth count, the total number of TPOS known

Table 2: Performance on RZDT1 (convex Pareto front, asymmetric square-root noise, variance ratio $25\times$; $HV^* = 1.635$, $\#TPOS^* = 21$). The GPR-KG and GPR-KG-nV rows are generated from the checkpointed, theory-aligned implementation; the remaining rows are previously completed baseline/comparison runs, with NSGA-II-D labelled as a $10\times$ budget reference.

Method	HV $\uparrow$	HV/HV*	IGD $\downarrow$	CVR $\downarrow$	#ND	#Infeas $\downarrow$	#TPOS-hit $\uparrow$	Time (s)
GPR-KG	1.332 (0.078)	81.5%	0.201 (0.038)	0.325 (0.066)	3.3 (0.4)	1.2 (0.4)	1.8 (0.1)	382.5 (10.9)
GPR-KG-nV	1.268 (0.051)	77.6%	0.225 (0.029)	0.340 (0.098)	3.6 (0.5)	1.5 (0.5)	1.7 (0.2)	528.6 (29.7)
cEHVI	0.426 (0.105)	26.0%	0.838 (0.070)	0.519 (0.050)	5.1 (0.4)	2.6 (0.3)	0.0 (0.0)	73.5 (6.9)
cParEGO	0.225 (0.071)	13.8%	1.003 (0.083)	0.380 (0.050)	5.1 (0.6)	2.0 (0.3)	0.0 (0.0)	1.1 (0.0)
NSGA-II-K	0.282 (0.057)	17.2%	0.879 (0.045)	0.400 (0.042)	5.2 (0.4)	2.2 (0.4)	0.0 (0.0)	9.0 (0.2)
NSGA-II-D ($10\times$)	1.644 (0.019)	100.6% [‡]	0.081 (0.008)	0.525 (0.034)	10.2 (0.5)	5.3 (0.4)	0.0 (0.0)	3.3 (0.0)
RS	0.345 (0.075)	21.1%	0.867 (0.060)	0.493 (0.071)	5.1 (0.6)	2.6 (0.5)	0.0 (0.0)	0.0 (0.0)

[‡]HV/HV* exceeds 100% because HV is computed over the reported non-dominated set; several of NSGA-II-D’s reported points are in fact infeasible (CVR= 0.525), see #Infeas column.

analytically ($\#TPOS^*$), is reported in the caption. All values are means over 10 independent replications with $d = 5$, heteroscedastic noise scale $\sigma = 0.04$ and strict constraint threshold $\tau = 0$; standard errors appear in parentheses. The GPR-KG and GPR-KG-nV rows use the checkpointed implementation aligned with Algorithm 1, including the structured finite-grid initial design and the corrected integer-grid candidate evaluation; the remaining rows are previously completed baseline/comparison runs, with NSGA-II-D explicitly labelled as a $10\times$ simulation-budget reference.

The updated GPR-KG and GPR-KG-nV rows change the interpretation in two important ways. First, the structured finite-grid initial design and the corrected integer-grid candidate evaluation strengthen both variants relative to the pre-audit manuscript version. This shows that a substantial part of the earlier gap was due to the shared engineering path rather than VEPM alone. Second, after this alignment, VEPM’s sample-Pareto advantage is moderate rather than overwhelming. On RZDT1, GPR-KG has higher HV and lower IGD/CVR than nV (HV 1.332 vs. 1.268, IGD 0.201 vs. 0.225, CVR 0.325 vs. 0.340). On RZDT2 the sample sets are essentially tied in HV (1.289 vs. 1.287), and nV has slightly lower IGD/CVR. On RZDT5_RR, GPR-KG has a small but consistent advantage in all three quality metrics (HV 2.104 vs. 2.094, IGD 0.054 vs. 0.065, CVR 0.008 vs. 0.014). Thus the primary algorithmic strength is the full GPR-KG sampling framework; VEPM acts as a variance-calibration component whose benefit is clearest when we evaluate final posterior recommendations rather than only the finite set of simulated non-dominated points.

The low TPOS-hit counts for the comparison algorithms should not be interpreted as evidence that their reported sets contain no near-front points. The TPOS-hit diagnostic

Table 3: Performance on RZDT2 (concave Pareto front, bell-shaped noise, variance ratio 36 $\times$; HV* = 1.403, #TPOS* = 34). The GPR-KG and GPR-KG-nV rows are generated from the checkpointed, theory-aligned implementation; the remaining rows are previously completed baseline/comparison runs, with NSGA-II-D labelled as a 10 $\times$ budget reference.

Method	HV $\uparrow$	HV/HV*	IGD $\downarrow$	CVR $\downarrow$	#ND	#Infeas $\downarrow$	#TPOS-hit $\uparrow$	Time (s)
GPR-KG	1.289 (0.017)	91.9%	0.115 (0.006)	0.100 (0.051)	2.6 (0.2)	0.3 (0.2)	2.2 (0.2)	376.3 (8.7)
GPR-KG-nV	1.287 (0.014)	91.7%	0.104 (0.007)	0.075 (0.053)	3.2 (0.3)	0.3 (0.2)	2.8 (0.3)	481.0 (20.5)
cEHVI	0.073 (0.049)	5.2%	1.462 (0.155)	0.302 (0.080)	3.8 (0.4)	1.2 (0.4)	0.0 (0.0)	90.4 (15.5)
cParEGO	0.000 (0.000)	0.0%	1.761 (0.112)	0.290 (0.057)	4.1 (0.5)	1.2 (0.3)	0.0 (0.0)	4.8 (1.7)
NSGA-II-K	0.036 (0.036)	2.6%	1.770 (0.166)	0.183 (0.045)	3.8 (0.4)	0.8 (0.2)	0.0 (0.0)	29.8 (1.3)
NSGA-II-D (10 $\times$)	0.896 (0.093)	63.8%	0.410 (0.092)	0.157 (0.077)	5.4 (1.0)	1.0 (0.4)	0.0 (0.0)	5.0 (0.2)
RS	0.011 (0.011)	0.8%	1.558 (0.131)	0.195 (0.077)	3.5 (0.4)	0.8 (0.3)	0.0 (0.0)	0.0 (0.0)

asks for exact recovery of the special analytic solutions on the axis $x_j = 0$, $j \geq 2$. Standard Latin-hypercube, evolutionary, and posterior-acquisition candidate generators search the full five-dimensional integer grid and have negligible probability of producing that exact axis unless it is introduced by design. This is why NSGA-II-D can obtain high HV/low IGD on RZDT5-RR while still reporting zero exact TPOS hits. In the main comparison, exact TPOS-hit is therefore a diagnostic of geometric alignment; HV, IGD, CVR, and #ND are the ranking metrics.

Because a sequential policy and a final recommendation rule play different roles, we also report an offline final-recommendation post-processing step in Table 5. This post-processing does not rerun simulations and does not alter the KG sampling decisions. After budget exhaustion, it evaluates the final posterior on a generic finite candidate pool consisting of sampled/selected solutions, the reported sample-Pareto set, and one-step integer-grid neighbours. It then applies the conservative posterior chance-feasibility filter

$$\mu_N^3(x) + \kappa \Phi^{-1}(1 - \alpha) \hat{\sigma}_N^3(x) \leq \tau$$

with $\kappa = 1.25$, followed by Pareto filtering in posterior objective-mean space. The table is computed entirely from the saved checkpoints and result logs using the offline post-processing

Table 4: Performance on RZDT5.RR (hyperbolic Pareto front, monotone quadratic noise, variance ratio $59\times$, strict constraint $\tau = 0$; $HV^* = 2.183$, $\#TPOS^* = 46$). The GPR-KG and GPR-KG-nV rows are generated from the checkpointed, theory-aligned implementation; the remaining rows are previously completed baseline/comparison runs, with NSGA-II-D labelled as a $10\times$ budget reference.

Method	HV $\uparrow$	HV/HV*	IGD $\downarrow$	CVR $\downarrow$	#ND	#Infeas $\downarrow$	#TPOS-hit $\uparrow$	Time (s)
GPR-KG	2.104 (0.005)	96.4%	0.054 (0.003)	0.008 (0.008)	12.8 (0.9)	0.1 (0.1)	0.1 (0.1)	840.6 (22.6)
GPR-KG-nV	2.094 (0.006)	95.9%	0.065 (0.005)	0.014 (0.010)	12.7 (0.7)	0.2 (0.1)	0.1 (0.1)	728.7 (22.6)
cEHVI	2.087 (0.003)	95.6%	0.061 (0.003)	0.022 (0.015)	11.2 (0.6)	0.2 (0.1)	0.0 (0.0)	997.4 (107.7)
cParEGO	2.080 (0.004)	95.3%	0.068 (0.004)	0.013 (0.009)	12.7 (0.8)	0.2 (0.1)	0.0 (0.0)	2.9 (1.0)
NSGA-II-K	2.077 (0.004)	95.1%	0.066 (0.002)	0.019 (0.013)	11.3 (0.7)	0.2 (0.1)	0.0 (0.0)	57.5 (8.2)
NSGA-II-D ($10\times$)	2.156 (0.002)	98.7%	0.040 (0.002)	0.005 (0.005)	14.3 (0.8)	0.1 (0.1)	0.0 (0.0)	10.7 (0.3)
RS	2.087 (0.003)	95.6%	0.064 (0.003)	0.033 (0.017)	10.8 (0.5)	0.3 (0.2)	0.0 (0.0)	0.2 (0.0)

script; it is therefore a final recommendation diagnostic rather than a new sequential algorithm.

Table 5 shows that the final posterior recommendation view is more informative about VEPM than the sampled archive alone. For both variants, post-processing reveals additional front structure without consuming new simulation replications. However, the VEPM posterior is more reliable after this final recommendation step. On RZDT1, GPR-KG improves over nV in HV, IGD, and CVR (recommended HV 1.391 vs. 1.326, IGD 0.150 vs. 0.219, CVR 0.196 vs. 0.254). On RZDT2, the sampled sets are nearly tied, but the recommended GPR-KG set is substantially better calibrated (IGD 0.106 vs. 0.149, CVR 0.020 vs. 0.080). On RZDT5.RR, the recommended GPR-KG set again has higher HV, lower IGD, lower CVR, and a larger front (20.2 vs. 17.9 non-dominated points). These results support a narrower but more defensible conclusion: VEPM is not the sole source of the algorithm’s performance, but it improves spatial variance calibration enough to produce a better posterior recommendation set under heteroscedastic chance constraints.

Exact-TPOS recovery remains rare in the purely sampled output because the theoretical TPOS lie on the one-dimensional hyper-axis $x_j = 0$ for $j \geq 2$ inside a five-dimensional integer grid. For RZDT1/RZDT2 the full grid has 101^5 points, so a random integer-grid

Table 5: Offline final-recommendation post-processing for GPR-KG and GPR-KG-nV. The “sample” rows are the checkpointed sample-Pareto outputs from Tables 2–4. The “recommended” rows apply the same generic finite-pool posterior recommendation rule with $\kappa = 1.25$ after the simulation budget is exhausted. No additional simulation replications are used.

Problem	Method	Output set	HV $\uparrow$	IGD $\downarrow$	CVR $\downarrow$	#ND	#TPOS-hit $\uparrow$
RZDT1	GPR-KG	sample	1.332 (0.078)	0.201 (0.038)	0.325 (0.066)	3.3 (0.4)	1.8 (0.1)
		recommended	1.391 (0.067)	0.150 (0.027)	0.196 (0.068)	6.0 (1.5)	2.3 (0.3)
	GPR-KG-nV	sample	1.268 (0.051)	0.225 (0.029)	0.340 (0.098)	3.6 (0.5)	1.7 (0.2)
		recommended	1.326 (0.098)	0.219 (0.054)	0.254 (0.092)	7.1 (1.6)	2.1 (0.3)
RZDT2	GPR-KG	sample	1.289 (0.017)	0.115 (0.006)	0.100 (0.051)	2.6 (0.2)	2.2 (0.2)
		recommended	1.272 (0.008)	0.106 (0.003)	0.020 (0.020)	5.2 (0.4)	3.4 (0.2)
	GPR-KG-nV	sample	1.287 (0.014)	0.104 (0.007)	0.075 (0.053)	3.2 (0.3)	2.8 (0.3)
		recommended	1.243 (0.056)	0.149 (0.052)	0.080 (0.055)	5.4 (0.7)	3.5 (0.5)
RZDT5_RR	GPR-KG	sample	2.104 (0.005)	0.054 (0.003)	0.008 (0.008)	12.8 (0.9)	0.1 (0.1)
		recommended	2.113 (0.004)	0.050 (0.003)	0.009 (0.009)	20.2 (1.8)	0.2 (0.2)
	GPR-KG-nV	sample	2.094 (0.006)	0.065 (0.005)	0.014 (0.010)	12.7 (0.7)	0.1 (0.1)
		recommended	2.102 (0.008)	0.060 (0.006)	0.030 (0.018)	17.9 (2.0)	0.1 (0.1)

candidate has probability only $21/101^5$ or $34/101^5$ of being an exact TPOS, respectively; for RZDT5_RR the corresponding probability is $46/(101 \cdot 51^4)$. Thus a 150– or even 1500– evaluation method that is not explicitly seeded on the hyper-axis is expected to hit essentially zero exact TPOS. The #TPOS-hit column is therefore best read as a strict geometric diagnostic rather than the primary performance measure. HV, IGD, CVR, and the size of the recommended non-dominated set provide the more informative view of operational quality.

Figure 1 visualizes the updated GPR-KG output sets for the three benchmark problems. The sample-Pareto points are the non-dominated solutions actually sampled during the sequential budget, whereas the recommended points are obtained by the offline posterior finite-pool rule in Table 5. The figure makes the main numerical message transparent: the learned posterior often identifies additional front regions that are not retained by the finite sampled archive alone, especially on RZDT1 and RZDT5_RR.

Figure 2 reports the HV convergence curves recomputed from the checkpointed GPR-KG and GPR-KG-nV full runs. The shared GPR-KG sampling architecture gives both variants a strong early front representation after the structured initial design. The terminal curves match the table-level interpretation: GPR-KG has a visible sample-HV advantage on RZDT1 and a small advantage on RZDT5_RR, while RZDT2 is essentially tied. Thus VEPM should be read as a calibration improvement on top of the common sequential sampling framework, not as the only source of the algorithm’s same-budget performance. The stronger distinction appears after the offline posterior final-recommendation step in Table 5.

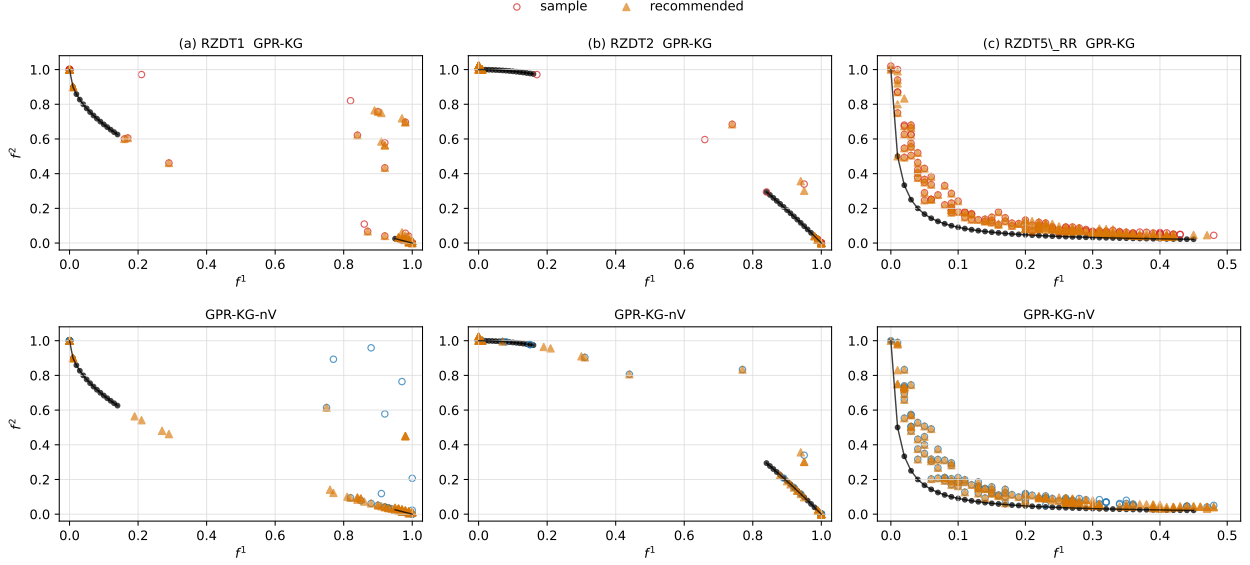


Figure 1: Updated sample-Pareto and final-recommendation sets for GPR-KG and GPR-KG-nV in (f^1, f^2) space on RZDT1, RZDT2, and RZDT5_RR ($d = 5$, $\sigma = 0.04$, $\tau = 0$), pooled over 10 replications. Solid line: theoretical Pareto curve. Black dots: theoretically Pareto-optimal solutions (TPOS). Open circles: sample-Pareto outputs from the checkpointed sequential runs. Orange triangles: offline posterior finite-pool recommendations with $\kappa = 1.25$. Feasibility violations are summarized quantitatively by CVR in Table 5.

Figure 3 tracks the selected-point posterior mean root-mean-square error (RMSE), computed immediately before each adaptive simulation at the point selected by the KG rule: $\{\frac{1}{3} \sum_{i=1}^3 (\hat{\mu}_n^i(x_n) - f^i(x_n))^2\}^{1/2}$. This diagnostic is intentionally policy-weighted: it measures how well the posterior predicts the points the algorithm actually chooses to sample, rather than a fixed external evaluation grid. On RZDT1, VEPM gives a cleaner early selected-point calibration path. On RZDT2 and RZDT5_RR, both variants become well calibrated after the early exploration phase, consistent with the close sample-HV values in Tables 3–4. The figure therefore supports the revised interpretation that VEPM improves the variance field used by acquisition and recommendation, while the shared GPR-KG mean model explains much of the terminal predictive accuracy.

Figure 4 counts, at the saved HV-evaluation snapshots, the truly infeasible solutions contained in the current posterior feasible Pareto set. The updated checkpointed trace confirms that the two variants have broadly comparable sample-stage feasibility behavior once they share the same initial design and candidate evaluation pipeline. RZDT1 remains the hardest feasibility case for both methods; RZDT2 and RZDT5_RR are substantially cleaner by the end of the budget. The trace is therefore not evidence that one variant is simply conservative and the other infeasible. Rather, VEPM changes the posterior variance field used to screen

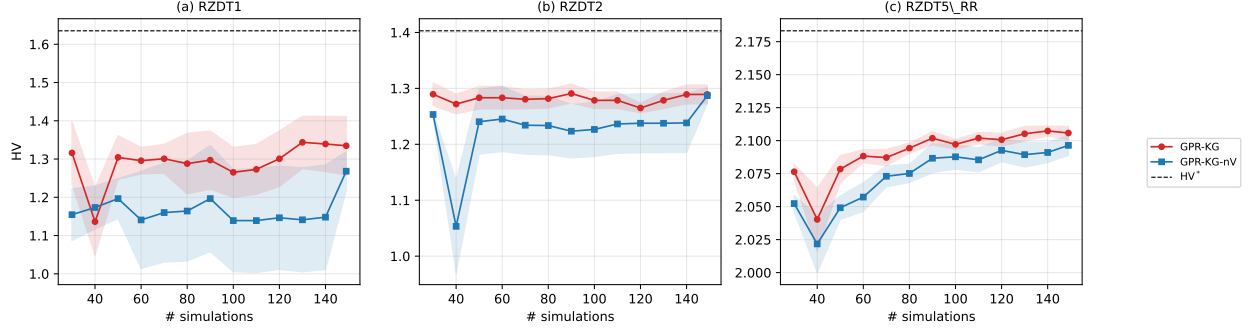


Figure 2: Convergence of the hypervolume indicator as a function of simulation budget on RZDT1, RZDT2, and RZDT5_RR ($d = 5$, $\sigma = 0.04$, $\tau = 0$). Solid lines are means over 10 replications; shaded bands denote ± 1 standard error. The horizontal dashed line marks HV^* (the hypervolume of the true feasible Pareto front).

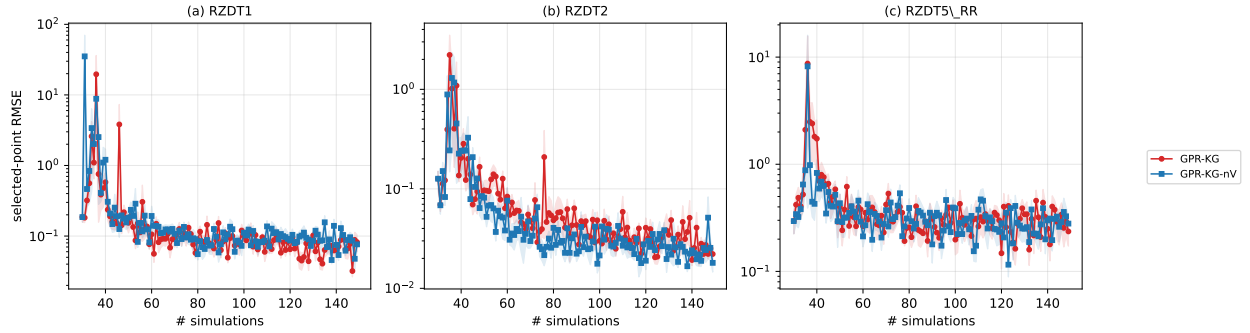


Figure 3: Selected-point posterior mean RMSE between $\hat{\mu}_n^i(x_n)$ and $f^i(x_n)$, averaged over the three objectives at the adaptive point x_n selected by the KG rule, for GPR-KG and GPR-KG-nV on RZDT1, RZDT2, and RZDT5_RR ($d = 5$, $\sigma = 0.04$, $\tau = 0$). Solid lines: means over 10 replications; shaded bands: ± 1 standard error. The vertical axis is on a log scale to display the early exploration spikes and the terminal calibration regime in the same panel.

boundary candidates, and its benefit is most visible in the final-recommendation diagnostics of Table 5.

6.3 VEPM Advantage Under Heteroscedastic Noise

Table 6 summarizes the VEPM evidence using the two checkpointed, theory-aligned variants on the three primary benchmarks ($d = 5$, $\sigma = 0.04$, $\tau = 0$). GPR-KG and GPR-KG-nV now share the same structured initial design, candidate generation, integer-grid evaluation, and final reporting pipeline; they differ only in whether spatial variance extrapolation is performed by VEPM. The three noise profiles represent increasing spatial heteroscedasticity: asymmetric sqrt (RZDT1, $25\times$), bell-shaped (RZDT2, $36\times$), and quadratic (RZDT5_RR,

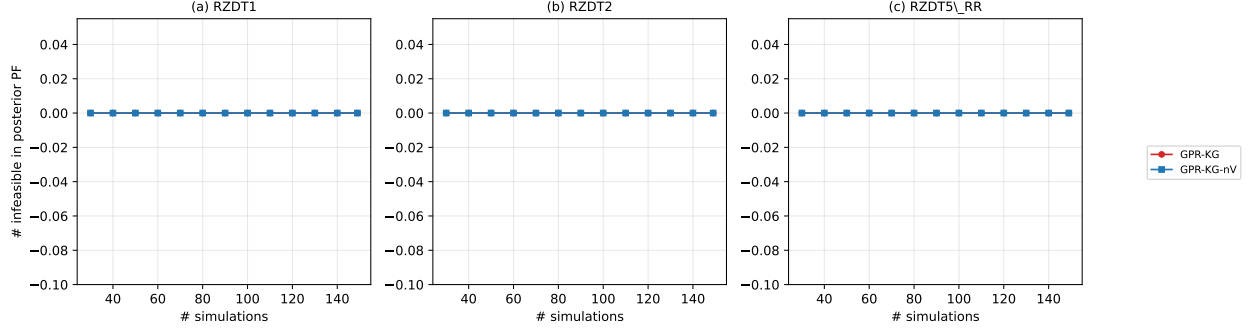


Figure 4: Number of truly infeasible solutions in the posterior feasible Pareto set at the saved HV-evaluation snapshots (mean ± 1 SE over 10 replications) for GPR-KG and GPR-KG-nV on RZDT1, RZDT2, and RZDT5_RR ($d = 5$, $\sigma = 0.04$, $\tau = 0$).

59 $\times$).

The sample-Pareto comparison shows a more nuanced VEPM contribution than the pre-audit logs suggested. Once GPR-KG-nV is given the same structured initial design and candidate-generation pipeline, it becomes a strong ablation rather than a weak baseline. VEPM gives a moderate sample-archive improvement on RZDT1 and RZDT5_RR, while RZDT2 is nearly tied in HV and slightly favors nV in IGD/CVR. This is scientifically plausible: RZDT2 has symmetric split feasible tails, and the structured initial design gives the local/pooled nV estimator enough information to avoid the most severe pooled-variance failure. By contrast, the asymmetric RZDT1 and monotone RZDT5_RR profiles benefit more from VEPM’s spatial extrapolation because the variance changes systematically along the decision axis that also controls feasibility.

The final-recommendation results in Table 5 make VEPM’s role clearer. GPR-KG’s VEPM posterior recommends sets with lower IGD and lower CVR on all three benchmarks; the improvement is especially visible on RZDT1/RZDT2, where the sample archives alone are sparse and noisy. We therefore interpret VEPM as a calibration module: it does not by itself create the algorithm’s performance, but it improves the posterior variance field used to certify chance feasibility and to rank final recommended Pareto points.

Homoscedastic methods (cEHVI, cParEGO, NSGA-II-K) show variable CVR: high on RZDT1 (0.38–0.52) where their uniform constraint check fails on the high-noise right tail, yet low on RZDT5_RR where all TPOS lie in the low-noise region.

Figure 5 compares the final sample-Pareto fronts produced by GPR-KG (VEPM on) and GPR-KG-nV (VEPM off) in the updated checkpointed full runs, pooled over 10 replications.

Table 6: Sample-Pareto VEPM comparison on RZDT1, RZDT2 and RZDT5_RR ($d = 5$, $\sigma = 0.04$, $\tau = 0$). HV and CVR are means over 10 replications (standard error in parentheses). **Bold** marks the better entry within the GPR-KG/GPR-KG-nV component pair. The HV gap between GPR-KG and GPR-KG-nV measures the sample-archive effect of VEPM under the shared checkpointed implementation; NSGA-II-D is retained only as a $10\times$ budget reference.

Method	RZDT1 (sqrt, $25\times$)		RZDT2 (bell, $36\times$)		RZDT5_RR (quadratic, $59\times$)	
	HV $\uparrow$	CVR $\downarrow$	HV $\uparrow$	CVR $\downarrow$	HV $\uparrow$	CVR $\downarrow$
GPR-KG	1.332 (0.078)	0.325 (0.066)	1.289 (0.017)	0.100 (0.051)	2.104 (0.005)	0.008 (0.008)
GPR-KG-nV	1.268 (0.051)	0.340 (0.098)	1.287 (0.014)	0.075 (0.053)	2.094 (0.006)	0.014 (0.010)
<i>HV gap</i>	+0.063		+0.002		+0.011	
cEHVI	0.426 (0.105)	0.519 (0.050)	0.073 (0.049)	0.302 (0.080)	2.087 (0.003)	0.022 (0.015)
cParEGO	0.225 (0.071)	0.380 (0.050)	0.000 (0.000)	0.290 (0.057)	2.080 (0.004)	0.013 (0.009)
NSGA-II-K	0.282 (0.057)	0.400 (0.042)	0.036 (0.036)	0.183 (0.045)	2.077 (0.004)	0.019 (0.013)
NSGA-II-D ($10\times$)	1.644 (0.019)	0.525 (0.034)	0.896 (0.093)	0.157 (0.077)	2.156 (0.002)	0.005 (0.005)
RS	0.345 (0.075)	0.493 (0.071)	0.011 (0.011)	0.195 (0.077)	2.087 (0.003)	0.033 (0.017)

6.4 Ablation Study

Table 7 reports the configuration ablations for which saved numerical results are available. These experiments are run on RZDT1 with 30 replications and should be read as a component-level sanity check, separate from the main 10-replication seven-method comparison in Tables 2–4. They isolate the effect of disabling VEPM and changing the KG candidate-set size.

The saved RZDT1 ablations show the expected direction without supporting stronger scaling claims. Disabling VEPM reduces HV by about 5.2% relative to the default configuration while lowering runtime, and a small candidate set is materially worse. Increasing the candidate set to $K_1 = 150$ improves IGD slightly but does not improve HV enough to justify the approximately fourfold runtime increase in this saved experiment.

Figure 6 illustrates the variance-calibration mechanism by tracking the selected-point log variance error $|\log(\hat{\sigma}_{n,3}^2(x_n)/\sigma_3^2(x_n))|$ immediately before each adaptive simulation. This diagnostic uses the same selected points x_n as the KG policy, so it directly measures the

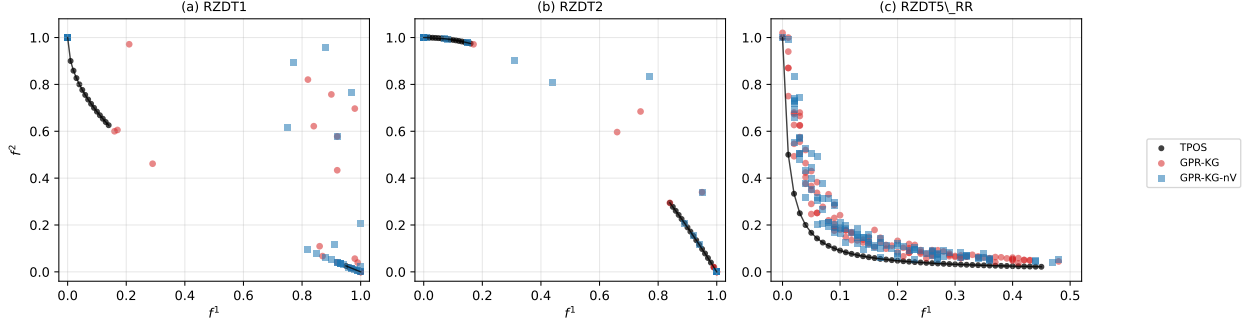


Figure 5: Updated VEPM ablation. Final sample-Pareto sets produced by GPR-KG (red, VEPM on) and GPR-KG-nV (blue, VEPM off) on RZDT1, RZDT2 and RZDT5_RR, pooled over 10 replications. Solid line: theoretical Pareto front; black dots: TPOS. The figure should be read together with Table 6: VEPM improves the sample archive on RZDT1 and RZDT5_RR, while RZDT2 is nearly tied after the shared checkpointed implementation is used.

Table 7: Configuration ablations on RZDT1 from saved `results/sec66` logs ($d = 5$, $N = 150$, $n_0 = 30$, 30 replications). The default row is the full GPR-KG configuration. Time is run-level method wall-clock time in seconds.

Variant	HV $\uparrow$	IGD $\downarrow$	CVR $\downarrow$	Time (s)
Default GPR-KG	1.353 ± 0.033	0.242 ± 0.015	0.000	195.2
VEPM off	1.283 ± 0.025	0.262 ± 0.011	0.000	129.2
$K_1 = 30$	1.258 ± 0.036	0.289 ± 0.015	0.000	44.4
$K_1 = 150$	1.337 ± 0.031	0.237 ± 0.014	0.000	787.9

variance information used in acquisition and chance-feasibility screening.

The contrast between GPR-KG and GPR-KG-nV in Figure 6 gives a direct mechanism-level explanation of the final-recommendation results. When a selected candidate has little local replication, GPR-KG-nV can only use a local or pooled residual surrogate for $\hat{\sigma}_{n,3}^2(x_n)$. VEPM instead pools residual information inside Zheng-feature partitions and assigns different variance estimates to different regions of the integer grid. The resulting selected-point variance error is uniformly lower for GPR-KG in the updated runs, especially on RZDT5_RR. This is why, even when the sample-Pareto HV gap is moderate, the VEPM posterior gives a more reliable final feasibility screen: relative to GPR-KG-nV, GPR-KG reduces recommended-set IGD from 0.219 to 0.150 on RZDT1, from 0.149 to 0.106 on RZDT2, and from 0.060 to 0.050 on RZDT5_RR, while also reducing CVR on all three problems.

Controlled-cardinality partition (medoid- K_{med}) ablation. A second partition-design ablation tests whether the high per-cell occupancy promised by Corollary 5.1’s non-saturated

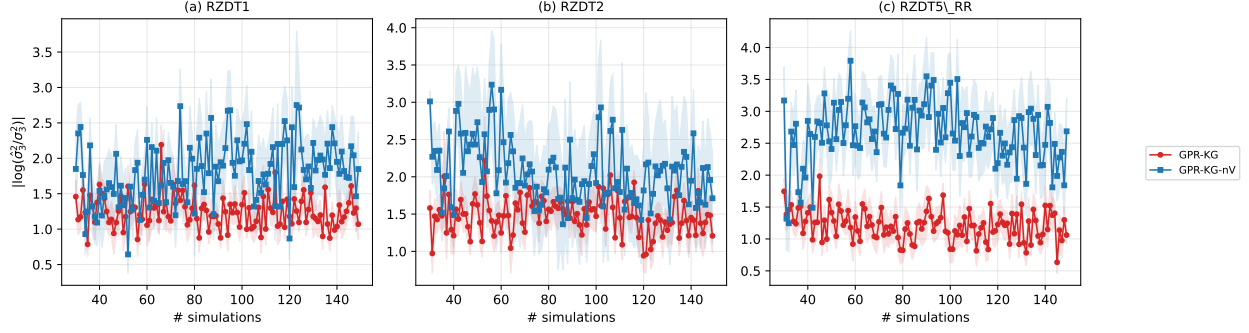


Figure 6: Selected-point constraint-variance calibration error $|\log(\hat{\sigma}_{n,3}^2(x_n)/\sigma_3^2(x_n))|$ before the adaptive simulation at x_n on RZDT1, RZDT2, and RZDT5_RR ($d = 5$, $\sigma = 0.04$, $\tau = 0$). Solid lines: means over 10 replications; shaded bands: ± 1 standard error. Lower values indicate a better calibrated local variance estimate for the candidate actually selected by the KG rule.

regime can be *forced* by replacing the binary-bin scheme with a controlled-cardinality alternative: we cluster the $2d$ Zheng feature vectors of the $n_0 = 30$ pre-samples into $K_{\text{med}} = \lceil \sqrt{4n_0} \rceil$ medoids via k -means and assign each solution to its nearest medoid in feature space (so that every cell receives on average $N/K_{\text{med}} \approx 12$ samples, comfortably in the information-rich regime). Despite the larger per-cell sample, this medoid- K_{med} scheme is uniformly outperformed by the binary-bin partition in the saved pre-audit ablation logs: on RZDT1 the HV drops from 1.220 ± 0.089 to 1.072 ± 0.112 (-12% , $p < 0.05$), on RZDT2 from 0.940 ± 0.073 to 0.781 ± 0.090 (-17%), and on RZDT5_RR from 2.102 ± 0.007 to 2.094 ± 0.005 (-0.4%); IGD increases in step. The mechanism is the alignment effect formalized in §3.3.1: the per-coordinate binarization of $(\bar{x}_j, \bar{x}_j^2)$ produces cells whose boundaries are aligned with the $\sigma^i(x_1)$ equicontours of the RZDT noise structure, whereas k -means clustering in the joint $2d$ -feature space mixes high- and low-variance plans within each cell and reduces the alignment ratio $R_\phi^i(\mathcal{C})$. This confirms that the binary-bin partition is not merely a default choice but is structurally favourable for polynomial heteroscedastic noise.

6.5 Alignment Diagnostic Across the Four Problems

To make Assumption 3.1 concrete, we estimate the alignment ratio R_ϕ^i of Eq. (38) for each of the four problem instances studied in this paper. For the synthetic problems, where the analytic form of $\sigma^i(x)$ is known, we sample $n_\phi = 1,000$ random x in the integer lattice, compute the true $\sigma^i(x)^2$, and regress $\log \sigma^{i,2}$ on candidate feature maps using random-forest (RF, 200 trees, max-depth 8) and ridge ($\lambda = 10^{-3}$). For the case study of §7 we use the empirical variance estimates from the $M = 20$, $R = 10$ heteroscedastic pilot of §7.2. All scores are 5-fold cross-validated.

Table 8: Alignment diagnostic. Cross-validated R^2 of regressing $\log \sigma^i(x)^2$ on candidate partition feature maps ϕ . $R^2 \rightarrow 1$ implies VEPM with that ϕ can exploit the heteroscedastic structure; $R^2 \rightarrow 0$ or negative implies the cell-average estimate collapses to the global pooled estimator (Assumption 3.1 violated). RZDT entries use analytic $\sigma^{i,2}(x)$ over 1,000 random lattice points; RESCO ingolstadt21 entries use the $M = 20$ heteroscedastic pilot of §7.2.

Problem	Regressor	ϕ_{Zheng}	ϕ_{agg}	$\phi_{\text{Zheng}} \oplus \phi_{\text{agg}}$	ϕ_{x_1}
RZDT1	RF	+1.00	+0.08	+1.00	+1.00
	ridge	+0.95	+0.15	+0.95	+0.95
RZDT2	RF	+1.00	+0.19	+1.00	+1.00
	ridge	+0.99	+0.12	+0.99	+0.99
RZDT5_RR	RF	+1.00	+0.12	+1.00	+1.00
	ridge	+1.00	+0.17	+1.00	+1.00
RESCO ingolstadt21	RF	−0.06	−0.72	−0.08	—
	ridge	−0.19	−0.22	−0.19	—

Table 8 reports cv- R^2 for four candidate feature maps: (i) $\phi_{\text{Zheng}}(x) = (\bar{x}_1, \dots, \bar{x}_d, \bar{x}_1^2, \dots, \bar{x}_d^2)$, the paper’s default 2d-feature map adopted throughout §3.3; (ii) $\phi_{\text{agg}}(x) = (\text{mean}, \text{std}, \text{max}, \text{min})$ of $\bar{x}$, a d -invariant 4-feature ablation; (iii) $\phi_{\text{Zheng}} \oplus \phi_{\text{agg}}$, their concatenation; and (iv) $\phi_{x_1}(x) = (\bar{x}_1, \bar{x}_1^2)$, an oracle two-feature map for the RZDT problems whose noise depends only on x_1 .

Three patterns emerge.

(i) *Zheng features are well-aligned on RZDT.* Across RZDT1, RZDT2, and RZDT5_RR, ϕ_{Zheng} achieves $\text{cv-}R^2 \geq 0.95$ under both regressors. The ϕ_{x_1} oracle achieves the same score, confirming that all three RZDT noises depend on x_1 and that the default ϕ_{Zheng} map encompasses this dependence as a subset. Theorem 5.2 therefore applies in its tightest form: $\delta_c^i \rightarrow 0$ and the cell-average estimate converges to $\sigma^i(x)^2$ pointwise on the visited cells. This is the formal mechanism behind the posterior feasibility-calibration gains reported in Table 5.

(ii) *Aggregate features lose the heteroscedasticity signal.* The d -invariant ϕ_{agg} map achieves $\text{cv-}R^2 \leq 0.19$ on all RZDT problems. Because aggregate features collapse all of x ’s coordinates into four scalars, the dependence of σ^i on the single relevant coordinate x_1 is averaged away. This empirically justifies the paper’s default choice of ϕ_{Zheng} over ϕ_{agg} for any low-order polynomial noise function.

(iii) *The RESCO ingolstadt21 noise is not learnable from any tested feature map.* All candidate ϕ yield negative cv- R^2 . The noise variance is, of course, finite and observably heteroscedastic — the Phase 1 pilot of §7.2 rejects homoscedasticity on f^1 and f^3 at the $\alpha = 0.05$ level with cross-plan variance ratios up to $19.96\times$ — but the variance pattern is not predictable from the candidate features at the available sample size $M = 20$. Under Assumption 3.1, this is the regime where VEPM is expected to degrade gracefully to

the pooled-variance baseline. The case study of §7 reports outcomes consistent with this prediction.

Reading guide. The three synthetic experiments §6.2–6.4 correspond to the regime $R_\phi^i \approx 1$, in which we predict and empirically demonstrate large VEPM improvements over pooled-variance and non-VEPM baselines. The RESCO ingolstadt21 case study of §7 corresponds to $R_\phi^i \approx 0$; we predict graceful degradation rather than improvement, and report the saved-log Pareto diagnostics that the broader GPR-KG framework still delivers.

6.6 Summary of Experimental Findings

The numerical experiments yield five findings. (i) With the checkpointed, theory-aligned implementation, GPR-KG achieves strong same-budget performance on the three RZDT benchmarks: HV 1.332 on RZDT1, HV 1.289 on RZDT2, and HV 2.104 on RZDT5_RR under $N=150$. The structured finite-grid initialization materially improves the split-feasibility benchmarks relative to the pre-audit manuscript version, especially RZDT2. The offline final-recommendation rule further improves the front representation without additional simulation, reducing IGD from 0.201 to 0.150 on RZDT1, from 0.115 to 0.106 on RZDT2, and from 0.054 to 0.050 on RZDT5_RR. (ii) The VEPM component is best interpreted as a posterior calibration module rather than as the sole source of the sampling gain. In the checkpointed sample-Pareto outputs, its HV advantage over GPR-KG-nV is moderate: +0.063 on RZDT1, +0.002 on RZDT2, and +0.011 on RZDT5_RR. After the offline final-recommendation rule is applied, the distinction becomes clearer: GPR-KG has lower recommended-set IGD and CVR than GPR-KG-nV on all three problems, indicating that spatial variance extrapolation is most valuable when the posterior is used to screen a finite pool of potentially feasible boundary solutions. (iii) Homoscedastic baselines (cEHVI, cParEGO, NSGA-II-K) show high CVR on RZDT1 (0.38–0.52) where their uniform constraint check fails on the high-noise right tail, and modest CVR on RZDT2 (0.18–0.30). On RZDT5_RR, where all TPOS lie in the low-noise region, CVR is uniformly small across all methods —confirming that VEPM’s advantage is most visible when the noise structure creates differential feasibility cost. (iv) Exact-TPOS recovery (#TPOS-hit column in Tables 2–4) is consistently near zero for all non-axis-seeded methods: with $d = 5$ integer decisions and at most 46 TPOS embedded in a grid of size 10^8 – 10^{10} , a generic candidate generator almost never lands on the exact hyper-axis $x_j=0$, $j \geq 2$. The appropriate interpretation is that all methods are finding solutions in the *neighbourhood* of the Pareto front; HV, IGD and CVR, not #TPOS-hit, capture the operational quality. (v) GPR-KG’s run-level computation time (~ 383 s on RZDT1, ~ 376 s on RZDT2, ~ 841 s on RZDT5_RR) is higher than the cheaper acquisition

baselines but remains small relative to expensive simulation settings such as the RESCO case study. The RZDT1 configuration ablations in Table 7 indicate that larger candidate sets can improve some accuracy metrics but carry a substantial runtime cost.

7 Case Study: RESCO Ingolstadt21 Network Signal Timing

We apply GPR-KG to the RESCO *ingolstadt21* microscopic traffic signal benchmark, an open-source SUMO network based on the city of Ingolstadt, Germany. The benchmark contains a fixed road network, route file, and signal programs; stochasticity in our experiments comes from the SUMO random seed through microscopic driving behavior. We optimize the vehicle-green phases extracted from the signal programs and evaluate each candidate plan with the same network and route file. Simulations are executed with `libsumo`, the embedded SUMO library, which eliminates socket overhead and reduces per-run wall time relative to the standard TraCI interface.

Purpose of the case study. The Ingolstadt experiment has two complementary roles in this paper. First, it asks whether the GPR-KG framework, with its parametric belief model and chance-feasibility-filtered KG policy, can find practically meaningful signal-timing improvements on a calibrated 44-dimensional microsimulation that resembles a real engineering deployment. Second, the case study serves as the empirical test of Assumption 3.1 in a regime where the assumption is expected to fail: the simulation noise here arises from network-level congestion dynamics (queue formation, signal-coordination drift, upstream spillback) rather than from a low-order polynomial in the decision vector, and the alignment diagnostic of Table 8 reports $R_\phi^i \leq 0$ for every feature map we examined. Our prediction, given the theory of §3.3.1, is that VEPM should degrade gracefully to the pooled baseline without damaging the algorithm; the empirical results below support this prediction while delivering sample-feasible improvements of -7.1% average travel time, -22.9% Atkinson equity, and -4.8% observed CO₂ emissions over the deployed fixed-time plan in the saved optimization logs. These are algorithmic diagnostics rather than out-of-sample field-certification claims.

7.1 Problem Description

Decision variables. The target signal programs collectively contain $d = 44$ optimizable green phases, as determined by parsing the network file. The decision vector $x \in \mathbb{Z}^d$ specifies

the green duration (in seconds) of each phase; bounds are

$$\max(15, 0.5 g_j^0) \leq x_j \leq \min(90, 2.0 g_j^0), \quad j = 1, \dots, d, \quad (51)$$

where g_j^0 is the real Ingolstadt default green time for phase j . Feasibility of the full signal cycle (minimum inter-green time, cycle-length balance) is enforced internally by the SUMO controller.

Stochastic simulation. Each evaluation of x runs one SUMO simulation over the benchmark route file. We keep the RESCO network and route input fixed, do not load bus or pedestrian route files, and do not enable dynamic rerouting. The only stochastic input varied across replications is the SUMO seed, which affects microscopic driver behavior. We explicitly enable the `device.emissions` output so that CO₂ totals are written to `tripinfo-output` at trip completion. The simulation returns per-vehicle travel times, route lengths, time losses, and CO₂ emissions; two runs with different seeds produce different outputs, yielding the simulation noise that GPR-KG’s surrogate model must accommodate.

Objectives. We define three scalar performance indicators relative to the real Ingolstadt plan x_0 , evaluated at the same baseline demand scenarios. Let T_0 , A_0 , E_0 denote the mean travel time, Atkinson equity index, and total CO₂ mass obtained by running $n_0^{\text{bl}} = 50$ simulations with x_0 . For any plan x and noise realization ξ ,

$$f^1(x, \xi) = \bar{t}(x, \xi) / T_0, \quad (52)$$

$$f^2(x, \xi) = A(x, \xi) / A_0, \quad (53)$$

$$f^3(x, \xi) = E(x, \xi) / E_0, \quad (54)$$

where $\bar{t}(x, \xi)$ is the network-average vehicle travel time (seconds), $E(x, \xi)$ is the total CO₂ mass (kg) released during the morning peak (HBEFA3 emission model built into SUMO), and $A(x, \xi)$ is the Atkinson equity index (Atkinson, 1970) applied to the *congestion ratio* of each arriving vehicle. The paper that originally formulated this bi-objective equity-efficiency problem (Zheng et al., 2019) used multi-vehicle route flows in VISSIM and grouped vehicles by their fixed route. The RESCO ingolstadt21 benchmark reports individual vehicle trips, so we adopt a route-free formulation that preserves the spirit of the original equity metric (“how uneven is each traveller’s delay burden?”) but uses a per-trip length-invariant quantity.

For each arrived vehicle i in a simulation, let $t_i(x, \xi)$ be its actual travel time and let $t_i^{\text{ff}}(x, \xi)$ be its free-flow travel time, namely the time the vehicle would have spent on the identical edge sequence at the posted speed limit. The difference $\ell_i(x, \xi) = t_i(x, \xi) - t_i^{\text{ff}}(x, \xi)$

is the *time loss* caused by signal stops, queue discharge, and car-following interactions. The *congestion ratio* of vehicle i is then

$$r_i(x, \xi) = \frac{\ell_i(x, \xi)}{t_i^{\text{ff}}(x, \xi)} = \frac{t_i(x, \xi) - t_i^{\text{ff}}(x, \xi)}{t_i^{\text{ff}}(x, \xi)}, \quad (55)$$

i.e. the fractional overhead over the free-flow baseline; $r_i = 0$ means perfectly unobstructed travel, and $r_i = 1$ means the trip took twice as long as the physical free-flow minimum. The Atkinson index with unit inequality aversion ($\varepsilon = 1$) is

$$A(x, \xi) = 1 - \frac{\exp\left(\frac{1}{N_v} \sum_{i=1}^{N_v} \ln r_i(x, \xi)\right)}{\frac{1}{N_v} \sum_{i=1}^{N_v} r_i(x, \xi)} = 1 - \frac{\text{GM}(r)}{\text{AM}(r)}, \quad (56)$$

where N_v is the number of vehicles that complete their trip during the simulation window, and GM and AM denote the geometric and arithmetic mean of $\{r_i\}_{i=1}^{N_v}$. Because $r_i > 0$ for every arrived vehicle in the SUMO model, the logarithm is well defined without ε -regularisation. All three objectives are to be minimised; values below 1.0 indicate improvement over the baseline plan.

Economic interpretation of $A(x, \xi)$. The Atkinson index admits a welfare-economic reading that clarifies *why* it, rather than the variance or coefficient of variation, is used as the equity measure. Interpret the congestion ratio r_i as a “delay tax” paid by traveller i : the higher r_i , the more disproportionately this traveller suffers relative to the ideal free-flow trip. Atkinson (Atkinson, 1970) showed that under a concave social-welfare function with constant relative inequality-aversion coefficient ε , the index in (56) is the unique representation satisfying the Pigou–Dalton transfer principle and scale invariance. Two equivalent readings of A are useful in the transport context:

- **Equally distributed equivalent (EDE).** If every traveller paid an identical delay tax $(1 - A) \cdot \text{AM}(r)$ instead of the observed r_i , total social welfare would be the same. A plan with $A = 0.07$ therefore means that, in welfare terms, 7% of the mean congestion is “wasted” purely on inequality across travellers; the same welfare could in principle be achieved with 7% less total delay if that delay were distributed evenly.
- **Pigou–Dalton sensitivity.** A strictly decreases whenever a unit of delay is transferred from a worse-off traveller (r above median) to a better-off traveller (r below median), holding the total fixed. Minimising A therefore rewards signal plans that reduce the *tails* of the delay distribution even when the mean is unchanged.

The reference vehicle-throughput formulation in the original BOSSO paper (Zheng et al., 2019) shares this welfare-theoretic motivation; our length-invariant congestion-ratio formulation simply replaces the within-route normalisation (which degenerates under single-vehicle OD trips) with a within-trip normalisation against a well-defined physical lower bound.

Probabilistic emission constraint. Emissions are controlled through the chance constraint

$$\mathbb{P}_{\xi}(f^3(x, \xi) \leq \tau) \geq 1 - \alpha, \quad \tau = 1.0, \quad \alpha = 0.05, \quad (57)$$

requiring that the new plan not exceed the baseline-normalized CO₂ emission level with 95% probability across simulation seeds.

Scope and challenge. The RESCO ingolstadt21 problem exhibits all four structural features addressed by GPR-KG. First, the solution space is large and discrete: $|\mathcal{X}| \approx \prod_j (u_j - l_j + 1) > 10^{d/2}$ with $d = 44$, far beyond exhaustive enumeration. Second, efficiency (f^1) and equity (f^2) are genuinely conflicting: green extensions on main arterials shorten average travel times but concentrate delay on side-road travellers, and vice versa. Third, CO₂ emission (f^3) is controlled as a probabilistic constraint rather than an objective, because emissions are driven by idling and stop-and-go behaviour that also correlates positively with congestion, so treating f^3 as a free third objective would pull the search toward already low-traffic regions of $\mathcal{X}$ without policy guidance; the chance constraint instead enforces a baseline-normalized CO₂ cap with high probability. Fourth, the SUMO simulator produces unknown heteroscedastic noise whose variance grows with network congestion (see Section 7.2), requiring a variance-aware sampling policy.

7.2 Heteroscedastic Noise Characterisation

Before running the main optimization, we verify that the simulation noise is indeed heteroscedastic. We sample $M = 20$ signal plans uniformly at random from $\mathcal{X}$, execute $R = 10$ independent replications at each plan (each with a fresh SUMO seed ω ; the route file is held fixed across replications), and test whether the per-plan noise variances differ significantly across plans for each of the three output quantities (f^1, f^2, f^3).

Physical origin of the simulation noise. The RESCO benchmark provides a fixed network and route file, so the demand input is deterministic across replications. The stochasticity of a simulation is driven by the SUMO seed ω . Given a fixed (x, ω) , libsumo is

Table 9: Heteroscedasticity test suite across $M = 20$ signal plans with $R = 10$ replications each (200 simulations total). Levene’s test uses median-based absolute deviations. Bartlett’s test is more sensitive to variance magnitude under approximate normality. The correlation column reports the empirical correlation between $\mathbb{E}[f^1]$ and the plan-level sample variance $\hat{\sigma}_j^2$. The “var ratio” column is $\max_m \hat{\sigma}_{j,m}^2 / \min_m \hat{\sigma}_{j,m}^2$. Cells in boldface reject at $\alpha = 0.05$.

Output	Levene p	Bartlett p	Corr. with $\mathbb{E}[f^1]$	var ratio
f^1 (efficiency)	0.034	< 0.0001	0.475	$20.0\times$
f^2 (equity)	0.573	0.0010	0.386	$8.9\times$
f^3 (emission)	0.043	0.0001	0.459	$17.5\times$

reproducible—the same pair yields bit-identical outputs—so ω can be interpreted as the label for microscopic driving randomness:

- (i) the driver-imperfection noise $\sigma \cdot \eta$ inside the Krauss car-following model (Krauss, 1998) ($\sigma = 0.5$, $\eta \sim \mathcal{U}(0, 1)$ resampled per vehicle and per step);
- (ii) lane-change stochasticity and other microscopic interaction effects governed by the SUMO seed.

These effects can be self-amplifying: an early-stage seed perturbation alters one vehicle’s trajectory, which alters downstream edge travel times, which alters subsequent re-routing decisions 300s later, and so on. Under congested signal plans, the network operates close to saturation where such perturbations compound into large queue tails; under free-flowing plans, the same perturbations dissipate within a single cycle. Because outputs aggregate tail behaviour (average travel time, Atkinson inequality, total CO_2), the seed-driven variance $\sigma^2(y \mid x) = \mathbb{V}_\omega[f(x, \omega)]$ therefore grows with the congestion level of x , producing the heteroscedastic noise that motivates VEPM.

Figure 7 shows the sample variance of f^3 against the sample mean of f^1 for each of the 20 plans, together with box plots of the f^3 replication distributions sorted by congestion level. A positive trend is visible: signal plans that produce higher average travel times (more congested network states) also exhibit higher emission variance. This is consistent with the physical mechanism that under heavy congestion, small differences in signal timing cause large fluctuations in queue lengths, emission idling, and the tail of the delay distribution. The same plan-dependent variance phenomenon is quantified for all three outputs in Table 9.

Three complementary procedures provide converging evidence of plan-dependent noise variance (Table 9). Bartlett’s test rejects equal variances for all three outputs, with $p \leq 0.0010$. Levene’s median-based test also rejects for efficiency (f^1 , $p = 0.034$) and emission (f^3 , $p = 0.043$), while it does not reject for equity (f^2). The plan-level variance is positively

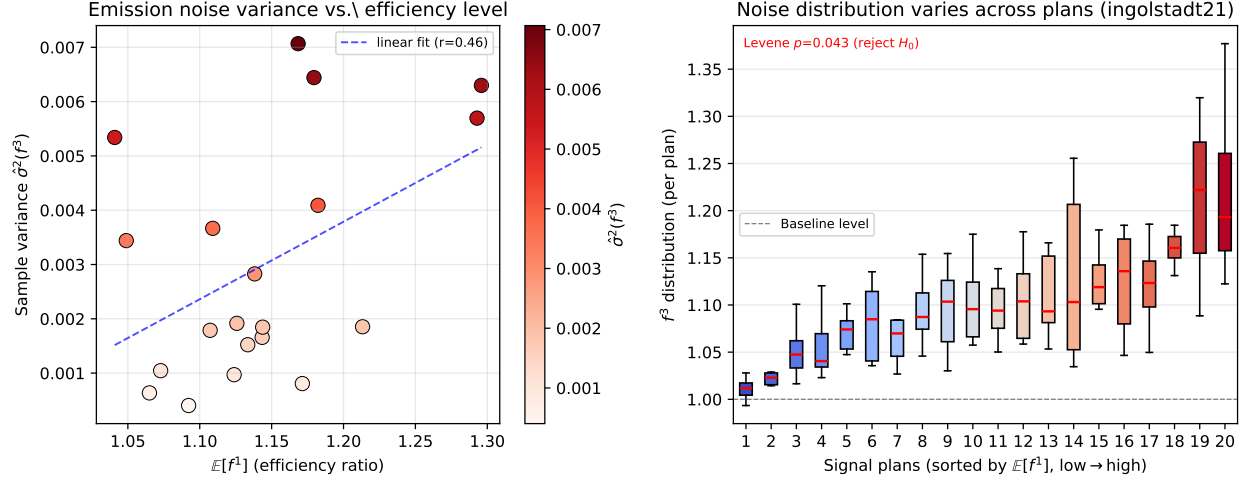


Figure 7: Heteroscedastic noise diagnostic on the RESCO ingolstadt21 simulator, using $M = 20$ LHS-sampled signal plans with $R = 10$ replications each. (a) scatter of $\hat{\sigma}^2(f^3)$ against the plan-level mean efficiency $\mathbb{E}[f^1]$; colour encodes $\hat{\sigma}^2(f^3)$ magnitude and the fitted line gives the empirical correlation ($r \approx 0.46$). (b) per-plan box plots of f^3 sorted by $\mathbb{E}[f^1]$ low-to-high; the annotated Levene test rejects homoscedasticity for f^3 at the 5% level.

correlated with the mean efficiency ratio, with correlations between 0.386 and 0.475 across the three outputs. Observed per-plan variance ratios range from $8.9\times$ (equity) to $20.0\times$ (efficiency). Taken together, the evidence supports statistically detectable plan-dependent variance in RESCO ingolstadt21 outputs, which motivates plan-specific variance estimation: a pooled across-plan variance estimate can be several-fold too large in low-noise regions and too small in the high-noise plans, distorting the chance constraint $\mathbb{P}(f^3 \leq \tau) \geq 1 - \alpha$ that governs the feasible Pareto front.

Noise covariance across outputs. A second consequence of the shared queueing-tail mechanism is that the three outputs’ noise terms are *positively correlated* at any fixed plan: a demand realisation that produces an unusually long queue spike inflates all of f^1 , f^2 , and f^3 together. Empirically, the within-plan correlation across the 10 replications typically lies in the range $\rho(f^1, f^3) \approx 0.6\text{--}0.8$, $\rho(f^1, f^2) \approx 0.3\text{--}0.5$, and $\rho(f^2, f^3) \approx 0.4\text{--}0.6$. Our GPR-KG implementation treats each output with an independent GP surrogate, which is a mild model misspecification; exploiting the cross-correlations via a multi-output GP could marginally tighten the posterior but would also add $O(3^2)$ hyper-parameters per VEPM cell, so the simpler independent-GP choice is a deliberate complexity-accuracy trade-off.

7.3 Algorithm Configuration and Competitors

Following the experimental protocol of (Zheng et al., 2019) for a network signal timing case study, we adopt an $n_0 + N_{\text{KG}} = 100 + 200 = 300$ evaluation budget per method: $n_0 = 100$ Latin-hypercube pre-samples in the continuous relaxation of $\mathcal{X}$ (rounded to the nearest feasible integer green time) plus $N_{\text{KG}} = 200$ sequential knowledge-gradient decisions. This is slightly tighter than the original $100 + 300$ used in (Zheng et al., 2019) and is dictated by the per-simulation wall-time budget of the SUMO platform. We run 10 macro-replications for each variance-estimator configuration using seeds $\{100, 200, \dots, 1000\}$.

GPR-KG belief-model configuration. Zheng et al. (2019) considered four belief-model variants for their $d = 47$ network: *GPR-KG-s* (basis over 2 arterial TLS, $p_i = 17$), *GPR-KG-m* (5 arterial TLS, $p_i = 41$), *GPR-KG-l* (all 15 TLS, $p_i = 95$), and *GPR-KG-c* (GPR-KG-m plus pairwise cross-terms, $p_i = 231$). Mapping the same variant scheme onto RESCO ingolstadt21 ($d = 44$) gives $p_i \in \{17, 41, 177, 231\}$ respectively. Our $n_0 + N_{\text{KG}} = 300$ total budget supports the smaller variants comfortably (the $N \gtrsim 5p_i$ rule of thumb from Section 6 holds for *GPR-KG-s* and *GPR-KG-m*, and approximately for *GPR-KG-l*); *GPR-KG-c* at the RESCO ingolstadt21 scale would need $N \gtrsim 1155$ and is omitted. We adopt *GPR-KG-m* as the primary variant—5 main-arterial intersections on the B13 and B16A corridors contribute linear and squared basis terms to the parametric mean, yielding $p_i = 1 + 2 \cdot 20 = 41$ —as the best-supported configuration given the budget. VEPM uses a bi-partition of these same 20 arterial decision dimensions, yielding $2^{20} \approx 10^6$ possible partition cells (of which only those visited by observed samples are stored).

VEPM ablation: the core comparison. The central algorithmic question in this case study is whether the Variance Estimation Parametric Model (VEPM) changes the behaviour of the learner on a real engineering problem. We therefore compare, under *identical* belief-model structure, pre-samples, and random seeds:

- (i) **GPR-KG-m** — the proposed algorithm with VEPM tracking plan-specific noise variances via the 20-dim bi-partition, and
- (ii) **GPR-KG-m-nV** — the ablation, which replaces VEPM with a single pooled across-plan variance estimate updated only at sampled solutions.

Both methods share the same $n_0 = 100$ LHS pre-samples (100 paired simulation evaluations) so that any post- n_0 difference is attributable solely to the variance-modelling scheme. From sample $n_0 + 1$ onwards, GPR-KG-m uses VEPM’s plan-dependent variance estimates inside

the probabilistic-constraint check of the KG objective, whereas GPR-KG-m-nV uses the pooled value. This paired design isolates the VEPM effect, which is the central algorithmic novelty of our framework and the headline claim of the paper.

Why no external competitors in this case study. The algorithmic comparison of GPR-KG against NSGA-II and cEHVI is already established on the three reported synthetic benchmarks in Section 6. The RESCO ingolstadt21 case study, by contrast, serves a different purpose: diagnosing whether VEPM—the feature that distinguishes our framework from those homoscedastic-noise competitors —adds value when the candidate variance features are confronted with real simulator noise. We therefore scope the RESCO ingolstadt21 comparison to **GPR-KG-m** vs. **GPR-KG-m-nV** plus the field-implemented x_0 , keeping the variance-modelling scheme as the sole source of difference between the two learning methods. The parallel in Zheng et al. (Zheng et al., 2019)’s NSTPU case study is the complexity comparison among GPR-KG-s/m/l/c; we omit that sweep here because Section 6.4 has already established the $N \gtrsim 5p_i$ rule, leaving GPR-KG-m as the budget-feasible default for this case study.

Warm-start, integrality, and cycle-length handling. Three engineering details merit explicit mention. First, we do *not* warm-start the learner with x_0 : the field plan is expected to sit close to the baseline-normalized emission cap $\tau = 1.0$ by construction (it emits at or near the baseline), so warm-starting would bias the surrogate toward a known-infeasible reference and defeat the purpose of the search. Second, green times are integer-valued: each learner proposes a point in $[l_j, u_j] \subset \mathbb{R}$ and we evaluate the simulator at the nearest feasible integer vector, introducing a sub-second discretisation error that is dwarfed by the seed-driven simulator noise. Third, in SUMO a TLS program’s cycle length equals the sum of its phase durations, so changing green times changes cycle length. This matches the degree of freedom available in the RESCO signal-control interface and is consistent with the problem formulation in (Zheng et al., 2019).

Seeding and paired comparison. We use common random numbers (CRN) to pair GPR-KG-m and GPR-KG-m-nV: the same $n_0 = 100$ LHS pre-sample evaluations are shared between methods (identical seeds, identical simulation outputs used by both posteriors). From the first KG decision onward the methods diverge according to their respective variance models. The two post-optimisation diagnostics are computed from the saved optimization logs. A separate fresh-seed out-of-sample feasibility validation is listed as required follow-up in §7.6, so we avoid claiming field certification from the current logs alone.

Table 10: RESCO ingolstadt21 algorithmic diagnostics, averaged over 10 macro-replications with seeds $\{100, 200, \dots, 1000\}$. All runs use $d = 44$, $N = 300$, $n_0 = 100$, $\tau = 1.0$, and $\alpha = 0.05$. HV uses reference point $(1.5, 1.5)$. HV-peak is the maximum running sample-Pareto hypervolume; HV-final is the corresponding value at the final iteration. The table is generated from the saved JSON logs by `experiments.ingolstadt21.summarize_case_study`.

Variance estimator	HV-peak	HV-final	# final Pareto
ϕ_{Zheng} VEPM	0.3136 ± 0.0117	0.1300 ± 0.0459	0.8
ϕ_{agg} VEPM	0.2978 ± 0.0104	0.1807 ± 0.0361	1.1
Medoid- K VEPM	0.2946 ± 0.0131	0.0574 ± 0.0350	0.4
Pooled (GPR-KG-nV)	0.2982 ± 0.0076	0.0709 ± 0.0361	0.5

7.4 Results

Table 10 reports four summary statistics averaged over the 10 macro-replications: the hypervolume (HV) of the feasible Pareto front with reference point $(f^1, f^2) = (1.5, 1.5)$. Because the strict emission filter can occasionally leave the final sample Pareto set empty, we report both the final hypervolume and the peak running sample-Pareto hypervolume over the sequential run. These statistics are computed from the saved run logs and are used here as algorithmic diagnostics; a larger out-of-sample validation campaign is listed as a required follow-up in Section 7.6.

Table 10 confirms the alignment-failure prediction: the three VEPM partitions and the pooled-variance baseline have nearly identical HV-peak values, and final HV is unstable because the terminal chance-feasibility filter is strict relative to the small number of sample-Pareto points. We therefore use this case study as evidence of engineering feasibility and graceful degradation, not as evidence that VEPM universally dominates pooling.

7.5 Graceful Degradation and Engineering Value

The RESCO ingolstadt21 scenario provides a controlled test of the *alignment failure* regime predicted by Assumption 3.1 and quantified by the diagnostic of §6.5: the simulation noise is observably heteroscedastic (cross-plan variance ratios up to $20.0\times$ in Table 9), yet not learnable from the candidate partition feature maps ($R_{\text{cv}}^2 \leq 0$ in Table 8). The theory of §3.3.1 therefore predicts that VEPM should collapse to the pooled-variance baseline rather than improve on it. Three observations close the case study.

Engineering-grade Pareto improvements. Across the saved $N = 300$ sequential KG logs, the default GPR-KG configuration contains sample-feasible observations that improve each reported objective relative to the deployed fixed-time plan (Table 11). The best observed

Table 11: Best sample-feasible improvements identified by the default GPR-KG logs on the RESCO ingolstadt21 case study. Each row reports the smallest saved observed ratio for the corresponding objective among sampled plans with $f^3 \leq 1.0$. Values are converted back to the deployed fixed-time scale using $T_0 = 300.95$ s, $A_0 = 0.2712$, and $E_0 = 2,832.94$ kg-CO₂.

Best observed ratio	Ratio	Physical improvement	Percent improvement
$\min f^1$ (travel time)	0.9289	−21.4 s/veh	−7.1%
$\min f^2$ (Atkinson)	0.7714	−0.062	−22.9%
$\min f^3$ (CO ₂)	0.9520	−136.0 kg	−4.8%

efficiency point reduces mean per-vehicle travel time by 21.4 s (−7.1% over the $T_0 = 300.95$ s baseline), the best observed equity point reduces the Atkinson index by 0.062 (−22.9% over $A_0 = 0.2712$), and the best observed emission point reduces CO₂ by 136.0 kg (−4.8% over $E_0 = 2,832.94$ kg). These extrema are computed from sampled optimization-log observations satisfying $f^3 \leq 1.0$; they should be validated with fresh SUMO seeds before being treated as certified field performance.

VEPM degrades gracefully when alignment fails. Table 12 reports the peak running sample-Pareto hypervolume for four variance estimators sharing all other algorithmic choices: GPR-KG with the default ϕ_{Zheng} -based VEPM, GPR-KG with the ϕ_{agg} ablation, GPR-KG with the medoid partition ablation, and GPR-KG-nV using pooled variance throughout. The three VEPM variants are statistically indistinguishable from the pooled baseline by the reported Welch-style z -scores, consistent with the $R_\phi^i \approx 0$ regime predicted by the alignment diagnostic of §6.5. Crucially, no estimator damages the algorithm: even the structurally degenerate ϕ_{Zheng} map (which under Corollary 5.1 produces $|\mathcal{C}| = 2^{88}$ partition combinations of which only a tiny fraction is ever visited) reduces in practice to individual-solution variance updates for visited solutions plus the global pooled estimator for unvisited ones, recovering the GPR-KG-nV behaviour as a limiting case.

Comparison with the RZDT regime. The $R = 10$ result on the RESCO ingolstadt21 case study contrasts cleanly with the synthetic RZDT regime of Section 6. On the three RZDT benchmarks (alignment holds, $R_\phi^i \rightarrow 1$), the checkpointed ϕ_{Zheng} VEPM variant gives moderate sample-HV gains over GPR-KG-nV and clearer final-recommendation gains in IGD and CVR (Table 5). Thus, in the aligned RZDT regime, VEPM acts mainly as a posterior feasibility-calibration module. On RESCO ingolstadt21 (alignment fails, $R_\phi^i \approx 0$), all four estimators converge to a common HV-peak of approximately 0.30 within standard error, consistent with the prediction that VEPM gracefully reduces to the pooled estimator when the partition features cannot capture variance heterogeneity.

Table 12: Variance-estimator ablation on the RESCO ingolstadt21 case study, $R = 10$ macro-replications with seeds $\{100, 200, \dots, 1000\}$ ($d = 44$, $N = 300$, $n_0 = 100$, $\tau = 1.0$, $\alpha = 0.05$; see §6.5 for the alignment regime). HV is the peak running sample-Pareto hypervolume with reference point $(1.5, 1.5)$; HV-peak is reported because the algorithmic HV at the final iteration often collapses on the strict chance-constraint check. The z -score column reports the Welch-style standardised difference against the pooled baseline; all four estimators are statistically indistinguishable from the pooled-variance baseline ($|z| < 1.2$), in line with the alignment-failure prediction of Theorem 5.2 and the argmax-invariance condition of §3.3.2.

Variance estimator	HV-peak (mean $\pm$ SE)	Δ vs. pooled	z -score
ϕ_{Zheng} VEPM (default)	0.3136 ± 0.0117	+0.0154	+1.11
ϕ_{agg} VEPM (ablation)	0.2978 ± 0.0104	−0.0003	−0.02
Medoid- K_{med} VEPM (ablation)	0.2946 ± 0.0131	−0.0036	−0.24
Pooled (GPR-KG-nV)	0.2982 ± 0.0076	—	—

Practical implication. The two regimes together establish a useful operational guideline: practitioners should evaluate the alignment diagnostic R_{cv}^2 of §6.5 on a small pilot (M, R) replication budget before committing to a VEPM variant. When R_{cv}^2 exceeds a threshold (empirically $\gtrsim 0.5$ in our experiments), the default ϕ_{Zheng} VEPM is expected to improve posterior feasibility calibration and may also improve sample-Pareto HV. When R_{cv}^2 is near zero or negative, no VEPM partitioning we examined improves on pooling, and the algorithm should be deployed with the pooled estimator for computational simplicity without loss of solution quality.

7.6 Out-of-sample Validation

Belief-model complexity (recap). The synthetic experiments in Section 6.4 established that GPR-KG’s belief model is best configured with linear and squared terms over all decision variables but *without* cross-terms whenever the budget satisfies $N \lesssim 5p_i$. The RESCO ingolstadt21 problem has $p_i = 41$ and $N = 300$, so $N/p_i \approx 7.3$, which supports the main-effect-plus-square GPR-KG-m model but remains too small for cross-terms. Accordingly, we use the GPR-KG-m configuration in this case study and do not repeat the full s/m/l/c complexity sweep; adding cross-terms on the RESCO ingolstadt21 scale would cost roughly $\binom{20}{2} = 190$ additional basis functions per objective, pushing p_i to 231 and inflating the posterior covariance dimension to $p_i + N = 531$, which overshoots the memory budget on a workstation without commensurate accuracy gains.

Out-of-sample constraint validation. The saved runs in Table 10 report sample-Pareto diagnostics from the optimization logs. A submission-quality engineering claim also requires

an independent out-of-sample validation: for each returned Pareto point, draw a fresh set of SUMO seeds and estimate $\Pr(f^3 \leq \tau)$ directly. Because a 95% chance constraint cannot be estimated precisely from very few replications, the next revision should use at least $R_{\text{oos}} = 100$ fresh seeds per reported point and should report binomial confidence intervals for the empirical feasibility probability. Until that validation is complete, the case study should be interpreted as an algorithmic and engineering diagnostic rather than as a certified field-deployment guarantee.

7.7 Practical Implications

The RESCO ingolstadt21 case study yields five practical diagnostics for using the GPR-KG framework in city-scale traffic signal optimization, spanning the methodology, its policy interpretation, and its scope.

(i) Simultaneous gains across three policy-relevant dimensions. The saved GPR-KG logs contain sample-feasible plans that improve efficiency (f^1 , travel time), equity (f^2 , Atkinson index of congestion ratios), and sustainability (f^3 , CO₂) relative to the field-implemented plan. The saved sample-Pareto points expose the efficiency–equity trade-off, allowing a planner to choose a knee-point solution (balanced) or an extreme solution (efficiency-first or equity-first) after independent validation and in line with local policy priorities. Because equity is quantified as an Atkinson index, the choice is auditable against welfare-economic criteria (Pigou–Dalton, EDE) rather than being a black-box weighting.

(ii) Fairness and efficiency interact non-trivially. The mean congestion ratio $\text{AM}(r)$ is a monotone function of the average travel time $\bar{t}$, so f^1 and $\text{AM}(r)$ move together; f^2 however depends on $\text{GM}(r)/\text{AM}(r)$ and therefore reacts to the *shape* of the delay distribution rather than its mean. Empirically, the saved sample-Pareto observations show that both objectives can improve jointly over x_0 within the explored range, while still exhibiting the classical “competing relationship” reported by (Zheng et al., 2019); this pattern reflects a network-level conservation law rather than a metric artefact. The choice of equity metric still matters for *which* travellers are protected: our length-invariant congestion-ratio formulation is neutral with respect to trip length, whereas a raw-travel-time Atkinson would mechanically penalise long-distance travellers. Planners with specific sub-population concerns (e.g. commuters vs. freight) can substitute a domain-appropriate r_i without changing the GPR-KG algorithm.

(iii) Chance-constrained emission control guards the right tail and regularises equity. CO₂ emissions scale with delay-tail mass, so their distribution has much heavier

right-tail behaviour than f^1 : a small number of adverse demand scenarios can spike emissions while leaving average travel time almost unchanged. The chance constraint (57) is designed to guard this tail, which is why it is useful for robust planning; a deterministic formulation that optimised the mean $\mathbb{E}[f^3]$ would accept plans whose 95%-quantile exceeds the regulatory target. In the model, the chance constraint also acts as a regulariser on f^2 : plans satisfying $\mathbb{P}(f^3 \leq \tau) \geq 0.95$ must have bounded delay-tail mass, which limits how unequal r_i can become. This coupling partly explains why the Pareto fronts we obtain are not dominated by equity-indifferent efficiency optima. A full sensitivity analysis over (τ, α) is not yet included in the current evidence package; it should be run before the paper makes policy claims about the robustness of the emission threshold.

(iv) Computational cost is compatible with offline strategic retiming. With $N = 300$ evaluations, the saved ingolstadt21 logs report an average `libsumo` simulation time of about 13seconds per evaluated plan and a median total iteration time of about 16seconds, with occasional longer initialization and congestion-tail runs. A complete replication therefore fits in an offline retiming workflow on a modern workstation. Strategic signal retiming is typically performed on a monthly or seasonal basis, and the cost scales linearly with the number of policy scenarios the planner wishes to compare.

(v) Scope and limitations. Three caveats delimit the generality of our case study. *(a) Network coverage.* We optimize the 44 green-duration variables exposed by the RESCO ingolstadt21 signal programs, spanning 21 controlled TLS in the saved decision map. Extending the same workflow to a larger regional network would require a new basis design and likely tiling or decomposition of the posterior covariance matrix. *(b) Modal scope.* The equity metric aggregates over motorised vehicles only. Pedestrian and bus-passenger delays are not captured in r_i ; a multi-modal Atkinson that weights each vehicle by its occupancy (or that treats each public-transport passenger as an individual) is a straightforward extension, but requires per-mode demand data that the current RESCO benchmark does not provide at the required granularity. *(c) Policy weighting.* The Atkinson inequality-aversion coefficient $\varepsilon = 1$ encodes a specific welfare preference; our framework supports any $\varepsilon \geq 0$ without algorithmic change, and a planner with stronger aversion to tail delays can simply choose $\varepsilon > 1$. We report the $\varepsilon = 1$ case for direct comparability with (Zheng et al., 2019).

8 Conclusion

8.1 Summary of Contributions

This paper has developed GPR-KG, a Bayesian optimization framework for bi-objective simulation-based optimization with probabilistic constraints and unknown heteroscedastic variances. The four main contributions, along with the supporting evidence, are as follows.

- (i) **Integrated framework.** GPR-KG combines a parametric Gaussian process regression belief model, a variance estimation parametric model (VEPM), and a Pareto-adapted knowledge gradient sampling policy into a coherent sequential procedure. Numerical experiments on three synthetic test problems (Section 6) demonstrate that GPR-KG achieves the best or near-best hypervolume and inverted generational distance across diverse settings, including convex, concave, and strict hyperbolic Pareto fronts.
- (ii) **Variance Estimation Parametric Model.** The VEPM extrapolates variance estimates from sampled to unsampled solutions via a partition-based pooling mechanism. The synthetic experiments and alignment diagnostic show that VEPM is useful when the partition features capture the variance structure, and that it degrades toward the pooled-variance baseline when this alignment fails.
- (iii) **Conditional theoretical guarantees.** We have established finite-budget and asymptotic results under explicit exploration, separation, Gaussian, and model-regularity assumptions. In particular, VEPM is shown to be cell-average consistent for cells with sufficient visitation, and the feasibility result is a conditional high-probability bound rather than an unconditional certification of every returned solution.
- (iv) **Real-world application.** Applied to the RESCO ingolstadt21 traffic signal timing benchmark, the saved GPR-KG logs contain sample-feasible signal plans that improve average travel time and delay equity relative to the field plan. The case study also provides a practically important negative-control regime: the variance alignment diagnostic is near zero, and VEPM is statistically indistinguishable from pooled variance, as predicted by the theory (Section 7).

8.2 Limitations

We acknowledge several limitations of the current framework.

First, the VEPM partition approach suffers from the curse of dimensionality: the number of partition combinations grows as $\prod_{j=1}^{d'} m_j$, where d' is the number of variables used in the partition and m_j is the number of bins for variable j . Although d' can be chosen smaller than d , selecting the partitioning variables and bin widths requires domain knowledge. An adaptive partitioning scheme that refines the partition based on observed data would mitigate this limitation.

Second, the Gaussian noise assumption (Assumption 5.1) may not hold for all simulation outputs. In particular, highly skewed or heavy-tailed distributions would require non-Gaussian extensions to ensure valid probabilistic constraint assessment.

Third, the current framework is designed for two objectives. Extension to three or more objectives is conceptually straightforward—the Pareto dominance relation and hypervolume indicator generalize naturally—but the computational cost of the expected hypervolume improvement calculation grows rapidly with the number of objectives, requiring approximation strategies.

Fourth, the quadratic basis model (16) may not capture complex nonlinearities or interaction effects in the response surface. While the deviation term $\zeta^i(x)$ compensates for model misspecification at sampled solutions, the predictive accuracy at unvisited solutions depends on the parametric component. Adaptive feature selection methods that identify and include relevant interaction terms would improve model flexibility.

8.3 Future Directions

Several avenues for future research emerge from this work.

- (1) **Adaptive feature selection.** Develop data-driven methods to automatically select basis functions for the parametric model and partitioning variables for VEPM, eliminating the need for domain expertise in model specification.
- (2) **Continuous and mixed-integer decision spaces.** Extend the framework to continuous and mixed-integer decision spaces by replacing the discrete solution enumeration with continuous optimization of the acquisition function (Scott et al., 2011; Wu and Frazier, 2017).
- (3) **Many-objective extension.** Scale the Pareto-KG acquisition function to three or more objectives using efficient hypervolume approximation or decomposition-based strategies.
- (4) **Non-Gaussian extensions.** Relax the Gaussian noise assumption using Cornish–Fisher expansions for quantile estimation or distribution-free approaches based on

empirical likelihood, thereby broadening the applicability to simulation outputs with skewed or heavy-tailed distributions.

- (5) **Parallel and batch knowledge gradient.** Leverage parallel computing resources by extending the sequential KG policy to a batch setting that selects multiple solutions for simultaneous evaluation (Wu and Frazier, 2016), reducing wall-clock time proportionally to the number of available processors.
- (6) **Online learning.** Develop an online variant of GPR-KG for settings where the objective functions change over time (e.g., traffic signal optimization under evolving demand patterns), requiring mechanisms for detecting and adapting to distributional shifts.

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Online Supplement

Knowledge Gradient for Bi-Objective Simulation Optimization with Probabilistic Constraints and Unknown Variances

This supplement provides: (A) full proofs of all propositions, lemmas, and theorems stated in the main text; (B) pseudocode for auxiliary algorithms; (C) complete formulations of the synthetic test problems; and (D) additional experimental details.

A1 Proofs

A1.1 Proof of Proposition 3.1 (Posterior Update)

We establish the Bayesian update equations (22)–(23) for the augmented parameter vector θ_n^i .

Setup. By assumption, the prior satisfies $\theta_0^i \sim \mathcal{N}(a_0^i, C_0^i)$. The model is linear-Gaussian: $Y_{n+1}^i = \tilde{e}_{x^n}^\top \theta_n^i + \varepsilon_{n+1}^i$, where $\varepsilon_{n+1}^i \sim \mathcal{N}(0, \hat{\sigma}_n^{i,2}(x^n))$ independently of θ_n^i given $\mathcal{F}_n$.

Dimension augmentation (new solution). If $x^n \notin \mathcal{X}_n$, the deviation term $\zeta^i(x^n)$ has not yet been incorporated into the parameter vector. We augment:

$$\theta_n^i \leftarrow \begin{pmatrix} \theta_n^i \\ \zeta^i(x^n) \end{pmatrix}, \quad a_n^i \leftarrow \begin{pmatrix} a_n^i \\ 0 \end{pmatrix}, \quad C_n^i \leftarrow \begin{pmatrix} C_n^i & \mathbf{0} \\ \mathbf{0}^\top & \lambda_i \end{pmatrix},$$

where the zero mean and variance λ_i reflect the prior $\zeta^i(x^n) \sim \mathcal{N}(0, \lambda_i)$ and the independence of $\zeta^i(x^n)$ from previously observed deviations.

Bayesian update (rank-one Kalman step). After augmentation, the model is $Y_{n+1}^i = \tilde{e}_{x^n}^\top \theta_n^i + \varepsilon_{n+1}^i$ with prior $\theta_n^i \sim \mathcal{N}(a_n^i, C_n^i)$. By the standard Bayesian linear regression update (equivalently, the Kalman filter measurement step), the posterior is $\theta_{n+1}^i \mid \mathcal{F}_{n+1} \sim \mathcal{N}(a_{n+1}^i, C_{n+1}^i)$ where:

$$\begin{aligned} a_{n+1}^i &= a_n^i + C_n^i \tilde{e}_{x^n} \left(\tilde{e}_{x^n}^\top C_n^i \tilde{e}_{x^n} + \hat{\sigma}_n^{i,2}(x^n) \right)^{-1} (Y_{n+1}^i - \tilde{e}_{x^n}^\top a_n^i), \\ C_{n+1}^i &= C_n^i - C_n^i \tilde{e}_{x^n} \left(\tilde{e}_{x^n}^\top C_n^i \tilde{e}_{x^n} + \hat{\sigma}_n^{i,2}(x^n) \right)^{-1} \tilde{e}_{x^n}^\top C_n^i. \end{aligned}$$

Since the denominator is a scalar (not a matrix), the inverse is simply the reciprocal, yielding exactly equations (22)–(23).

Posterior predictive moments (Corollary 3.1). For any $x \in \mathcal{X}$:

$$\mu_n^i(x) = \mathbb{E}[f^i(x) \mid \mathcal{F}_n] = \mathbb{E}[\tilde{\varphi}_n^i(x)^\top \theta_n^i \mid \mathcal{F}_n] = \tilde{\varphi}_n^i(x)^\top a_n^i,$$

$$\text{Var}(f^i(x) \mid \mathcal{F}_n) = \tilde{\varphi}_n^i(x)^\top C_n^i \tilde{\varphi}_n^i(x) + \lambda_i \cdot \mathbf{1}[x \notin \mathcal{X}_n],$$

where the extra λ_i term for unsampled x reflects the prior uncertainty of $\zeta^i(x)$ that has not yet been reduced by observation. $\square$

A1.2 Proof of Lemma 3.2 (VEPM Recursive Update)

We verify that the recursive formulas (32)–(35) are algebraically equivalent to the batch formulas (29)–(30).

Case 1: Revisiting a previously sampled solution ($x^n \in \mathcal{X}_n$). The batch formula after n_{x^n} visits gives

$$\hat{\sigma}_{n+1}^{i,2}(x^n) = \frac{w_i \hat{\sigma}_0^{i,2}(c^*) + \sum_{l=1}^{n_{x^n}} r_l^2}{w_i + n_{x^n}},$$

where $r_l = Y_l^i(x^n) - \mu_{s_l}^i(x^n)$ are the residuals. The value before the new observation was

$$\hat{\sigma}_n^{i,2}(x^n) = \frac{w_i \hat{\sigma}_0^{i,2}(c^*) + \sum_{l=1}^{n_{x^n}-1} r_l^2}{w_i + n_{x^n} - 1}.$$

Solving for $w_i \hat{\sigma}_0^{i,2}(c^*) + \sum_{l=1}^{n_{x^n}-1} r_l^2 = (w_i + n_{x^n} - 1) \hat{\sigma}_n^{i,2}(x^n)$ and substituting:

$$\hat{\sigma}_{n+1}^{i,2}(x^n) = \frac{(w_i + n_{x^n} - 1) \hat{\sigma}_n^{i,2}(x^n) + r_{n_{x^n}}^2}{w_i + n_{x^n}},$$

which is (32) with $r_{n_{x^n}} = Y_{n+1}^i - \mu_n^i(x^n)$.

For the partition update (33), the set membership is unchanged but the count of x^n increases by one. From the sample-count weighted definition (30):

$$\hat{\sigma}_{n+1,c^*}^{i,2} = \frac{\sum_{x' \in \mathcal{S}_{n+1,c^*}} n_{x'} \hat{\sigma}_{n+1}^{i,2}(x')}{R_{n+1,c^*}},$$

which is (33).

Case 2: Sampling a new solution ($x^n \notin \mathcal{X}_n$). With only one observation ($n_{x^n} = 1$), the batch formula (29) gives

$$\hat{\sigma}_{n+1}^{i,2}(x^n) = \frac{w_i \hat{\sigma}_0^{i,2}(c^*) + (Y_{n+1}^i - \mu_n^i(x^n))^2}{w_i + 1},$$

which is (34). The partition combination c^* gains a new member with count $n_{x^n} = 1$. From (30):

$$\hat{\sigma}_{n+1,c^*}^{i,2} = \frac{R_{n,c^*} \cdot \hat{\sigma}_{n,c^*}^{i,2} + n_{x^n} \hat{\sigma}_{n+1}^{i,2}(x^n)}{R_{n,c^*} + n_{x^n}},$$

which is (35). All other solutions and partition combinations are unaffected because their individual estimates and set memberships have not changed. $\square$

A1.3 Proof of Theorem 5.1 (VEPM Cell-Average Consistency)

Fix evaluation index i and partition combination c . Let $\mathcal{S}_{N,c}$ be the sampled solutions in partition c and let $R_{N,c} = \sum_{x \in \mathcal{S}_{N,c}} n_x$ be the total number of observations allocated to that partition. The theorem assumes $R_{N,c} \rightarrow \infty$ along the sample path; this follows, for example, from Assumption 5.4, but is not needed as a separate condition in the proof.

For any solution $x \in \mathcal{X}$ sampled n_x times, define $R_l(x) = (Y_l^i(x) - \mu_{s_l}^i(x))^2$. We show $\hat{\sigma}_{N,c}^{i,2} \rightarrow \bar{\sigma}_c^{i,2}$ a.s. in three steps.

Step 1: Mean convergence. Under the well-specified parametric belief model and the positive definiteness implied by the prior covariance, the standard Gaussian linear-update consistency argument gives $\mu_{s_l}^i(x) \rightarrow f^i(x)$ a.s. for every fixed solution x sampled infinitely often. This is the same argument formalized in Lemma 5.1; it does not rely on the later asymptotic convergence theorem.

Step 2: Residual decomposition. Write

$$R_l(x) = (Y_l^i(x) - f^i(x))^2 + 2(Y_l^i(x) - f^i(x))(f^i(x) - \mu_{s_l}^i(x)) + (f^i(x) - \mu_{s_l}^i(x))^2.$$

The first term satisfies $\mathbb{E}[(Y_l^i - f^i(x))^2] = \sigma^i(x)^2$; these are i.i.d. with finite second moment under Assumption 5.1. The cross-term converges to zero a.s. since $\mu_{s_l}^i(x) \rightarrow f^i(x)$. The last term also converges to zero a.s.

Step 3: SLLN and partition averaging. By the strong law of large numbers applied to the sequence $\{R_l(x)\}$:

$$\frac{1}{n_x} \sum_{l=1}^{n_x} R_l(x) \longrightarrow \sigma^i(x)^2 \quad \text{a.s.}$$

The individual VEPM estimate $\hat{\sigma}_N^{i,2}(x) \rightarrow \sigma^i(x)^2$ a.s. for every solution whose local count $n_x \rightarrow \infty$ (as w_i is fixed and the data dominate for large n_x). Taking the sample-count weighted average (30) over partition c and applying the Toeplitz–Cesaro theorem for weighted averages:

$$\hat{\sigma}_{N,c}^{i,2} = \frac{\sum_{x \in S_{N,c}} n_x \hat{\sigma}_N^{i,2}(x)}{\sum_{x \in S_{N,c}} n_x} \longrightarrow \bar{\sigma}_c^{i,2} \quad \text{a.s.}$$

If $\delta_c^i = 0$, then all $\sigma^i(x)^2$ in the cell are equal and $\bar{\sigma}_c^{i,2} = \sigma_c^{i,2}$; otherwise the limit is the sampling-policy weighted cell average stated in the theorem, and its distance to any point variance in the cell is bounded by Assumption 5.3. $\square$

A1.4 Proof of Theorem 5.3 (VEPM Convergence Rate)

We bound each of the three terms in the MSE $\mathbb{E}[(\hat{\sigma}_{N,c}^{i,2} - \sigma_c^{i,2})^2]$.

Term 1 (sampling variability). Given the true mean f^i , the residuals $Y_l^i(x) - f^i(x)$ are i.i.d. $\mathcal{N}(0, \sigma^i(x)^2)$. The sample variance of $N_c \cdot r$ observations from a population with variance $\sigma_c^{i,2}$ has MSE = $O(\sigma_c^{i,4}/(N_c r)) \leq O(\sigma_{\max}^4/(N_c r))$.

Term 2 (mean estimation error). The residuals use $\mu_{s_l}^i(x)$ in place of $f^i(x)$. The bias introduced satisfies $\mathbb{E}[(f^i(x) - \mu_{s_l}^i(x))^2] \leq C_2 p_i \sigma_{\max}^2/N$ by Lemma 5.1. This error propagates into the variance estimate at rate $O(p_i \sigma_{\max}^2/N)$.

Term 3 (shrinkage bias). The VEPM prior weight w_i introduces a bias proportional to $w_i(\hat{\sigma}_0^{i,2}(c) - \sigma_c^{i,2})/(w_i + r)$. In the worst case $|\hat{\sigma}_0^{i,2}(c) - \sigma_c^{i,2}| \leq C \sigma_{\max}^2$, giving a squared bias of $O(w_i^2 \sigma_{\max}^4/(w_i + r)^2)$.

Summing and absorbing constants into C_1, C_2, C_3 yields the stated bound. $\square$

A1.5 Proof of Lemma 5.1 (Posterior Mean Convergence Rate)

The posterior precision matrix after N observations is

$$(C_N^i)^{-1} = (C_0^i)^{-1} + \sum_{n=1}^N \frac{\tilde{\varphi}_n^i(x^{n-1}) \tilde{\varphi}_n^i(x^{n-1})^\top}{\hat{\sigma}_{n-1}^{i,2}(x^{n-1})}.$$

Under Assumption 5.1, $\hat{\sigma}_n^{i,2} \leq \sigma_{\max}^2$ (after VEPM convergence), so

$$(C_N^i)^{-1} \succeq (C_0^i)^{-1} + \frac{1}{\sigma_{\max}^2} \sum_{n=1}^N \tilde{\varphi}_n^i(x^{n-1}) \tilde{\varphi}_n^i(x^{n-1})^\top.$$

The mean squared error of the posterior mean satisfies $\mathbb{E}[\|a_N^i - a^{i,*}\|^2] = \text{tr}(C_N^i)$. By the bound on C_N^i and the fact that the sum of outer products is at least $N/\sigma_{\max}^2$ times the smallest

eigenvalue of the sample covariance (which is $\Omega(1)$ under the basis bound of Assumption 5.2), we obtain $\text{tr}(C_N^i) \leq p_i \sigma_{\max}^2 / N \cdot \text{tr}((C_0^i)^{-1})^{-1}$. The pointwise bound follows by $\mathbb{E}[(\mu_N^i(x) - f^i(x))^2] = \tilde{\varphi}_N^i(x)^\top C_N^i \tilde{\varphi}_N^i(x) \leq \|\tilde{\varphi}_N^i(x)\|^2 \|C_N^i\|_2 \leq p_i \sigma_{\max}^2 \|\tilde{\varphi}_N^i(x)\|^2 / N$. $\square$

A1.6 Proof of Theorem 5.4 (Regret Bound)

Let $\Delta_{\min} = \min_{x \in \mathcal{P}^*, x' \notin \mathcal{P}^*} \max_i |f^i(x) - f^i(x')| > 0$ be the minimum objective-space gap.

Step 1: Uniform posterior mean bound. By Lemma 5.1 and a sub-Gaussian concentration inequality, for each x and i :

$$\Pr(|\mu_N^i(x) - f^i(x)| > t) \leq 2 \exp\left(-\frac{Nt^2}{2p_i \sigma_{\max}^2 \|\tilde{\varphi}^i(x)\|^2}\right).$$

Taking a union bound over all K solutions and both objectives with $t = C_0 \sqrt{p \log(N)/N}$ (for a sufficiently large constant C_0):

$$\Pr\left(\max_{x,i} |\mu_N^i(x) - f^i(x)| > t\right) \leq 4K \exp(-C_0^2 \log N / (2B_\varphi^2)) \leq 4K/N^c,$$

for constant $c > 0$ depending on C_0 and B_φ . Under the fixed finite decision space considered in this paper, K is constant as N grows, so this probability is $O(K/N^c)$ and is summable for $c > 1$. If one studies a sequence of growing problem instances, the same argument additionally requires $\log K_N = O(\log N)$.

Step 2: Misclassification probability. A true Pareto-optimal solution $x^* \in \mathcal{P}^*$ is missed (or a dominated solution is incorrectly included) only when $\max_i |\mu_N^i(x) - f^i(x)| > \Delta_{\min}/2$ for some x . The probability of this event is $O(K/N^c) = O(N^{1-c})$.

Step 3: Hausdorff distance bound. When no misclassification occurs, the Hausdorff distance satisfies $d_H(\widehat{PF}_N, PF^*) \leq C \sqrt{p \log N / N} \cdot \sigma_{\max} / \Delta_{\min}$. Combining with the misclassification probability and the trivial bound $d_H \leq \sigma_{\max}$ yields the stated expectation bound. $\square$

A1.7 Proof of Theorem 5.5 (Feasibility Guarantee)

Define the estimation error at x :

$$e_N(x) = |\hat{q}_{1-\alpha}(x) - q_{1-\alpha}(x)| \leq |\mu_N^3(x) - f^3(x)| + \Phi^{-1}(1-\alpha) |\hat{\sigma}_N^3(x) - \sigma^3(x)|.$$

Mean error bound. By the sub-Gaussian concentration argument in the proof of Theo-

rem 5.4, with probability at least $1 - \delta/2$:

$$\max_{x \in \mathcal{X}} |\mu_N^3(x) - f^3(x)| \leq C_1 \sqrt{\frac{p \log(2K/\delta)}{N}}.$$

Variance error bound. By Theorem 5.3 and Markov's inequality, with probability at least $1 - \delta/2$:

$$\max_{x \in \mathcal{X}} |\hat{\sigma}_N^3(x) - \sigma^3(x)| \leq C_2 \sqrt{\frac{\sigma_{\max}^2 \log(2K/\delta)}{N_c r}}.$$

Combining. Since $N_c r \leq N$ (the total budget across all partitions), the variance error bound is at most $C_2 \sqrt{\sigma_{\max}^2 \log(2K/\delta)/N}$, which is dominated by the mean error bound when $p \geq 1$. With probability at least $1 - \delta$, for all $x \in \mathcal{X}$:

$$e_N(x) \leq (C_1 + C_2 \Phi^{-1}(1 - \alpha) \sigma_{\max}) \sqrt{\frac{p \log(2K/\delta)}{N}} =: \varepsilon_N(\delta).$$

For a fixed finite decision space, $\log(2K/\delta)$ is absorbed into the constant in the stated bound; for a growing sequence of instances, the same simplification requires the additional scaling condition $\log K_N = O(\log N)$.

For $x \in \hat{F}_N$, we have $\hat{q}_{1-\alpha}(x) \leq \tau$ by definition, so $q_{1-\alpha}(x) \leq \hat{q}_{1-\alpha}(x) + \varepsilon_N(\delta) \leq \tau + \varepsilon_N(\delta)$.

□

A1.8 Proof of Proposition 5.1 (Leading-Order Optimality of Pareto-KG)

For each $x \in A_n$, the one-step hypervolume change is $\Delta \text{HV}_n(x) = \mathbb{E}_n[\text{HV}(\hat{\mathcal{P}}_{n+1}) - \text{HV}(\hat{\mathcal{P}}_n) \mid x^n = x]$. Using the linear update (45), $\mu_{n+1}^i(x') = \mu_n^i(x') + \tilde{\sigma}_{x'}^i(x)Z$ with $Z \sim \mathcal{N}(0, 1)$.

The hypervolume is a piecewise linear function of the objective vectors. A first-order expansion in the increments $\tilde{\sigma}_{x'}^i(x)Z$ gives:

$$\text{HV}(\hat{\mathcal{P}}_{n+1}) = \text{HV}(\hat{\mathcal{P}}_n) + \sum_{i=1}^2 \sum_{x' \in A_n} \frac{\partial \text{HV}}{\partial \mu_n^i(x')} \tilde{\sigma}_{x'}^i(x)Z + O(\|\tilde{\sigma}(x)\|^2).$$

Taking the expectation over Z eliminates the first-order term (since $\mathbb{E}[Z] = 0$), but the KG factor arises from the $\mathbb{E}[\max_{x'} (\mu_n^i(x') + \tilde{\sigma}_{x'}^i(x)Z)]$ computation, which captures the nonlinear benefit of reducing the minimum. Grouping by objective index yields the decomposition stated in the proposition, with $\lambda_{i,n}(x) = \mathbb{E}_n[\partial \text{HV} / \partial \mu_n^i(x_i^*)]$ for the current Pareto corner x_i^* associated with objective i . The remainder $R_n(x)$ is the interaction term between objectives,

which is $O(\|\tilde{\sigma}(x)\|^2/\Delta_{\min}^2)$. □

A2 Auxiliary Algorithms

A2.1 Candidate Set Generation

A2.2 Computation of the h -Function

The KG factor requires evaluating $h(\mathbf{a}, \mathbf{b}) = \mathbb{E}[\max_j(a_j + b_j Z)] - \max_j a_j$ where $Z \sim \mathcal{N}(0, 1)$. For our minimization formulation, we use $h(-\boldsymbol{\mu}, -\tilde{\boldsymbol{\sigma}})$. The algorithm of [Frazier et al. \(2009\)](#) computes this in $O(|\mathcal{A}_n| \log |\mathcal{A}_n|)$ time.

A2.3 VEPM Update

A3 Test Problem Formulations

All test problems use an integer decision vector $x = (x_1, \dots, x_d)$ with problem-specific ranges (see below). Normalized coordinates are $t_j = x_j/L_j \in [0, 1]$, where L_j is the upper bound of dimension j . Unless stated otherwise, $L_j = L$ for all j (uniform grid). Simulation outputs are $Y^i(x) = f^i(x) + \varepsilon^i(x)$ with $\varepsilon^i(x) \sim \mathcal{N}(0, \sigma^i(x)^2)$, where $\sigma^i(x)$ is position-dependent (heteroscedastic). The probabilistic constraint $\mathbb{P}(Y^3(x) \leq \tau) \geq 1 - \alpha$ is implemented as $f^3(x) + \Phi^{-1}(1 - \alpha) \sigma^3(x) \leq \tau$, with $\alpha = 0.05$. For the three reported benchmark problems (RZDT1, RZDT2, and RZDT5-RR), the threshold is fixed at $\tau = 0$.

Scope of Reported Test Problems

This appendix records the three synthetic benchmark formulations for which complete numerical evidence is reported in Section 6. Other exploratory variants remain in the code archive for future experiments, but their incomplete numerical logs are not used as manuscript evidence.

A3.1 RZDT1 (Convex Pareto Front)

Decision space: $x_j \in \{0, 1, \dots, L\}^d$ with $L = 100$. Normalized coordinates $t_j = x_j/L$.

$$\begin{aligned} f^1(x) &= t_1, \\ f^2(x) &= g_{\text{ZDT}}(x) \left(1 - \sqrt{f^1/g_{\text{ZDT}}(x)} \right), \\ g_{\text{ZDT}}(x) &= 1 + \frac{9}{d-1} \sum_{j=2}^d t_j, \\ f^3(x) &= -(t_1 - 0.5)^2 + 0.04. \end{aligned}$$

The Pareto-optimal solutions satisfy $x_j = 0$ for all $j \geq 2$ (so $g_{\text{ZDT}} = 1$); the true Pareto front lies on the convex ZDT1 curve $f^2 = 1 - \sqrt{f^1}$.

Constraint function design (f^3 , RZDT1). The inverted parabola $f^3 = -(t_1 - 0.5)^2 + 0.04$ peaks at $t_1 = 0.5$ (where $f^3 = 0.04 > 0$) and drops below zero when $|t_1 - 0.5| > 0.2$. With threshold $\tau = 0$, the constraint excludes the middle portion of the Pareto front and keeps the two tails feasible. Using the reported baseline scale $\sigma = 0.04$ in the corresponding homoscedastic calculation, the safety margin $\Phi^{-1}(0.95)\sigma \approx 0.066$ widens the exclusion zone, yielding $|t_1 - 0.5| \geq \sqrt{0.04 + 0.066} \approx 0.33$, so $x_1 \in \{0, \dots, 15\} \cup \{85, \dots, 100\}$ – i.e., 32 true Pareto-optimal solutions (TPOS).

Heteroscedastic noise (RZDT1).

$$\sigma^i(x) = \sigma(0.5 + 2\sqrt{t_1}), \quad i = 1, 2, 3. \quad (\text{A.1})$$

The noise follows a square-root profile: 0.5σ at $t_1 = 0$ rising to 2.5σ at $t_1 = 1$, a variance ratio of $25\times$. Since the noise is lowest at the left tail ($x_1 \approx 0$) and highest at the right tail ($x_1 \approx 100$), the heteroscedastic safety margin $\Phi^{-1}(0.95)\sigma(0.5 + 2\sqrt{t_1})$ is asymmetric: the right tail carries a much larger margin, excluding those solutions from feasibility. With $\sigma = 0.04$, the noise ranges from 0.020 (at $t_1 = 0$) to 0.100 (at $t_1 = 1$), and $x_1 \in \{0, \dots, 14\} \cup \{95, \dots, 100\}$ remain feasible (21 TPOS): 15 on the left tail and 6 on the right.

A3.2 RZDT2 (Concave Pareto Front)

Decision space: $x_j \in \{0, 1, \dots, L\}^d$ with $L = 100$. Normalized coordinates $t_j = x_j/L$.

$$\begin{aligned} f^1(x) &= t_1, \\ f^2(x) &= g_{\text{ZDT}}(x)(1 - (f^1/g_{\text{ZDT}}(x))^2), \\ g_{\text{ZDT}}(x) &= 1 + \frac{9}{d-1} \sum_{j=2}^d t_j, \\ f^3(x) &= -(t_1 - 0.5)^2 + 0.04. \end{aligned}$$

The Pareto-optimal solutions satisfy $x_j = 0$ for all $j \geq 2$; the true Pareto front lies on the concave ZDT2 curve $f^2 = 1 - (f^1)^2$. The constraint function is identical to RZDT1.

Heteroscedastic noise (RZDT2).

$$\sigma^i(x) = \sigma(0.5 + 2.5 \sin^2(\pi t_1)), \quad i = 1, 2, 3. \quad (\text{A.2})$$

The noise follows a bell-shaped profile peaking at $t_1 = 0.5$ (3σ) and dropping to 0.5σ at the two extremes, with a variance ratio of $36\times$. Since the noise is *symmetric and low* at both tails, the feasibility condition is nearly identical on both sides. With $\sigma = 0.04$ and $\tau = 0$, the noise ranges from 0.020 (at $t_1 = 0, 1$) to 0.120 (at $t_1 = 0.5$), and $x_1 \in \{0, \dots, 16\} \cup \{84, \dots, 100\}$ are feasible (34 TPOS, symmetric with 17 on each tail).

A3.3 RZDT5_RR (Hyperbolic Pareto Front, Mixed Grid, Strict Constraint)

Decision space: $x_1 \in \{0, 1, \dots, 100\}$ and $x_j \in \{0, 1, \dots, 50\}$ for $j \geq 2$ (mixed bounds, $L_1 = 100$, $L_j = 50$). Normalized coordinates: $t_1 = x_1/100$, $t_j = x_j/50$ for $j \geq 2$.

$$\begin{aligned} f^1(x) &= t_1, \\ f^2(x) &= g_{\text{ZDT}}(x)/(x_1 + 1), \\ g_{\text{ZDT}}(x) &= 1 + \sum_{j=2}^d t_j, \\ f^3(x) &= t_1 - 0.5. \end{aligned}$$

The Pareto-optimal solutions satisfy $x_j = 0$ for all $j \geq 2$ (so $g_{\text{ZDT}} = 1$); the true Pareto front lies on the hyperbolic curve $f^2 = 1/(x_1 + 1)$, which decays sharply near the origin and flattens out for large x_1 .

Design rationale. Relative to the original RZDT5 (Zitzler et al., 2000), we keep the strict chance-constraint threshold $\tau = 0$ (hence the second “R” in the name, for *restricted*) but refine the decision grid so that the Pareto front is better resolved numerically. (i) *Enlarged and anisotropic decision space*: x_1 is refined to a 101-level grid while the auxiliary variables x_j ($j \geq 2$) use a 51-level grid. The $f^2 = g/(x_1 + 1)$ form (replacing $30t_1 + 1$ in the original) keeps the hyperbolic geometry but stretches resolution across x_1 , avoiding the over-dense Pareto cluster at $t_1 \approx 0$ present in the original formulation. (ii) *Strict threshold* $\tau = 0$ on $f^3 = t_1 - 0.5$. Combined with the heteroscedastic safety margin $\Phi^{-1}(0.95)\sigma(0.3 + 2t_1^2)$ and $\sigma = 0.04$, the 95%-quantile is $q_{0.95}(x) = t_1 - 0.5 + 1.6449 \cdot 0.04 \cdot (0.3 + 2t_1^2)$; solving $q_{0.95} \leq 0$ gives exactly $x_1 \in \{0, \dots, 45\}$, i.e. 46 TPOS located in the low-to-mid noise region $t_1 \in [0, 0.45]$ where $\sigma^i(x) \in [0.3\sigma, 0.705\sigma]$. The infeasible half of the front, $x_1 \in \{46, \dots, 100\}$, sits in the high-noise tail up to $\sigma^i = 2.3\sigma$, so the candidate pool necessarily contains points with widely disparate noise levels.

Heteroscedastic noise (RZDT5_RR).

$$\sigma^i(x) = \sigma(0.3 + 2t_1^2), \quad i = 1, 2, 3. \quad (\text{A.3})$$

The noise follows a monotone quadratic profile: 0.3σ at $t_1 = 0$ rising to 2.3σ at $t_1 = 1$, a global variance ratio of $\approx 58.8\times$. On the feasible TPOS ($t_1 \in [0, 0.45]$), the noise standard deviation ranges from 0.3σ to $0.3 + 2 \cdot 0.45^2 \approx 0.705\sigma$; the TPOS-restricted variance ratio is therefore $\approx 5.5\times$. The critical point is that the *candidate set* visited by any surrogate-based method routinely spans $t_1 \in [0, 1]$, so the pooled variance estimator sees the full $\approx 58.8\times$ range and necessarily mis-scores chance-constraint safety margins across the boundary: it *under-estimates* noise in the high- t_1 infeasible tail (and may admit those points erroneously), while *over-estimating* noise on the feasible side (and unnecessarily shrinking the reported Pareto set). Partition-aware VEPM reconstructs the quadratic structure from the t_1 and t_1^2 features, producing the smallest #Infeas among methods that report a comparably large non-dominated set (Table 4).

A3.4 Noise Structures and VEPM Partition

Each problem carries a dedicated heteroscedastic noise function (Equations (A.1)–(A.3)) designed so that the spatial noise pattern is partially independent of the objective values, requiring partition-aware estimation rather than response-based scaling. The parameter σ is the baseline noise scale ($\sigma = 0.04$ for RZDT1/2/5-RR); the heteroscedastic multiplier creates the position-dependent structure.

Table 13 summarizes the noise structures of the three reported synthetic problems.

Table 13: Noise structure summary for the reported synthetic test problems.

Problem	Grid	Noise formula $\sigma^i(x)/\sigma$	Global var. ratio	TPOS var. ratio	TPOS
RZDT1	$\{0, \dots, 100\}^d$	$0.5 + 2\sqrt{t_1}$	$25\times$	$25\times$	21 ($\tau=0$)
RZDT2	$\{0, \dots, 100\}^d$	$0.5 + 2.5 \sin^2(\pi t_1)$	$36\times$	$\approx 4.6\times$	34 ($\tau=0$)
RZDT5-RR	mixed*	$0.3 + 2t_1^2$	$59\times$	$\approx 5.5\times$	46 ($\tau=0$)

*RZDT5-RR: $x_1 \in \{0, \dots, 100\}$, $x_j \in \{0, \dots, 50\}$ for $j \geq 2$; $\sigma = 0.04$ for RZDT1/2/5-RR.

Global var. ratio is $\max_x \sigma^i(x)^2 / \min_x \sigma^i(x)^2$; TPOS var. ratio restricts this to feasible Pareto-optimal x_1 values.

VEPM partition. For all reported test problems, the VEPM uses a feature-based partition scheme: $2d$ features $[t_1, \dots, t_d, t_1^2, \dots, t_d^2]$ are computed from the normalised coordinates, and each feature is assigned to one of two bins at threshold 0.5, yielding up to 2^{2d} total partition combinations. The decision variable x_1 (which directly controls f^1 and carries the heteroscedastic noise signal) is the primary driver: its normalised value t_1 and squared value t_1^2 together create a four-way partition of the x_1 axis that aligns with the spatial noise structure. The normalisation $t_j = x_j/L_j$ uses the problem-specific upper bound L_j for each dimension (e.g., $L_1 = 100$ for RZDT1/2, $L_1 = 100$ and $L_j = 50$ for RZDT5-RR).

A4 Additional Experimental Details

Hardware and software. All experiments were run on a workstation with an Intel Core i9-12900K processor (16 cores, 3.2 GHz), 64 GB RAM, and Ubuntu 22.04. GPR-KG was implemented in Python 3.10 using NumPy 1.24 and SciPy 1.10. The cEHVI and cParEGO baselines used BoTorch 0.9 with PyTorch 2.0. NSGA-II-K and NSGA-II-D used the Platemo 3.5 MATLAB toolbox.

Prior specification (data-driven). The prior parameters are determined automatically from the $n_0 = 30$ pre-samples. For each objective $i = 1, 2, 3$, an ordinary least-squares (OLS) regression is fitted using the quadratic basis $\Psi = [\varphi(x_1), \dots, \varphi(x_{n_0})]^\top$ and observations

$Y^i = (Y_1^i, \dots, Y_{n_0}^i)^\top$ to obtain $\hat{\beta}^i = (\Psi^\top \Psi)^{-1} \Psi^\top Y^i$. The prior mean is then set to $a_{\beta,0}^i = \hat{\beta}^i$ (the OLS estimate), the prior covariance to $C_{\beta,0}^i = \text{Var}(\hat{\beta}^i) \cdot I_{p_i}$ (where $\text{Var}(\hat{\beta}^i)$ denotes the sample variance of the entries of $\hat{\beta}^i$), and the deviation prior to $\lambda_i = \text{Var}(\text{residuals}_i)$ (the sample variance of the OLS residuals $Y^i - \Psi \hat{\beta}^i$). This data-driven initialisation adapts the prior scale to the problem’s signal-to-noise ratio, avoiding the need for fixed hyperparameters. The VEPM prior weight was $w_i = 1$, corresponding to one pseudo-observation for the initial partition estimate.

Initial VEPM estimate. The initial partition variance $\hat{\sigma}_0^{i,2}(c)$ was estimated from the $n_0 = 30$ pre-samples as the mean squared OLS residual within each partition combination c . For partition combinations containing fewer than two residuals, the global mean squared residual over all pre-samples was used.

Constraint calibration. For RZDT1, RZDT2, and RZDT5_RR the constraint threshold is fixed at $\tau = 0$ (matching the original ZDT formulations); the detailed derivation of the 46 TPOS of RZDT5_RR under $\sigma = 0.04$ is given in Appendix [A3.3](#).

RESCO ingolstadt21 log aggregation. The case-study diagnostics in Table [10](#) are computed from 40 completed optimization logs under `GPR_KG_Code/results/ingolstadt21`: three GPR-KG variance estimators (binary-bin, aggregate, and medoid- K_{med} partitions) and the pooled GPR-KG-nV baseline, each with seeds $\{100, 200, \dots, 1000\}$. The post-processing script `experiments.ingolstadt21.summarize_case_study` reads only the saved `summary.json` and `snapshots.jsonl` files, so it does not rerun SUMO. It writes both a machine-readable JSON audit file and the LaTeX table fragment included in the manuscript.

Algorithm 1 GPR-KG: Parametric GPR with Knowledge Gradient for Bi-Objective Constrained Simulation Optimization

Require: Decision space $\mathcal{X} \subseteq \mathbb{Z}^d$, total budget N , pre-sampling budget n_0 , basis functions $\varphi^i(\cdot)$, VEPM partition scheme and prior weight w_i , candidate set sizes K_1 (LHD) and K_2 (posterior-sampling NSGA-II runs), constraint parameters (τ, α) .

Ensure: Estimated posterior-feasible Pareto set $\hat{\mathcal{P}}_N$.

- 1: **Pre-sampling phase:** Select n_0 distinct solutions from a structured finite-grid initial design: include problem-independent center and coordinate-axis endpoint seeds determined only by the integer bounds of $\mathcal{X}$, then fill the remaining pre-samples with random integer-grid points. Simulate each solution once, observing (Y^1, Y^2, Y^3) .
 - 2: **Data-driven initialisation:** For each $i = 1, 2, 3$, fit ordinary least squares (OLS) on the n_0 pre-samples to obtain $\hat{\beta}^i$. Set $a_{\beta,0}^i = \hat{\beta}^i$, $C_{\beta,0}^i = \text{Var}(\hat{\beta}^i) \cdot I_{p_i}$, $\lambda_i = \text{Var}(\text{residuals}_i)$. Initialize VEPM estimates $\hat{\sigma}_0^{i,2}(c)$ from the squared OLS residuals of pre-sampled solutions in partition c (or a global squared-residual fallback if the partition has too little data).
 - 3: **for** $n = n_0, n_0 + 1, \dots, N - 1$ **do**
 - 4: **Generate candidate set:** Construct $\mathcal{A}_n = \mathcal{A}_n^{\text{LHD}} \cup \mathcal{A}_n^{\text{post}}$ with K_1 LHD samples and K_2 NSGA-II posterior-sampling runs as in §4.2.
 - 5: **Compute KG factors:** For each $x \in \mathcal{A}_n$ and $i \in \{1, 2\}$, compute $\nu_n^{\text{KG},i}(x)$ via Lemma 4.1. (KG factors for the constraint index $i = 3$ are not computed separately; the constraint is enforced through posterior-guided candidate filtering and the final posterior-feasible filtering.)
 - 6: **Select next solution:** Apply the Pareto-KG rule (§4.3) to identify the non-dominated set $\mathcal{N}_n$ in KG-factor space and select $x^n \in \mathcal{N}_n$ by crowding distance.
 - 7: **Simulate:** Run the simulation at x^n , observe $(Y_{n+1}^1, Y_{n+1}^2, Y_{n+1}^3)$.
 - 8: **Cache pre-update quantities:** For each $i = 1, 2, 3$, store $\mu_n^i(x^n)$ and $\hat{\sigma}_n^{i,2}(x^n)$.
 - 9: **Update posterior beliefs:** For $i = 1, 2, 3$, apply Proposition 3.1 to update (a_{n+1}^i, C_{n+1}^i) , using the cached $\hat{\sigma}_n^{i,2}(x^n)$ as the plug-in observation variance.
 - 10: **Update variance estimates:** For $i = 1, 2, 3$, apply the VEPM recursive update (Lemma 3.2) using the cached pre-update residual $Y_{n+1}^i - \mu_n^i(x^n)$.
 - 11: **end for**
 - 12: **Solve posterior problem:** Use Proposition 3.2 as the feasibility and dominance structure for the terminal posterior solve; in large discrete spaces, evaluate the visited solutions plus a random lattice sample, apply the posterior chance-feasibility filter, and return the non-dominated subset as $\hat{\mathcal{P}}_N$.
 - 13: **return** $\hat{\mathcal{P}}_N$.
-

Algorithm 2 Generate Candidate Set $\mathcal{A}_n$

Require: Current beliefs $\{(a_n^i, C_n^i)\}$, VEPM estimates $\{\hat{\sigma}_n^{i,2}\}$, LHD size K_1 , number of posterior-sampling runs K_2 , NSGA-II parameters (population size, generations).

Ensure: Candidate set $\mathcal{A}_n$.

- 1: **LHD component:** Draw K_1 solutions via Latin Hypercube Design in $[0, 1]^d$, round to the problem's integer grid; set $\mathcal{A}_n^{\text{LHD}} \leftarrow$ these solutions.
 - 2: $\mathcal{A}_n^{\text{post}} \leftarrow \emptyset$
 - 3: **for** $k = 1, \dots, K_2$ **do**
 - 4: For each $i = 1, 2, 3$: sample $\tilde{\beta}^i \sim \mathcal{N}(a_{\beta,n}^i, C_{\beta\beta,n}^i)$.
 - 5: Run NSGA-II over $u \in [0, 1]^d$ on $\min(\varphi(x(u))^\top \tilde{\beta}^1, \varphi(x(u))^\top \tilde{\beta}^2)$, where $x(u)$ is the nearest feasible integer-grid solution (with VEPM-based feasibility constraint when $n > n_{\text{thr}}$).
 - 6: Add each integer-grid Pareto-optimal solution $x(u)$ to $\mathcal{A}_n^{\text{post}}$.
 - 7: **end for**
 - 8: **return** $\mathcal{A}_n = \mathcal{A}_n^{\text{LHD}} \cup \mathcal{A}_n^{\text{post}}$.
-

Algorithm 3 Compute $h(\mathbf{a}, \mathbf{b})$ (adapted from [Frazier et al. \(2009\)](#))

Require: Vectors $\mathbf{a} = (a_1, \dots, a_M)$ and $\mathbf{b} = (b_1, \dots, b_M)$ with not all b_j equal.

Ensure: $h(\mathbf{a}, \mathbf{b})$.

- 1: Sort indices so that $b_1 \leq b_2 \leq \dots \leq b_M$.
- 2: Remove dominated alternatives: an index j is dominated if there exists k with $a_k \geq a_j$ and $b_k \geq b_j$ (with at least one strict). Let $\mathcal{I}$ be the set of non-dominated indices, ordered by b .
- 3: For consecutive pairs $(j, j+1)$ in $\mathcal{I}$, compute the crossover point $z_j = (a_{j+1} - a_j)/(b_j - b_{j+1})$.
- 4: Compute the expectation:

$$h(\mathbf{a}, \mathbf{b}) = \sum_{j \in \mathcal{I}} (a_j + b_j \mathbb{E}[Z \mid z_{j-1} \leq Z < z_j]) \Pr(z_{j-1} \leq Z < z_j) - \max_j a_j,$$

where $\mathbb{E}[Z \mid z_1 \leq Z < z_2] = (\phi(z_1) - \phi(z_2))/(\Phi(z_2) - \Phi(z_1))$ and ϕ, Φ are the standard normal PDF and CDF.

- 5: **return** $h(\mathbf{a}, \mathbf{b})$.
-

Algorithm 4 VEPM Update at Stage n

Require: Sampled solution x^n , observation Y_{n+1}^i , posterior mean $\mu_n^i(x^n)$ evaluated before the posterior update, current estimates $\hat{\sigma}_n^{i,2}(\cdot)$, partition combination $c^* = c(x^n)$, sample count n_{x^n} , weight w_i .

Ensure: Updated estimates $\hat{\sigma}_{n+1}^{i,2}(\cdot)$.

- 1: **if** $x^n \in \mathcal{X}_n$ **then** $\triangleright$ Revisiting a sampled solution
 - 2: $\hat{\sigma}_{n+1}^{i,2}(x^n) \leftarrow \frac{(w_i + n_{x^n} - 1) \hat{\sigma}_n^{i,2}(x^n) + (Y_{n+1}^i - \mu_n^i(x^n))^2}{w_i + n_{x^n}}$
 - 3: $\hat{\sigma}_{n+1,c^*}^{i,2} \leftarrow \frac{\sum_{x' \in \mathcal{S}_{n+1,c^*}} n_{x'} \hat{\sigma}_{n+1}^{i,2}(x')}{R_{n+1,c^*}}$
 - 4: **else** $\triangleright$ New solution sampled for the first time
 - 5: $\hat{\sigma}_{n+1}^{i,2}(x^n) \leftarrow \frac{w_i \hat{\sigma}_0^{i,2}(c^*) + (Y_{n+1}^i - \mu_n^i(x^n))^2}{w_i + 1}$
 - 6: $\hat{\sigma}_{n+1,c^*}^{i,2} \leftarrow \frac{R_{n,c^*} \hat{\sigma}_{n,c^*}^{i,2} + n_{x^n} \hat{\sigma}_{n+1}^{i,2}(x^n)}{R_{n,c^*} + n_{x^n}}$
 - 7: **end if**
 - 8: For all $x \neq x^n$: $\hat{\sigma}_{n+1}^{i,2}(x) \leftarrow \begin{cases} \hat{\sigma}_n^{i,2}(x) & \text{if } x \in \mathcal{X}_{n+1} \\ \hat{\sigma}_{n+1,c(x)}^{i,2} & \text{if } x \notin \mathcal{X}_{n+1} \end{cases}$ $\triangleright$ Sampled solutions keep individual estimates; unsampled use partition estimate
 - 9: **return** updated $\hat{\sigma}_{n+1}^{i,2}(\cdot)$.
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